

Probabilistic Narrative Identification: Evidence from Tax Policy*

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Preliminary and incomplete.
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Abstract

This paper generalizes narrative identification by treating ambiguity in the historical record as identifying information. Traditional narrative approaches resolve documentary evidence through a binary classification, treating selected policy actions as fully exogenous. Instead, I interpret narrative evidence as probabilistic information about latent exogeneity and construct a probability that a policy action belongs to the exogenous class. Identification proceeds at the level of reduced-form moments: variation in narrative support disciplines outcome–policy comovement, allowing the orthogonal-class moment to be recovered. I construct this probability with a document-grounded language-model procedure that retrieves policy-specific passages, extracts stated motivations, and maps those motivations into exogeneity probabilities.

Empirically, I first validate the measure against the benchmark classifications of Romer and Romer (2010) and Cloyne (2013). I then construct a new dataset of routine U.S. federal tax actions from 1950 to 2024. The resulting aggregate tax responses are contractionary, closely aligned with the canonical narrative tax literature, and operate mainly through investment. Finally, I construct a new narrative dataset of French VAT changes from 1960 to 2024 and combine it with UK VAT changes from Cloyne’s dataset. The pooled UK–France evidence provides the indirect-tax counterpart: VAT increases contract output mainly through private consumption and generate a short-lived rise in inflation, consistent with tax pass-through.

Keywords: Narrative identification; fiscal policy; tax shocks; causal inference.

JEL Codes: C21; C26; E62; E32.

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Introduction

Evaluating the macroeconomic effects of fiscal policy is empirically challenging because policy actions typically respond endogenously to economic conditions. Legislated tax and spending changes reflect contemporaneous shocks, anticipated developments, and evolving policy objectives, confounding causal interpretation. A large literature has proposed identification strategies to address this problem, including structural VARs, external-instrument approaches, and narrative methods that exploit documentary evidence on policymakers’ stated motivations. Within this landscape, narrative identification has played a central role in settings where credible instruments are scarce and policy intent must be inferred from historical records. Influential applications span fiscal and monetary policy and include work by Friedman and Schwartz (1963), Mertens and Ravn (2012), Ramey (2011), and Romer and Romer (1989, 2010, 2023). These designs require researchers to determine which historical episodes provide sufficiently credible support for orthogonality, a consequential judgment when the documentary record contains mixed or incomplete evidence.

Traditional narrative identification resolves this judgment through a binary inclusion rule. Policy actions are retained when the documentary record is judged sufficiently supportive of the maintained orthogonality condition, and estimation proceeds with the resulting shock series. Actions that do not meet this evidentiary criterion do not enter the identifying moments, even though their documentary record may remain informative about latent class membership. Within the retained series, differences in the strength of documentary support are likewise not used directly. This selection strategy has proved powerful in practice. Fiscal measures, however, frequently reflect mixed motivations, and the documentary record often provides only partial or ambiguous support for orthogonality. A binary design therefore resolves this ambiguity before estimation and leaves variation in the strength of narrative support unused.

This paper recasts narrative identification through a two latent-type representation that makes explicit the role of classification uncertainty. Rather than asking whether a policy action is exogenous or not, the framework interprets documentary evidence as providing calibrated narrative support for assignment to an orthogonal latent class. Intermediate classifications therefore reflect uncertainty about latent exogeneity, not partial exogeneity. Identification proceeds by exploiting systematic variation in narrative support across policy actions. Under calibration, stronger support corresponds to a larger average share of actions in the orthogonal class. The resulting cross-moment variation disciplines the causal interpretation of policy–outcome comovement without requiring any observed action to be treated as perfectly exogenous. Identification further requires that, conditional on latent class membership, narrative support not systematically sort the relevant outcome–policy cross-moment.

This perspective shifts the identifying content of narrative analysis away from a zero-contamination requirement for a selected subsample and toward moment restrictions involving imperfect information about latent orthogonality. Causal inference rests on how outcome–policy comovement changes with the strength of narrative support, not on the recovery of observation-level structural shocks. Section 1 formalizes this argument and characterizes the conditions under which variation in narrative support identifies the outcome–policy moment associated with the orthogonal latent class.

To operationalize this perspective, the paper develops a document-grounded measure of narrative support for exogeneity. Conventional narrative identification applies broad coding principles to many policy-specific records, requiring numerous judgments about the relevant evidence and its interpre-

tation. The document-grounded construction concentrates those choices in a low-dimensional set of taxonomy, retrieval, calibration, and model choices that can be stated and scrutinized explicitly. Large language models then extract stated motivations from contemporaneous policy documents and map them into calibrated policy-level probabilities. Conditional on the specified choices, the mapping from documents to probabilities is deterministic and reproducible, making the measurement rule portable across policy settings. The resulting probabilities remain traceable to the underlying documentary record, permitting validation and diagnostics of the evidence supporting the identifying variation.

The empirical analysis begins by evaluating the document-based support measure against established narrative evidence. In edited Romer–Romer narratives, I remove the authors’ explicit classification language and ask whether the remaining evidence assigns systematically greater support to episodes that they classify as exogenous. I then turn to Cloyne’s UK tax dataset, where the full document-grounded construction can be evaluated against an independent expert classification at the policy-row level. In both settings, actions classified as exogenous receive higher narrative support on average than their endogenous counterparts. This indicates that the measure captures a systematic signal present in traditional narrative classifications. The original Romer–Romer sample is too small and sparse to yield a precise continuous support-indexed macro estimate on its own. Thresholding the support score can yield estimates resembling the canonical responses, but those estimates hinge on the level of documentary support that the researcher treats as sufficient for inclusion. As the threshold is tightened, the retained sample can become so sparse that the estimated response changes sharply and no longer provides a credible empirical basis for inference. Thresholding therefore requires the researcher to defend a particular point at which ambiguous narrative evidence becomes sufficiently credible to support a zero-contamination restriction. Reasonable changes in that judgment can overturn the empirical conclusion.

The main U.S. application revisits the macroeconomic effects of tax changes while relaxing the binary narrative requirement that the researcher select a credibility threshold and maintain a zero-contamination restriction for the resulting series. This requires substantially more policy variation than is available in the original Romer–Romer sample. I therefore construct a new dataset of routine federal tax measures drawn from the annual budget process, covering more than 1,000 actions outside the original sample. The denser policy record provides the variation in narrative support needed to estimate support-indexed moments and recover the orthogonal-class response from how outcome–tax comovement sorts across support levels. The resulting estimates imply that tax increases contract output and lower inflation, with the output response concentrated in investment. The aggregate response is closely aligned with Romer–Romer, so their central empirical finding survives when the binary zero-contamination requirement is relaxed and identification uses variation in narrative support across routine budget measures. A composition analysis attributes most of the aggregate response to individual and corporate income-tax changes.

Finally, I construct a new long-run narrative dataset of French VAT changes from 1960 to 2024 and combine it with UK VAT changes from Cloyne’s dataset. VAT changes are relatively infrequent and their documentary record often combines inflation, fiscal, structural, and harmonization motives. This makes binary narrative selection especially restrictive: a hard design either retains only a small set of unambiguous episodes or requires consequential binary decisions for borderline cases. The support-indexed moments use the policy-level support probabilities without requiring a definitive classification for each episode. The pooled UK–France estimates provide the indirect-tax counterpart to the U.S.

results: VAT increases raise prices on impact and contract output primarily through private consumption.

The paper makes three contributions. First, it develops a probabilistic formulation of narrative identification in which variation in narrative support identifies the orthogonal-class policy–outcome moment under explicit calibration and within-class stability restrictions. Second, it provides a reproducible document-based measurement rule that concentrates policy-specific judgment in a low-dimensional set of explicit choices while preserving traceability to the underlying evidence. Third, it validates the resulting support probabilities against the Romer–Romer and Cloyne classifications and applies the framework to new U.S. budget and French VAT data. The U.S. application preserves the central contractionary tax response under weaker binary-classification requirements and identifies its tax composition, while the VAT application extends narrative identification to an indirect-tax setting characterized by mixed motives.

Literature Review

Identification in Macroeconomics. The paper first situates itself within the literature on policy identification, which includes empirical strategies relying on reduced-form VARs identified through timing, sign, or other model-based restrictions (Blanchard & Perotti, 2002; Sims, 1980; Uhlig, 2005). A distinct line of work instead employs external instruments—variables correlated with the policy shock of interest but orthogonal to other structural innovations—to achieve identification (V. Ramey, 2016; Stock & Watson, 2018). Narrative approaches provide a primary methodology for constructing such instruments, using contemporaneous evidence on policymakers’ stated motives. Foundational examples include Friedman and Schwartz (1963) for monetary policy and Hamilton (1985) for oil shocks, with subsequent applications (Ramey, 2011; Ramey & Shapiro, 1998; Ramey & Zubairy, 2018; Romer & Romer, 1989, 2004, 2010) establishing narrative identification as a leading empirical tool. A key limitation of these series is their extreme sparsity: as documented by Barnichon and Mesters (2025), typical narrative shock measures contain roughly 80 percent zeros, which substantially weakens statistical power. Their estimator proposes an econometric remedy for this power–credibility trade-off, but the underlying sparsity reflects, in part, the difficulty of constructing narrative shocks with limited coverage of smaller policy actions.

This paper contributes by reinterpreting narrative classifications as probabilistic information about exogeneity rather than as binary inclusion decisions. Traditional narrative approaches implicitly impose a zero-contamination restriction on a selected subset of policy actions. In contrast, the framework developed here treats exogeneity as a latent property and interprets documentary evidence as probabilistic information about the likelihood that a given action reflects orthogonal policy variation. Identification proceeds through observable cross-moments: variation in narrative support across policy actions isolates the outcome response associated with policy changes that are most plausibly exogenous, without requiring a perfectly exogenous subsample or the recovery of observation-level structural shocks. In this sense, the approach preserves the core logic of narrative identification, using historical evidence to assess orthogonality in expectation, while relaxing the knife-edge assumption that selected actions are free of endogenous contamination.

Text and AI in Economic Analysis. A growing literature uses text as an empirical input in economics, turning unstructured documents into structured variables such as beliefs, expectations,

policy intent, or economic narratives. Foundational surveys emphasize that this requires defining the text unit, the target construct, the representation of text, and the validation design (Ash & Hansen, 2023; Gentzkow et al., 2019).

Recent LLM-based empirical work differs in how it treats the model output. One approach uses the LLM as a scalable research assistant, asking it to read documents and produce labels, summaries, rationales, or structured fields from text. This approach is useful when the goal is to reproduce or scale human coding of economic narratives (Clayton et al., 2025; Jamilov, 2025; Martell, 2025). A complementary approach treats the model as a source of statistical signals: classifier probabilities, token probabilities, embeddings, confidence measures, or model disagreement are retained and then calibrated, thresholded, or aggregated into economic variables (Buckmann & Hill, n.d.; Chen et al., 2026; Hartley, 2025).

This paper combines these two perspectives. The language model is used to organize contemporaneous fiscal narratives, but the final object is not a hard LLM label. Instead, model-generated textual signals are treated as noisy measurements of latent narrative exogeneity and are mapped into calibrated policy-level probabilities. This links the paper to recent concerns about validation and generated variables when AI-produced measurements enter econometric analysis (Battaglia et al., 2024; Carlson & Dell, 2025; Ludwig et al., 2025). Relative to existing LLM measurement papers, the contribution is to connect soft textual classification to narrative identification: the probability object is not only a confidence score or descriptive index, but an input into the moment restrictions that define the empirical object of interest.

Imperfect Moments and Measurement Robustness. The text and AI literature motivates why the constructed support index should be treated as a generated and potentially noisy variable. The moment-based formulation developed below determines how that noise enters the identification problem. Once narrative support enters the outcome–policy moment, residual classification error can be interpreted as a form of moment contamination. This connects the paper to econometric work on conditional moment restrictions (Chamberlain, 1987), plausibly exogenous or imperfect instruments (Conley et al., 2012; Nevo & Rosen, 2012), corrected moments under measurement error (Evdokimov & Zeleneev, 2022), and nonclassical measurement error with multiple proxies (Hu & Schennach, 2008). Since LLM-generated variables may violate classical-error assumptions, the analysis also relates to approximate or misspecified moment conditions (Armstrong & Kolesár, 2021; Bonhomme & Weidner, 2022; Hansen & Lee, 2021). Relative to this literature, the contribution is to show how the credibility problem in narrative identification can be written as a moment-contamination problem, making these robustness tools directly relevant.

Fiscal Policy Transmission and Tax Composition. A final strand of literature concerns the macroeconomic transmission of fiscal policy. Seminal VAR-based analyses (Auerbach & Gorodnichenko, 2012; Blanchard & Perotti, 2002; Mountford & Uhlig, 2009) show that exogenous fiscal expansions raise activity, with multipliers varying across instruments and business-cycle conditions. Narrative studies of government spending shocks (Ramey, 2011; Ramey & Zubairy, 2018) highlight the role of expectations, while analogous work on taxation (Cloyne, 2013; Mertens & Ravn, 2012, 2013; Romer & Romer, 2010) shows that tax multipliers depend on timing, composition, and economic context. This literature also emphasizes that tax composition matters: income, corporate, payroll, and consumption-

tax changes need not transmit through the same channels. That concern is central to the empirical exercises below. The Cloyne exercise evaluates whether the probability procedure recovers expert labels in a setting with rich tax-type heterogeneity, while the U.S. budget extension asks whether a dense set of routine federal tax measures can be filtered into macroeconomically meaningful identifying variation.

1 A Probabilistic Model of Narrative Identification

The objective of this paper is to estimate the dynamic causal effects of legislated tax changes on macroeconomic outcomes using narrative evidence from policy documents. Let $\Delta\tau_t$ denote the aggregate legislated tax change in period t , and let y_{t+h} denote an outcome of interest at horizon $h \geq 0$. The structural relation of interest is

$$y_{t+h} = \beta_h \Delta\tau_t + \varepsilon_{t+h},$$

where β_h is the causal response of the outcome to an orthogonal tax innovation and ε_{t+h} collects other macroeconomic disturbances. Identification requires the relevant orthogonality condition, $\mathbb{E}(\Delta\tau_t \varepsilon_{t+h}) = 0$, to hold over the estimation sample; it fails when tax policy responds endogenously to current or anticipated conditions.

Traditional narrative identification addresses this problem by reading contemporaneous policy documents and classifying each tax change as either admissible or inadmissible for identification. That approach has proved influential in empirical work on tax policy, as in Romer and Romer (2010), Mertens and Ravn (2013), and Cloyne (2013), but it hardens a fundamentally uncertain historical judgment into a binary decision. The broader point is not specific to tax narratives. Other narrative fiscal instruments, such as defense-news shocks in Ramey and Zubairy (2018), are also motivated by historical judgments about plausibly exogenous sources of variation. Although major military buildups are often driven by geopolitical events rather than domestic stabilization policy, the resulting narrative signal need not be treated as exogenous with probability one: some defense-spending news may still be correlated with political, strategic, or macroeconomic conditions relevant for the outcome under study. Narrative evidence is therefore better viewed as providing probabilistic, rather than certain, identifying information.

This section takes that view literally. Narrative evidence is summarized by a continuous index $p_t \in [0, 1]$ measuring the strength of support for an orthogonal interpretation of the tax change in period t . This shift moves the identifying argument away from hard classification and toward moment recovery: variation in p_t across observations is used to trace how contamination changes across the sample and to recover the orthogonal policy moment. The resulting estimator is therefore best understood as a moment estimator that adjusts for contamination rather than as a thresholding rule.

In that sense, the framework is closer in spirit to work on imperfect instruments and approximate moment conditions (Conley et al., 2012; Nevo & Rosen, 2012) and to recent analyses of inference under misspecified moment conditions (Armstrong & Kolesár, 2021; Bonhomme & Weidner, 2022; Hansen & Lee, 2021). The contribution here is to bring in a new source of auxiliary information: an LLM-based reading of policy documents that delivers an observation-level measure of narrative support. That measure disciplines how contamination varies across observations and thereby allows recovery of the

orthogonal policy moment.

1.1 Traditional Narrative Identification as Hard Classification

Traditional narrative identification maps documentary evidence into a binary inclusion rule. A tax change is retained for identification when the historical record is judged to support an exogenous interpretation, and it is discarded otherwise. Let $\mathbb{1}\{\Delta\tau_t \text{ is classified as exogenous}\}$ denote that decision.

Although narrative studies typically report only the binary classification, it is useful to represent that decision as thresholding an underlying soft assessment $p_t \in [0, 1]$:

$$\mathbb{1}\{\Delta\tau_t \text{ is classified as exogenous}\} = \mathbb{1}\{p_t \geq \bar{p}\},$$

for some evidentiary cutoff $\bar{p} \in (0, 1)$. The identifying assumption of the traditional approach is that, once this cutoff is applied, the retained observations are treated as orthogonal to other macroeconomic disturbances:

$$\mathbb{E}[\varepsilon_{t+h}\Delta\tau_t \mid p_t \geq \bar{p}] = 0, \quad \text{for } h \geq 0.$$

That is, narrative classification is used to construct a subset of tax changes that is taken to satisfy the exogeneity condition required for identification.

Under this interpretation, traditional narrative identification imposes two restrictions. First, it discards observations with intermediate narrative support even though these observations may still contain information about the relation between policy variation and macroeconomic outcomes. Second, once the threshold is crossed, residual classification uncertainty is ignored: retained observations are treated as providing orthogonal policy variation. The framework below relaxes both restrictions.

1.2 Probabilistic Narrative Support and Contaminated Moments

Let $\mathcal{D}_t$ denote the contemporaneous documentary record associated with tax change t . Let $u_t \in \{0, 1\}$ denote the latent class of the tax change, with $u_t = 1$ corresponding to the orthogonal class and $u_t = 0$ to the contaminated class. The documentary record is used to form a support index $p_t \in [0, 1]$ for the claim that $u_t = 1$. A useful benchmark interpretation is to view p_t as approximating the posterior probability that the tax change belongs to the orthogonal latent class, conditional on the documentary record $\mathcal{D}_t$. The identifying argument itself requires calibration of the scalar support score:

$$\mathbb{E}(u_t \mid p_t = p) = \Pr(u_t = 1 \mid p_t = p) = p. \tag{1}$$

That is, among observations assigned support level p , the average fraction belonging to the orthogonal class is p .

Within-class moment stability and exclusion restriction. The affine benchmark also requires within-class moment stability: conditional on latent class membership, variation in narrative support should not systematically sort the average outcome-tax moment over the estimation sample. For each

horizon $h \geq 0$ and all p in the support of p_t ,¹

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = j, p_t = p) = \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = j), \quad j \in \{0, 1\}. \quad (2)$$

In particular, even if orthogonality holds within the orthogonal class, the cross moment may still vary with narrative support through systematic variation in shock magnitude, tax composition, implementation timing, anticipation, or related features within class. The benchmark also imposes the orthogonality restriction for the latent orthogonal class,

$$\mathbb{E}(\Delta\tau_t \varepsilon_{t+h} \mid u_t = 1) = 0, \quad \text{for } h \geq 0.$$

Support-indexed moments. Under these restrictions, conditioning the outcome-tax moment on the support index gives:²

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p) = p \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1 - p) \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0). \quad (3)$$

Equation (3) makes the benchmark decontamination logic explicit. Under calibration and within-class moment stability, variation in p_t changes the posterior weight placed on the orthogonal and contaminated class moments. At the endpoint $p = 1$, the contaminated component receives zero weight, so the support-indexed moment coincides with the orthogonal-class moment. The object of interest is therefore $\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1)$. The corresponding normalized response is

$$\beta_h(1) = \frac{\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1)}{\mathbb{E}(\Delta\tau_t^2 \mid u_t = 1)}.$$

This normalizes the orthogonal-class numerator by the squared tax-change magnitude in the same latent class.

The expectations in this section should be interpreted as population counterparts of averages over the historical sample used in estimation. The framework does not require the macroeconomic environment to be stationary or identical across calendar time. More generally, class-conditional moments may vary across periods because of changes in tax composition, institutions, or macroeconomic regimes.

1.3 Identification from Variation in Narrative Support

The key source of identification is variation in p_t across dates. If p_t were constant, the data would reveal only one weighted average of the orthogonal and contaminated outcome-tax moments. With at least two distinct values of p_t , the weights vary and the orthogonal-class moment can be recovered.

To see this most clearly, suppose p_t takes two values, p_L and p_H , with $p_H \neq p_L$. Then (3) implies

$$\begin{cases} \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_L) = p_L \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1 - p_L) \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0), \\ \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_H) = p_H \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) + (1 - p_H) \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0). \end{cases}$$

¹Appendix A shows that the same two-way representation can summarize a richer contaminated component, including multiple latent sources with opposite signs, provided the average contaminated moment is stable conditional on p_t .

²By iterated expectations over u_t , $\mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p) = \Pr(u_t = 1 \mid p_t = p) \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1, p_t = p) + \Pr(u_t = 0 \mid p_t = p) \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0, p_t = p)$. The calibration condition in equation (1) gives $\Pr(u_t = 1 \mid p_t = p) = p$ and $\Pr(u_t = 0 \mid p_t = p) = 1 - p$; the within-class moment stability restriction in equation (2) then allows the conditioning on p_t to be dropped inside the class-conditional moments.

Solving this two-equation system gives the orthogonal-class moment:

$$\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) = \frac{(1 - p_L) \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_H) - (1 - p_H) \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p_L)}{p_H - p_L}.$$

Dividing this moment by $\mathbb{E}(\Delta\tau_t^2 \mid u_t = 1)$ gives the normalized response $\beta_h(1)$. The same system also identifies the contaminated-class moment, so both class-conditional moments are identified. The GMM formulation below preserves this affine case as the baseline, but writes the estimation problem in terms of the more general support-indexed moment.

1.4 Moment Conditions and a GMM Formulation

In practice, the moments are estimated after residualizing outcomes and tax changes on the controls included in the empirical specification.

More generally, let

$$m_h(p) \equiv \mathbb{E}(y_{t+h}\Delta\tau_t \mid p_t = p),$$

denote the support-indexed moment. For estimation, approximate this function by a finite-dimensional specification $m_h(p; \theta_h)$. Correct specification implies the conditional moment

$$\mathbb{E}[y_{t+h}\Delta\tau_t - m_h(p_t; \theta_h) \mid p_t] = 0. \quad (4)$$

Following the conditional-moment logic in Chamberlain (1987), equation (4) implies unconditional moments after multiplication by functions of p_t . In practice, choose a finite vector of functions $\phi(p_t)$ and impose $\mathbb{E}[(y_{t+h}\Delta\tau_t - m_h(p_t; \theta_h)) \phi(p_t)] = 0$. The basis $\phi(\cdot)$ determines how much of the conditional restriction is used in estimation. A parsimonious basis gives a finite GMM approximation; richer bases move the estimator closer to the full support-indexed restriction.

Affine benchmark. Under the calibration and within-class moment-stability restrictions above, reparameterizing equation (3) gives the affine benchmark

$$m_h(p; \theta_h) = \underbrace{\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0)}_{a_h} + p \underbrace{[\mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 1) - \mathbb{E}(y_{t+h}\Delta\tau_t \mid u_t = 0)]}_{b_h},$$

with $\theta_h = (a_h, b_h)'$. Thus $a_h + b_h$ is the orthogonal-class moment and $\beta_h(1) = (a_h + b_h) / \mathbb{E}(\Delta\tau_t^2 \mid u_t = 1)$. For this baseline, the minimal just-identified estimator uses the two moments generated by $\phi(p_t) = (1, p_t)'$; equivalently, it is the OLS projection of $y_{t+h}\Delta\tau_t$ on a constant and p_t .

1.5 Measurement Error in Narrative Support

The preceding analysis treats the support index as observed, while in practice it is generated from text and may contain residual error from model choice, prompting, summaries, and ambiguity in the documents. Since p_t indexes the conditional moment $m_h(p)$, such errors affect both the location of an observation on the support scale and the finite GMM moments used to estimate the support-indexed function. The value of the GMM formulation is that, after projecting the conditional restriction on functions of p_t , measurement error can be studied as a perturbation of standard moment conditions.

The first benchmark is a residual classical-error approximation. After calibration and aggregation of the generated scores, corrected-moment methods in the spirit of Evdokimov and Zeleneev (2022) can be applied to the finite projected moments to understand the attenuation created by noisy support scores. This benchmark is useful because it gives a disciplined correction for small errors in the implemented GMM approximation.

The second layer is robustness to nonclassical error. Recent work on AI-generated and unstructured-data regressors emphasizes that generated variables can be model-dependent, biased, and correlated with features of the underlying text (Battaglia et al., 2024; Carlson & Dell, 2025; Ludwig et al., 2025). Repeated generated measures can document the stability of p_t and, under stronger independence assumptions, connect to multiple-proxy approaches such as Hu and Schennach (2008). More agnostically, one can allow the projected GMM moments to hold only approximately and ask how large the violation must be before the estimated orthogonal response changes materially, following bounded or approximate moment approaches (Armstrong & Kolesár, 2021; Conley et al., 2012).

2 Measuring Narrative Support for Exogeneity

Section 1 takes the support index as given. This section defines the measurement object behind that index and the restrictions imposed when constructing it from contemporaneous policy documents. The object is a policy-level scalar $p_i \in [0, 1]$ summarizing the extent to which the stated motivations for policy action i support assignment to the orthogonal narrative class.

The procedure does not remove judgment from narrative identification. Rather, it shifts judgment from a sequence of case-by-case classification decisions to a low-dimensional set of explicit measurement choices. The researcher must specify the relevant narrative classes, the calibration of textual signals to those classes, and the aggregation rule from motivations to policy-level support. These choices become observable inputs to the empirical design: they can be reported, debated, varied, and subjected to robustness analysis, rather than remaining implicit in a binary classification.

I write p_i for the policy-level support score used in the paper. Although the empirical procedure starts from documents, I first define the probability object constructed from stated motivations, since this is the object that enters the identification argument. The construction proceeds in four steps. Starting from a policy description q_i and its contemporaneous document set $\mathcal{D}_i$, the measurement mapping selects relevant passages, extracts stated motivations, assigns support to each motivation, and aggregates these scores to the policy level:

$$(q_i, \mathcal{D}_i) \rightarrow \mathcal{R}_{K,L}(q_i; \mathcal{D}_i) \rightarrow \mathcal{M}_i \rightarrow \{p_{i\ell}\}_{\ell \in \mathcal{M}_i} \rightarrow p_i.$$

Here $\mathcal{R}_{K,L}$ denotes the retrieval rule, $\mathcal{M}_i$ the set of stated motivations recovered from the retrieved passages, $p_{i\ell}$ the support assigned to motivation ℓ , and p_i the resulting policy-level support score.

What is required is a semantic mapping from stated motivations into the latent classes relevant for identification, such as orthogonal versus contaminated, or exogenous versus endogenous. In the applications below, I adopt the mapping provided by the narrative tax literature, following Romer and Romer (2010)’s distinction between tax changes motivated by current economic conditions and those motivated by more orthogonal objectives, and the more detailed taxonomy proposed by Cloyne (2013). The contribution of the framework is therefore not to redefine the narrative notion of exogeneity, but

to quantify the extent to which the documentary record supports assigning a given policy action to that narrative class. The details of this tax-policy implementation are described in Section 3.

2.1 The Policy-Level Support Object

For a policy action i , the measurement object is the policy-level probability p_i that the contemporaneous documentary record supports interpreting the action as belonging to the orthogonal latent class. The primitive textual evidence is the set of stated motivations recovered from that record,

$$\mathcal{M}_i = \{M_{i1}, \dots, M_{iL_i}\}.$$

A motivation is a short statement of the reason given for the policy action. The motivation-level object is the support probability $p_{i\ell}$, and the policy-level object is a scalar aggregation $p_i \in [0, 1]$.

This construction preserves mixed narrative evidence. A policy action may contain several stated motivations, some pointing toward the orthogonal class and others toward the contaminated class. The object p_i records the strength of support for the orthogonal interpretation rather than forcing the action into a binary narrative label.

2.2 From Stated Motivations to Support Probabilities

Given stated motivations, the empirical task is to assign calibrated probability mass over the researcher-supplied narrative taxonomy. Let $\mathcal{G}$ denote the set of narrative categories and let $\mathcal{G}_{\text{exo}} \subset \mathcal{G}$ denote the categories interpreted as supporting the orthogonal class. The categories are an intermediate representation: the object needed for identification is the probability mass assigned to the orthogonal block.

The 1977 French VAT reform illustrates the mapping. This is a useful example because it is a major tax change: the reform reduced the standard VAT rate from 20 percent to 17.6 percent, merged it with the intermediate rate, and carried a reported fiscal impact of 8.5 billion francs, about 0.5 percent of GDP. It is also useful because the contemporaneous record contains both short-run stabilization motives and structural-reform motives. The retrieved documentary evidence is condensed into three stated motivations:

- **Mitigate the rise in prices after the end of the price freeze:** *the measure aims to limit the inflationary rebound likely to occur at the end of the price-freeze period by reducing the tax burden on consumption.*
- **Complete a structural reform:** *the measure is the final step in a reform undertaken since 1969 to merge the standard VAT rate with the intermediate rate and reduce the number of VAT rates.*
- **Fight inflation:** *the reduction of the rate is part of the broader government programme to combat inflation by easing indirect taxation.*

These motivations are then mapped into category probabilities and aggregated into a policy-level support probability.

Category probabilities. For each motivation, the measurement task is to construct calibrated probability mass over $\mathcal{G}$. In the baseline construction, these probabilities are obtained from language-model representations of the motivation, following recent work that uses model representations and classifier probabilities as statistical measurements of text (Buckmann & Hill, n.d.; Chen et al., 2026; Hartley, 2025). Let $h(M_{i\ell})$ denote the hidden-state representation of motivation $M_{i\ell}$, and let κ denote a calibrated statistical map from hidden states to category probabilities:

$$\hat{P}_{i\ell,c} := \kappa_c(h(M_{i\ell})), \quad c \in \mathcal{G}. \quad (5)$$

The vector $(\hat{P}_{i\ell,c})_{c \in \mathcal{G}}$ is normalized to sum to one. In the empirical implementation, these probabilities are averaged across calibrated specifications that differ in model choice and calibration design. Section 3.2 reports the concrete calibration datasets, models, and aggregation choices used in the applications, including the multinomial logistic calibration used in the baseline specification. Appendix D.3 records an alternative signal-extraction route based on incremental log probabilities from the same category prompts.

Aggregation to policy-level support. The motivation-level probability of supporting the orthogonal class is the calibrated probability mass assigned to exogenous categories:

$$\hat{p}_{i\ell} = \sum_{c \in \mathcal{G}_{\text{exo}}} \hat{P}_{i\ell,c}. \quad (6)$$

For a policy action with extracted motivations $\mathcal{M}_i = \{M_{i1}, \dots, M_{iL_i}\}$, the baseline policy-level support probability is the equal-weight average of motivation-level support:³

$$p_i = \frac{1}{L_i} \sum_{\ell=1}^{L_i} \hat{p}_{i\ell}. \quad (7)$$

Table 1: Robust calibrated category probabilities for the 1977 VAT reform example

	Category	M1	M2	M3	Overall policy prob.
Endo	Demand Management (DM)	0.75	0.02	0.69	0.65
	Spending-Driven (SD)	0.02	0.09	0.04	
	Supply Stimulus (SS)	0.01	0.01	0.01	
	Deficit Reduction (DR)	0.17	0.05	0.08	
Exo	Deficit Consolidation (DC)	0.02	0.22	0.05	0.35
	External (EXT)	0.01	0.05	0.05	
	Ideological (IL)	0.01	0.35	0.04	
	Long-Run (LR)	0.01	0.22	0.05	

Note: Each motivation column sums to 1 across the eight categories. The shaded cell marks the highest category within each motivation. The final column reports the policy-level probability mass assigned to the endogenous and exogenous blocks after averaging across motivations; the two block probabilities sum to 1.

³More general aggregation rules could replace the equal-weight average with $p_i = \sum_{\ell=1}^{L_i} \omega_{i\ell} \hat{p}_{i\ell}$, where $\sum_{\ell=1}^{L_i} \omega_{i\ell} = 1$. Such weights should be used only when the researcher has an auditable rule for ranking the relative importance of stated motivations.

Table 1 reports the calibrated category probabilities for the VAT example. The first and third motivations load mainly on Demand Management, so they contribute mostly to the endogenous or contaminated block. The second motivation loads more on categories in the exogenous block, including Ideological, Long-Run, and Deficit Consolidation. The resulting policy-level support probability is therefore interior: the displayed calculation assigns 0.35 of the probability mass to the exogenous block and 0.65 to the endogenous block.

2.3 Recovering Stated Motivations from Documents

The preceding subsection takes stated motivations as given. I now describe how they are recovered from the documentary record. Let q_i be a short policy description and let $\mathcal{D}_i$ denote the contemporaneous document collection associated with policy action i . Partition the document collection into segments of length L ,

$$\mathcal{D}_i = \{d_{i,1}, \dots, d_{i,N_i}\}.$$

A text representation $\psi(\cdot)$ maps the policy description and each document segment into a common semantic space. Relevance is measured by a distance

$$s_{i,k} = \text{dist}\{\psi(q_i), \psi(d_{i,k})\}, \quad d_{i,k} \in \mathcal{D}_i.$$

The retrieval operator selects the K closest passages,

$$\mathcal{R}_{K,L}(q_i; \mathcal{D}_i) = \arg \min_{d_{i,k} \in \mathcal{D}_i}^K s_{i,k} = \{d_i^{(1)}, \dots, d_i^{(K)}\}. \quad (8)$$

This semantic-distance retrieval step is part of the measurement mapping. The concrete embedding model, segment length, and retrieval depth are design choices described in Section 3.

The retrieved passages are then condensed into stated motivations. Formally, a summarization operator maps the retrieved evidence into a structured set of motivations,

$$\mathcal{S}\{\mathcal{R}_{K,L}(q_i; \mathcal{D}_i)\} = \mathcal{M}_i.$$

The purpose of this step is constrained evidence compression: the summarization operator is not allowed to infer motives absent from the retrieved passages, but only to express stated rationales in a compact and comparable form. In the VAT example, this retrieval and summarization step produces the three motivations used in the probability calculation above.

2.4 Measurement Error, Repeated Measures, and Traceability

The support score is generated from text and should be treated as a measured input rather than as an error-free observation. Conditional on the researcher’s measurement choices, however, the mapping from documents to support is deterministic and systematic. Let

$$\eta = (\eta_R, \eta_S, \eta_C, \eta_K, \eta_A)$$

collect the choices governing retrieval, summarization, semantic classification, calibration, and aggregation. A given admissible specification corresponds to a value $\eta^{(r)}$ and returns

$$\hat{p}_i^{(r)} = f(q_i, \mathcal{D}_i; \eta^{(r)}).$$

When repeated specifications are used, the reported support score is the average

$$p_i = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \hat{p}_i^{(r)}. \quad (9)$$

Differences across $\hat{p}_i^{(r)}$ capture sensitivity to measurement choices. Relevant evidence may be missed at retrieval, summarized motivations may omit or overstate parts of the documentary record, category probabilities may be miscalibrated, and aggregation may place the wrong weight on competing motivations. Agreement across specifications is informative, but it is not sufficient for validity: a stable score may still reflect generic language patterns rather than policy-specific evidence. For this reason, the next subsection reports diagnostics that assess whether extracted motivations are grounded in the retrieved policy-specific passages.

Traceability is an audit requirement. Each final support score should be linked to the policy description, source passages, extracted motivations, category probabilities, calibration choices, and aggregation rule used to transform textual evidence into probabilities. This subsection defines the measurement logic. Section 3 specifies the particular choices used in the applications. The point is not that a given language-model construction is definitive, but that the object entering the econometric analysis is auditable, replicable conditional on stated choices, and open to sensitivity analysis.

The object defined here is policy-level support. When the econometric analysis is conducted at quarterly frequency, these policy-level scores are aggregated into p_t using the amount-weighting rule described in Section 3.4.

2.5 Grounding and Placebo Diagnostics

The measurement procedure is useful only if the stated motivations are anchored in the retrieved policy-specific evidence. Similar values of p_i across measurement constructions do not by themselves establish this: model outputs may reflect generic linguistic regularities or pretraining priors. I therefore assess whether generated motivations depend on the passages that should contain the relevant policy rationale.

Marginal contributions. For a policy action i , let the retrieved passages be $\hat{R}_i = \{d_1, \dots, d_K\}$. I consider counterfactual evaluations in which subsets of these passages are withheld while holding all other inputs fixed. Let $v \in \{0, 1\}^K$ denote the corresponding inclusion vector, with $v_\ell = 1$ when passage d_ℓ is included. If $g(v)$ denotes the log-odds of the model output of interest under inclusion pattern v , the marginal contribution of passage d_ℓ at context $v_{-\ell}$ is

$$\Delta_\ell^g(v_{-\ell}) = g(v_\ell = 1, v_{-\ell}) - g(v_\ell = 0, v_{-\ell}).$$

In practice, I approximate $g(v)$ with a linear surrogate,

$$g(v) \approx b + \sum_{\ell=1}^K w_{\ell} v_{\ell},$$

estimated from random inclusion vectors. Positive coefficients indicate passages that support the generated motivation, while coefficients near zero indicate irrelevance. Table 2 reports estimated passage-level contributions for the 1977 French VAT reform.

Table 2: Attribution weights and placebo-test results across top ten retrieved passages

	Retrieved segments (1–10)									
	1	2	3	4	5	6	7	8	9	10
M_1	41.56	20.92	7.33	-0.24	-1.76	8.36	2.78	-0.69	-4.93	4.47
M_2	18.82	-4.13	15.29	-4.56	0.25	1.20	1.93	-3.47	-0.51	-3.26
M_3	2.88	8.99	1.19	0.76	-1.26	1.04	1.05	-1.40	-0.46	2.67

Note: Each entry reports the estimated marginal log-odds contribution of the corresponding passage to regenerating the motivation. Shaded cells indicate passages that pass the placebo-based grounding test at the 5% level. These entries exhibit positive marginal contributions that are statistically larger than would arise under random assignment of placebo labels. Further implementation details are provided in Section 2.5.

Placebo-based validation. Marginal contributions measure dependence, but a large positive coefficient does not by itself guarantee substantive relevance. For each measure, retrieved passages are partitioned into policy-relevant passages and placebo passages drawn from the same document but semantically unrelated to the measure. The diagnostic compares positive coefficients among relevant passages to the empirical distribution of positive coefficients among placebo passages. Small placebo p-values indicate that a passage’s contribution is unusually large relative to unrelated text. Appendix C.2 reports a complementary global test confirming that the observed grounding patterns are unlikely to arise by chance.

3 Constructing and Using Narrative Support

This section specifies the common empirical protocol used in the applications. The protocol has two economic roles. First, it maps a policy description and its contemporaneous documents into the policy-level support score defined in Section 2, $(q_i, \mathcal{D}_i) \mapsto p_i$. Second, it maps policy-level support and signed tax amounts into the support-indexed moments used to estimate the high-support response in Section 1.4, $(\tau_i, p_i) \mapsto \hat{\beta}_h(1)$. The choices below are therefore not ancillary settings. They determine how narrative evidence is translated into variation in p_i , and how that variation is used to recover the orthogonal policy moment. All applications use this common protocol unless otherwise stated.

3.1 Semantic Taxonomy for Tax-Policy Motivations

For tax policy, the generic semantic mapping is anchored in the narrative tax literature. Romer and Romer (2010) distinguish tax changes motivated by current economic conditions from those motivated by more orthogonal objectives. Cloyne (2013) refine this distinction into a set of tax-policy motivation

categories. I use this taxonomy because it translates the abstract latent classes in Section 1 into auditable economic rationales for tax changes.

The taxonomy determines the economic content of contamination. Demand management, spending-driven changes, supply stimulus, and short-run deficit reduction are treated as potentially contaminated because their stated rationales are explicitly tied to contemporaneous economic or fiscal conditions. Long-run, ideological, external, and consolidation motives are treated as supporting the orthogonal class because their stated rationales are less directly linked to current output disturbances.

Figure 1 recalls the Cloyne categories. Endogenous measures are those linked to contemporaneous economic or fiscal conditions, such as demand management, supply stimulus, deficit reduction, or spending-driven motives. Exogenous measures are justified by objectives less directly tied to the current state of the economy, such as long-run reform, ideological objectives, external constraints, or deficit consolidation. This taxonomy is a design choice for the tax applications, not part of the abstract measurement object.

Group	Sub category	Explanation and examples
Endogenous (N)	1. Demand management (DM)	- Targeting the aggregate level of demand e.g. to boost investment, consumption, growth, or curb inflation. - Specific help to households and individuals by stimulating disposable income. - Dealing with a balance of payments crisis via demand.
	2. Supply stimulus (SS)	- Certain help for businesses during a downturn (e.g. NIC cut). - Short term sector support (e.g. targeted tax cuts for a sector).
	3. Deficit reduction/BoP crisis (DR)	Direct measures to deal with a budget or external deficit contemporaneously caused.
	4. Spending-driven (SD)	Taxes which fund specific spending commitments.
Exogenous (X)	1. LR performance (LR)	- Measures to improve competitiveness, productivity, efficiency and long-run growth (but not taken to offset a shock). - Simplification and deregulation measures. - Long-term support for business or sectors of the economy.
	2. Ideological (IL)	- Long-term social or political goals, independent of their effect on performance and not to offset current shocks. - Some anti-avoidance measures (where no other motive is given).
	3. External (ET)	Court rulings and enforcement of directives.
	4. Deficit consolidation (DC)	- Measures to lower inherited deficit for reasons of economic philosophy or to offset current actions in the future. - Does not include actions forced on the government, or decisions contemporaneous motivated by a current shock.

Figure 1: Classification of Tax Measures Following Cloyne (2013)

Each extracted motivation is interpreted relative to category-specific semantic definitions. These definitions are written as prompts so that the category scheme is auditable and reproducible. In the baseline probability construction, they guide the synthetic calibration corpus rather than serving as final Yes/No classifications. The prompt templates are reported in Appendix D.2.

3.2 Calibration and Probability Construction

The support index enters the affine moment model in Section 1 as a probability object. Raw textual representations do not have this interpretation. Calibration is therefore required to place stated motivations on a common probability scale over the tax-policy taxonomy in Section 3.1. The object of

calibration is the category distribution

$$\Pr(c \mid M_{ij}), \quad c \in \mathcal{G},$$

for motivation j attached to policy action i . Given the set $\mathcal{E} \subset \mathcal{G}$ of categories interpreted as supporting the orthogonal class, the motivation-level support probability is

$$p_{ij} = \sum_{c \in \mathcal{E}} \Pr(c \mid M_{ij}).$$

Thus calibration does not define exogeneity directly. It maps textual evidence into probability mass over economically defined narrative categories, and the exogeneity-support score is the mass assigned to the categories that correspond to the orthogonal block.

The probability map is trained on a synthetic corpus of policy motivations rather than on expert-coded historical labels. The calibration task is defined at the level of individual motivations, whereas the Romer–Romer and Cloyne classifications are defined at the level of policy actions. These classifications therefore do not provide direct training labels for the motivation-level category probabilities required by the framework. The synthetic corpus instead supplies controlled examples of the eight narrative categories, allowing the calibration map to learn the semantic distinction between motivations associated with different narrative classes. Each example is generated to have one primary category, which serves as the training label. Some examples also contain secondary motives, but these are not used as fractional labels. They are included to expose the calibration map to the mixed policy language commonly found in fiscal documents. Because the Romer–Romer and Cloyne classifications are not used in training, they remain available as external validation benchmarks for the resulting probability scale.

The calibration corpus is designed to span the main semantic distinctions and ambiguities that arise in narrative tax classification. It contains twelve independently generated datasets, corresponding to four prompt families and three random seeds. The prompt families cover four types of motivations: canonical examples that anchor the center of each narrative category, institutional fiscal language resembling historical policy documents, boundary cases that separate nearby categories, and mixed-motive examples that combine a primary rationale with secondary considerations. Together, these examples expose the calibration map to both the core semantic content of the taxonomy and the types of ambiguity commonly encountered in fiscal narratives.

Particular attention is given to motivations that are difficult to classify, including cases in which similar language is used to justify conceptually distinct policy actions and cases that lie near the boundary between narrative categories. These are precisely the settings in which narrative classifications tend to be most sensitive to judgment.

Each synthetic set contains 60 examples in each of the eight categories, for a balanced set of 480 motivations. For each set k and each open-weight language model m , I compute hidden-state representations of the motivation text and train an eight-class multinomial logistic decoder. The regularization parameter is selected by stratified cross-validation within the synthetic set. The trained decoder returns a probability distribution over the eight categories for each real extracted motivation. Let $h_{ij}^{(m)}$ denote the hidden-state representation of motivation M_{ij} under model m . The set–model–

specific support probability is

$$p_{ij}^{(k,m)} = \sum_{c \in \mathcal{E}} \widehat{\text{Pr}}_{k,m}(c \mid h_{ij}^{(m)}).$$

For a policy action with several extracted motivations, I first average these probabilities across motivations,

$$p_i^{(k,m)} = \frac{1}{L_i} \sum_{j=1}^{L_i} p_{ij}^{(k,m)}.$$

The policy-level score then averages over the three hidden-state encoders and the twelve independently generated calibration sets,

$$p_i^{\text{rob}} = \frac{1}{12} \sum_{k=1}^{12} p_i^{(k)}, \quad p_i^{(k)} = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} p_i^{(k,m)}, \quad \mathcal{M} = \{\text{Gemma}, \text{Mistral}, \text{Qwen}\}.$$

The baseline empirical support score is p_i^{rob} . I also retain the standard deviation and the 10th and 90th percentiles of the set-level predictions as measures of calibration uncertainty.

Cross-validation within the synthetic corpus establishes that the narrative categories are internally recoverable from the hidden-state representations used in the calibration step. The external validity of the resulting probability scale is assessed separately through the Romer–Romer and Cloyne validation exercises in Section 4. The calibration design provides a probability scale over the maintained narrative taxonomy, averages probabilities across models and synthetic calibration sets, and records the resulting dispersion as a measurement diagnostic.

Measurement stability. The support score is a generated variable and should therefore be treated as a measured input rather than an error-free observation. Because it averages across independently generated calibration sets and three open-weight language models, I use dispersion across sets and models, together with synthetic holdout checks, to assess sensitivity to the calibration design. Appendix E reports the synthetic-corpus construction checks, holdout performance, agreement across calibration sets, and agreement across language models. The evidence indicates that the categories are internally learnable and that support scores are broadly stable, while the remaining variation is retained and reported as calibration uncertainty rather than treated as negligible.

3.3 Document Processing and Motivation Recovery

Retrieval settings. The first empirical restriction is evidentiary: the support score should be based only on contemporaneous passages that plausibly contain the stated rationale for the policy action. Retrieval therefore filters the model’s information set, restricting it to passages close to the policy description rather than allowing the classification to reflect generic knowledge about the policy or period. In the baseline specification, I use OpenAI’s *text-embedding-3-large* model to locate these passages. For each policy action i , the policy description is used as the query against the associated document collection. Documents are split into fixed-length chunks and the baseline retrieval keeps the ten most relevant chunks. Retrieval quality is checked on the French VAT corpus using a manually constructed *gold article* benchmark: each VAT measure is linked to the Senate-report article that discusses the corresponding budget-law provision. Across chunk lengths $L \in \{512, 1024, 2048, 4096\}$,

retrieval is highly stable. For $L = 2048$, at least one of the top-ten chunks overlaps the gold article in 97 percent of cases, and on average 5.5 of the ten retrieved chunks overlap the gold article.

Motivation extraction and representation. The retrieved passages are passed to the language models with a fixed prompt asking for the government’s stated motivations for the tax change. The prompt instructs the model to rely only on the retrieved excerpts, avoid adding interpretation, and return at most three motivations in a uniform bullet-point format. The dynamic fields in the prompt are the fiscal year, the tax type, and the policy description taken from the policy dataset.

The summarization and classification stages are performed with three open-weight instruction-tuned models: Gemma, Mistral, and Qwen. Stochastic elements of text generation are disabled. All retrieved passages, prompts, outputs, model identifiers, hidden-state files, and decoder settings are stored so that each final support score can be traced back to the source text, generated motivations, calibration set, and model calls. Appendix E.1 reports the exact Hugging Face repositories, pinned revisions, and runtime environment used for replication.

3.4 Support-Indexed Moment Estimation

This subsection gives the sample analogue of the support-indexed moment in Section 1.4. For each horizon, I first remove predictable variation in outcomes and tax changes using the country-specific controls. I then estimate the affine relation between the residual outcome–tax moment and narrative support over the full quarterly calendar, and report the high-support endpoint of that relation. The country-specific sections define the macro sample, the outcome variables, the horizon grid, and the exact control vector.

Let $\mathcal{T}_h$ denote the calendar sample for horizon h , and let $n_h = |\mathcal{T}_h|$. The sample contains all quarters for which the lead outcome y_{t+h} , the quarterly tax change, and the controls $W_{t,h}$ are observed. The controls are predetermined quarterly variables, typically lags of the outcome, other macro variables, and the tax-change series. To keep notation compact, I write p_t for the constructed quarterly support score; the final paragraph discusses the fact that this score is generated from text.

Policy-to-quarter aggregation. Let A_t denote the set of policy actions implemented in quarter t . Each action has a signed tax amount τ_i and a policy-level support score $p_i \in [0, 1]$. The signed quarterly tax change is

$$\tau_t = \sum_{i \in A_t} \tau_i.$$

Let $G_t \equiv \sum_{i \in A_t} |\tau_i|$ denote gross action-level tax variation. When $G_t > 0$, the quarterly support index is the absolute-tax-weighted average

$$p_t = \frac{\sum_{i \in A_t} |\tau_i| p_i}{\sum_{i \in A_t} |\tau_i|}.$$

When $G_t = 0$, I set $\tau_t = 0$ and $p_t = 0$. This convention is part of the full-calendar implementation. It means that the left endpoint of the affine projection pools quarters with no policy action and any tax-change quarters with very low narrative support. Hence the intercept should not be read as the mean outcome–tax moment of a separately observed contaminated policy class. It is the fitted value at $p = 0$ under the full-calendar affine specification. The high-support endpoint $p = 1$ is the paper-facing object.

Absolute-tax weighting makes p_t measure the average narrative support attached to the gross policy variation entering the quarterly tax change, while the signed sum τ_t preserves the net macroeconomic impulse. Keeping no-action quarters in the sample also preserves the quarterly spacing used for residualization and HAC inference.

Residualization. For each horizon h , I residualize the outcome lead and the quarterly tax change on the controls $W_{t,h}$:

$$\hat{\pi}_{y,h} = \arg \min_{\pi} \sum_{t \in \mathcal{T}_h} (y_{t+h} - \pi' W_{t,h})^2, \quad \hat{\pi}_{\tau,h} = \arg \min_{\pi} \sum_{t \in \mathcal{T}_h} (\tau_t - \pi' W_{t,h})^2.$$

The residualized variables are

$$\tilde{y}_{t,h} = y_{t+h} - \hat{\pi}_{y,h}' W_{t,h}, \quad \tilde{\tau}_{t,h} = \tau_t - \hat{\pi}_{\tau,h}' W_{t,h}.$$

The numerator and denominator variables are

$$x_{t,h} = \tilde{y}_{t,h} \tilde{\tau}_{t,h}, \quad z_{t,h} = \tilde{\tau}_{t,h}^2.$$

Thus $x_{t,h}$ is the residualized outcome–tax product that estimates the numerator moment, and $z_{t,h}$ estimates the corresponding tax-variation denominator.

Affine endpoint estimator. Let

$$R_t = (1, p_t)'$$

The affine numerator estimator is defined by the sample moment condition

$$\frac{1}{n_h} \sum_{t \in \mathcal{T}_h} R_t (x_{t,h} - R_t' \theta_{x,h}) = 0, \quad \theta_{x,h} = (\alpha_h, \lambda_h)'$$

Equivalently,

$$\hat{\theta}_{x,h} = \left(\sum_{t \in \mathcal{T}_h} R_t R_t' \right)^{-1} \sum_{t \in \mathcal{T}_h} R_t x_{t,h},$$

provided the full-calendar support matrix is nonsingular. This is the OLS projection of $x_{t,h}$ on a constant and p_t over the full quarterly sample. The fitted support-indexed numerator is

$$\widehat{M}_h(p) = (1, p) \hat{\theta}_{x,h}.$$

The high-support endpoint is

$$\widehat{M}_{A,h} = \widehat{M}_h(1) = (1, 1) \hat{\theta}_{x,h} = \hat{\alpha}_h + \hat{\lambda}_h.$$

The left endpoint,

$$\widehat{M}_{0,h} = \widehat{M}_h(0) = \hat{\alpha}_h,$$

is reported only as a diagnostic. Because $p_t = 0$ includes no-action quarters by construction, $\widehat{M}_{0,h}$ does not isolate a latent class of endogenous tax changes. It records the left endpoint implied by the

full-calendar affine projection. Since empirical support scores are typically distributed over the interior of the unit interval, $p = 1$ should likewise be interpreted as the high-support value implied by the affine support-indexed moment, not as a support level that must be observed in the data.

The denominator is estimated symmetrically. Let $\theta_{z,h} = (\alpha_{z,h}, \lambda_{z,h})'$. The sample moment condition is

$$\frac{1}{n_h} \sum_{t \in \mathcal{T}_h} R_t (z_{t,h} - R_t' \theta_{z,h}) = 0,$$

so that

$$\hat{\theta}_{z,h} = \left(\sum_{t \in \mathcal{T}_h} R_t R_t' \right)^{-1} \sum_{t \in \mathcal{T}_h} R_t z_{t,h}.$$

The high-support denominator is

$$\hat{Q}_{A,h} = (1, 1) \hat{\theta}_{z,h} = \hat{\alpha}_{z,h} + \hat{\lambda}_{z,h}.$$

The reported endpoint response is

$$\hat{\beta}_{A,h} \equiv \hat{\beta}_h(1) = \frac{\hat{M}_{A,h}}{\hat{Q}_{A,h}}.$$

This is the full-calendar sample counterpart of the normalized high-support response in Section 1.4. Estimation diagnostics report the rank of $\sum_t R_t R_t'$ and flag horizons for which $\hat{Q}_{A,h}$ is close to zero.

Stacked moment representation. For inference, I write residualization, numerator estimation, and denominator estimation as one just-identified moment system. Let

$$\eta_h = (\pi'_{y,h}, \pi'_{\tau,h}, \theta'_{x,h}, \theta'_{z,h})'$$

and define

$$\tilde{y}_{t,h}(\pi_y) = y_{t+h} - \pi_y' W_{t,h}, \quad \tilde{\tau}_{t,h}(\pi_\tau) = \tau_t - \pi_\tau' W_{t,h}.$$

The stacked score is

$$g_{t,h}(\eta_h) = \begin{pmatrix} W_{t,h} \tilde{y}_{t,h}(\pi_y) \\ W_{t,h} \tilde{\tau}_{t,h}(\pi_\tau) \\ R_t [\tilde{y}_{t,h}(\pi_y) \tilde{\tau}_{t,h}(\pi_\tau) - R_t' \theta_{x,h}] \\ R_t [\tilde{\tau}_{t,h}(\pi_\tau)^2 - R_t' \theta_{z,h}] \end{pmatrix}.$$

The estimator $\hat{\eta}_h$ solves

$$\frac{1}{n_h} \sum_{t \in \mathcal{T}_h} g_{t,h}(\hat{\eta}_h) = 0.$$

This representation incorporates residualization, numerator estimation, and denominator estimation in the same moment system. Because the baseline system is just identified, the GMM weighting matrix does not affect the point estimate. Richer support bases can be estimated as overidentified specifications, but the affine estimator above is the baseline sample analogue of the moment model in Section 1.4.

HAC inference. Let

$$\hat{g}_{t,h} = g_{t,h}(\hat{\eta}_h), \quad \hat{G}_h = \frac{\partial}{\partial \eta'} \left[\frac{1}{n_h} \sum_{t \in \mathcal{T}_h} g_{t,h}(\eta) \right] \bigg|_{\eta = \hat{\eta}_h}.$$

Write the ordered calendar sample as $\mathcal{T}_h = \{t_1 < \dots < t_{n_h}\}$. For $\ell \geq 0$, define

$$\hat{\Gamma}_{\ell,h} = \frac{1}{n_h} \sum_{r=\ell+1}^{n_h} \hat{g}_{t_r,h} \hat{g}'_{t_{r-\ell},h}.$$

The Newey–West long-run covariance estimator is

$$\hat{\Omega}_h = \hat{\Gamma}_{0,h} + \sum_{\ell=1}^{L_h} \left(1 - \frac{\ell}{L_h + 1}\right) (\hat{\Gamma}_{\ell,h} + \hat{\Gamma}'_{\ell,h}), \quad L_h = \max\{4, h\}.$$

The bandwidth is therefore at least four quarters and increases with the impulse-response horizon. The covariance matrix for the stacked estimator is

$$\hat{V}_{\eta,h} = \frac{1}{n_h} \hat{G}_h^{-1} \hat{\Omega}_h \hat{G}_h^{-1'}.$$

Under the standard weak-dependence and nonsingularity conditions for these sample moments, conditional on the constructed support sequence,

$$\sqrt{n_h}(\hat{\eta}_h - \eta_{h,0}) \Rightarrow \mathcal{N}(0, G_h^{-1} \Omega_h G_h^{-1'}).$$

Endpoint standard errors use the delta method. Let $c = (1, 1)'$ and write

$$b(\eta_h) = \frac{c' \theta_{x,h}}{c' \theta_{z,h}}.$$

The gradient has zero entries with respect to the residualization coefficients and

$$\nabla_{\theta_x} b(\hat{\eta}_h) = \frac{c}{\hat{Q}_{A,h}}, \quad \nabla_{\theta_z} b(\hat{\eta}_h) = -\frac{\hat{M}_{A,h}}{\hat{Q}_{A,h}^2} c.$$

Thus

$$\hat{V}_{\beta,h} = \nabla b(\hat{\eta}_h)' \hat{V}_{\eta,h} \nabla b(\hat{\eta}_h).$$

Equivalently, the endpoint influence score is

$$\hat{\psi}_{t,h} = -\nabla b(\hat{\eta}_h)' \hat{G}_h^{-1} \hat{g}_{t,h},$$

and the scalar HAC variance can be computed from $\hat{\psi}_{t,h}$. Reported confidence bands use the delta-method standard error $\hat{V}_{\beta,h}^{1/2}$ and include sampling variation in the endpoint denominator.

Generated support scores. The baseline inference conditions on the constructed support sequence $\{\hat{p}_t\}_{t \in \mathcal{T}_h}$. This treats the document-to-support mapping as fixed when computing the time-series un-

certainty of the macroeconomic estimator. If the first-stage error in the support score is asymptotically negligible, for example if

$$\max_{t \in \mathcal{T}_h} |\hat{p}_t - p_t| = o_p(n_h^{-1/2}),$$

then the HAC variance above is first-order valid for $\hat{\beta}_{A,h}$. If instead the first-stage construction has a first-order asymptotically linear component, its score can be appended to $g_{t,h}(\eta_h)$ and the derivative of the affine moments with respect to p_t included in $\hat{G}_h$. The baseline does not impose this stronger structure. It reports dispersion across calibration sets and language models, together with alternative support specifications, as measurement diagnostics.

Specification diagnostics. Appendix diagnostics examine whether endpoint estimates are sensitive to alternative support specifications, such as threshold, bin, rank-normalized, local, or spline variants. These diagnostics assess whether the high-support response is driven by the affine extrapolation to $p = 1$. They are robustness checks, not separate identification strategies.

4 External Validation of Narrative Support

This section validates the proposed narrative-support measure against external benchmarks. The empirical design separates two validation margins. First, using edited Romer–Romer narratives provided in their appendix, does the probability-construction stage assign higher support to episodes classified as exogenous in the canonical U.S. narrative series? Second, using Cloyne’s UK tax data, does the full document-grounded procedure align with an independent classification when retrieval, summarization, and motivation extraction are all performed from policy documents?

Expert narrative classifications inevitably involve difficult judgment calls, particularly for episodes with mixed or ambiguous motivations. Nevertheless, because the support measure is built from the same semantic distinction between categories that underlies the narrative literature, one would expect episodes classified as exogenous to receive stronger documentary support for exogeneity on average. Establishing this relationship is what makes the probability object informative for the budget-document application that follows.

4.1 Validation using the Romer–Romer narratives

This subsection evaluates whether the common procedure in Section 3 recovers the economic content of Romer and Romer (2010) when applied to the edited narratives used in the original study. The exercise holds fixed the curated Romer–Romer historical narratives and asks whether the procedure assigns systematically higher exogeneity support to RR-exogenous episodes and whether the resulting probability-based shock series reproduces the benchmark macroeconomic response.

By conditioning on the Romer–Romer narratives, this benchmark abstracts from document retrieval while preserving the summarization, motivation decomposition, and probability aggregation stages of the framework. It therefore tests whether the probability object carries identifying content similar to the original RR narratives.

4.1.1 Setup

The Romer–Romer replication material documents 50 major federal tax changes in the United States between 1945 and 2007. For each episode, the authors provide a detailed narrative discussion of the motivations underlying the reform, drawing on contemporaneous sources such as presidential speeches, budget documents, and Congressional records. Each narrative concludes with a classification of the tax change as either exogenous or endogenous.

To construct the benchmark validation task, I extract the narrative text describing each policy episode and remove the portion in which the authors explicitly state and justify their classification. In particular, I delete the sentence reporting the final classification along with the two immediately preceding sentences, which often contain direct cues revealing the intended label (see Appendix F.1.1 for examples). The resulting edited narratives form the input to the validation exercise.

4.1.2 From edited narratives to policy-level probabilities

The edited RR narrative does not enter the benchmark as a single paragraph to be mapped directly into a binary label. Instead, each edited narrative is first summarized into a set of stated motivations. Because a single RR episode can contain several rationales, the same policy may generate multiple motivations, each of which is scored separately. These motivation-level outputs are then mapped into exogeneity probabilities and aggregated into a policy-level structural probability.

The construction follows the cross-implementation aggregation described in Subsection 2.4; see in particular equation (9). Throughout this benchmark, the object of interest is therefore the policy-level narrative support measure p_i , rather than any single construction-specific estimate $\hat{p}_i^{(r)}$.

4.1.3 Probability validation against RR classifications

Table 3 reports the original Romer–Romer binary label and asks whether RR-exogenous episodes receive systematically higher policy-level probabilities. The mean probability rises from 0.40 for RR-endogenous episodes to 0.62 for RR-exogenous episodes, and the share above tighter thresholds also rises.

Table 3: Policy-Level Structural Probabilities by Original RR Classification

RR class	N	Mean p	Share $p \geq 0.5$	Share $p \geq 0.75$
RR endogenous	24	0.40	20.8%	0.0%
RR exogenous	26	0.62	69.2%	34.6%
Difference (exo - endo)	–	0.23	48.4 pp	34.6 pp

Note: Entries are computed from the 50 edited RR narratives. p is the robust policy-level structural probability averaged across independently generated synthetic motivation calibration sets.

Under the calibration interpretation in Section 1, a support value (p) means that, among policy actions assigned that value, the average share belonging to the orthogonal latent class is (p), while the remaining share ($1-p$) belongs to the contaminated latent class. The mean probability of 0.62 among RR-exogenous episodes therefore indicates stronger support for orthogonality than among RR-endogenous episodes, while still leaving substantial uncertainty about latent class membership. This is consistent with the view that the binary RR label aggregates episodes with heterogeneous degrees

of narrative support. Appendix F.2.5 discusses representative RR cases and shows how intermediate probabilities often correspond to identifiable ambiguities in the original narrative motives, including current-macro motives, spending-package contamination, and deficit or social-insurance financing. Appendix F.2 reports a probabilistic label-prediction exercise showing that support probabilities are systematically related to the original RR classification, together with a training-leakage test designed to detect potential memorization of the RR narratives by the language model that might bias the results.

4.1.4 Romer–Romer through support-indexed moments

I next apply the support-indexed moment estimator from Section 1.4 to the Romer–Romer tax episodes. For each horizon, the controlled outcome–tax moment is projected on narrative support and evaluated at the high-support endpoint ($p=1$). This places the Romer–Romer narratives directly into the identification framework developed in Section 1 and provides a natural benchmark for assessing whether the extracted support measure preserves the identifying content of the original classification.

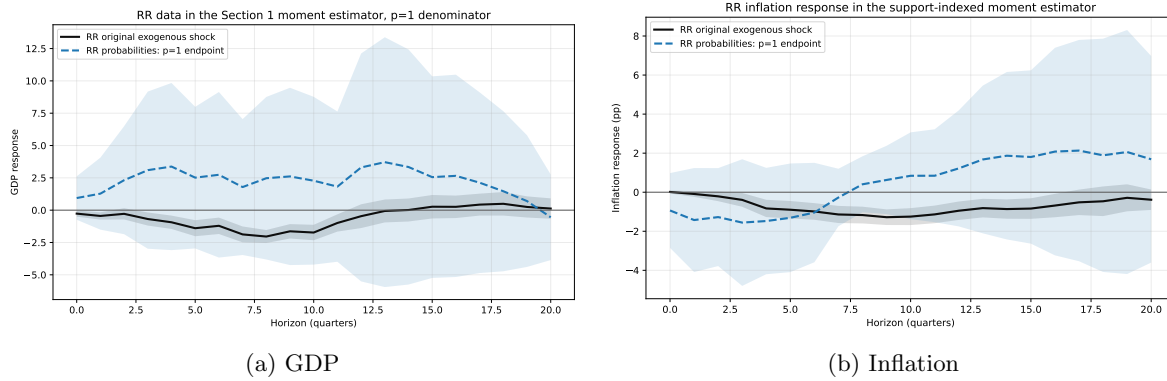


Figure 2: Romer–Romer Panel Under the Support-Indexed Moment Estimator

Note: The figure reports the RR policy panel estimated with the support-indexed moment estimator in Section 1.4. The black line uses the original RR exogenous indicator in the same moment estimator. The blue line uses the probability-based affine endpoint at $p = 1$, normalized by the affine denominator evaluated at $p = 1$. Shaded areas denote 68% HAC confidence bands.

Figure 2 compares the support-indexed moment estimator of Section 1.4 with the same estimator applied to the original RR exogenous classification. The probability-based estimates do not fully recover the benchmark RR responses. This does not appear to reflect a failure of the support measure itself: Table 3 showed that RR-exogenous episodes receive systematically higher support probabilities than RR-endogenous episodes. Rather, the result suggests that the 50 RR episodes contain too little variation in narrative support for the continuous moment-sorting approach to recover the RR benchmark precisely.

Appendix F.2.3 illustrates this point by replacing the continuous support measure with threshold indicators $\mathbb{1}\{p_i \geq \bar{p}\}$. This corresponds to resolving narrative ambiguity through a binary classification rule, as in traditional narrative identification. For thresholds around ($\bar{p} = 0.5$) or (0.6), the resulting responses are substantially closer to the RR benchmark. The comparison becomes sensitive, however, to the particular cutoff chosen by the researcher. Appendix F.2.5 shows that higher thresholds exclude not only episodes with weak support for exogeneity, but also historically important episodes

whose RR classification is plausible yet narratively ambiguous. This sensitivity to the treatment of intermediate-support observations is one reason for using support probabilities directly in the baseline framework rather than imposing a single binary cutoff. Appendix F.2.2 reports the exact Romer–Romer distributed-lag replication as a complementary benchmark closest to the original published specification.

4.2 External Label Validation: Cloyne (2013) UK Tax Changes

The Romer–Romer exercise above isolates the probability-construction stage by starting from curated narrative text. I next ask whether the same probability object remains informative in a more demanding setting, where the relevant policy documents must first be reconstructed. Cloyne (2013) provides a useful benchmark for this purpose. The dataset records individual UK tax changes, their revenue effects and implementation dates, and an expert narrative classification of each change as endogenous or exogenous.

The validation starts from Cloyne’s row-level tax workbook. The reconstruction preserves the 2,462 tax-change rows in the original file and maps them into 85 narrative events. For each event, I build an event-level source base from contemporaneous UK fiscal and parliamentary materials, apply the retrieval and summarization procedure described above, and extract up to three stated motivations for each policy row. These motivations are then mapped into structural exogeneity probabilities. Because Cloyne’s classification is also defined at the policy-row level, it provides a direct external label against which to compare the resulting probabilities.

Cloyne’s classification provides an external benchmark against which to evaluate the probability object. The comparison asks whether rows classified as exogenous by Cloyne receive systematically higher support probabilities than rows classified as endogenous. Table 4 presents the results in the same spirit as the Romer–Romer validation.

Table 4: Policy-Level Structural Probabilities by Cloyne Classification

Cloyne class	N	Mean p	Share $p \geq 0.5$	Share $p \geq 0.75$
Cloyne endogenous	795	0.69	88.2%	40.5%
Cloyne exogenous	1,306	0.76	95.8%	60.3%
Difference (exo - endo)	–	0.08	7.6 pp	19.8 pp

Note: Entries are computed from 2,101 policy rows with usable Cloyne labels and complete probability inputs. “Cloyne exogenous” denotes rows with Cloyne’s major classification X . p is the policy-level structural probability used in the UK validation exercise. No UK macroeconomic outcome enters the probability construction.

The Cloyne validation delivers a positive but more modest separation than the Romer–Romer exercise. The probabilities are high in both classes, making the (0.5) cutoff weakly discriminating. The ordering is nevertheless monotone: Cloyne-exogenous rows have an average probability of 0.76, compared with 0.69 for Cloyne-endogenous rows, and the share above the (0.75) threshold is 19.8 percentage points higher. Appendix F.2.6 reports a one-variable probabilistic label-prediction exercise that confirms the same monotone relationship.

The overlap between the two distributions is informative about the measurement problem. Many Cloyne-endogenous rows receive high probabilities because the reconstructed source text often con-

tains mixed package-level motivations, or because Cloyne’s policy-row classification aggregates several motives within a fiscal package. This is precisely the type of setting in which a continuous support measure is useful: it preserves variation in documentary support that a binary expert label must resolve discretely.

Taken together, the Romer–Romer and Cloyne exercises validate the two measurement margins needed for the rest of the paper. The edited Romer–Romer benchmark shows that the probability-construction stage preserves the ranking implicit in a canonical U.S. narrative series and carries macroeconomic content inside the paper’s support-indexed moment estimator. The Cloyne benchmark shows that the full document procedure remains aligned with an external expert classification when the source base must be reconstructed from policy documents. I now use the validated object to study tax transmission. The application asks not only whether document-based shocks reproduce familiar aggregate responses, but whether high-support tax changes transmit through the margins associated with the tax base they affect.

5 The Main End-to-End Application: Tax Composition in U.S. Budget Documents

The preceding section shows that the narrative-support measure is aligned with expert notions of exogeneity in the Romer–Romer and Cloyne benchmarks. This section applies the full framework to new U.S. policy data. The application has two roles. First, it expands coverage from the major tax acts identified by Romer and Romer to a much broader universe of federal tax measures reported in annual budget documents. Second, because the budget rows retain tax-instrument detail, the application can ask which statutory margins account for the aggregate response. The resulting dataset therefore extends narrative identification to policy actions that lie largely outside the existing RR narrative tax series and makes it possible to decompose the high-support aggregate endpoint by tax composition.

5.1 U.S. tax data and coverage

Existing narrative coverage. The narrative tax dataset of Romer and Romer (2010) identifies major federal tax actions in the United States over the postwar period. It focuses on large and salient legislative episodes, including comprehensive tax reforms and major standalone acts such as income tax reforms, Social Security legislation, and federal highway funding acts. These episodes generate sizable and clearly identifiable changes in aggregate tax liabilities. The resulting series is therefore well suited to measuring large, infrequent tax shocks, but it covers only part of legislated federal tax activity and records tax changes at an aggregated tax-instrument level.

Budget-based extension. I extend this coverage by constructing a dataset of legislated federal tax measures from the annual *Budget of the United States Government* over 1950–2024. For each fiscal year, I extract reported changes in federal taxation and record them at the highest level of detail available, including a disaggregation by tax instrument whenever possible. I apply the same extraction procedure to the major tax acts identified by Romer and Romer (2010), yielding a finer tax-level decomposition than that available in the extension of Mertens and Ravn (2013).

The end-to-end application focuses on the budget-based portion of this broader dataset. The annual budget volumes provide a large set of routine tax measures that mostly fall outside the canonical RR benchmark while retaining the contemporaneous policy text needed to construct narrative support. This sample is therefore the natural setting for applying the framework to policy actions not covered by existing narrative tax series.

The separation between the RR series and the budget-based series is not mechanical. RR records enacted tax changes, including both standalone legislation and budget or reconciliation laws. The annual budget volumes record tax measures as they appear in the executive budget process, including proposed legislation, enacted provisions whose budget effects are reported in later volumes, and extensions or accelerations of earlier laws. As a result, some RR episodes also appear in the budget data. This overlap is useful, because it provides a limited set of cases in which the two sources can be compared at the provision level. But it is not enough to merge the two datasets by law name, date, or aggregate amount. A single RR episode may correspond to several budget rows, and a budget row may describe a later fiscal treatment of an earlier RR law rather than the original enactment itself. Appendix F.1.2 documents the clean examples found so far and explains why combining the two series requires a dedicated provision-level crosswalk.

Table 5 reports the composition of the resulting budget dataset. It aggregates all tax changes explicitly proposed in the yearly *Budget of the United States Government* over the sample period and classifies them into six major federal tax categories. These categories are taken directly from the budget documents and correspond to the government’s own classification of the proposed measures rather than to an ex post classification constructed by the researcher.

For each measure, the dataset records the implementation date reported in the budget documents and assigns the measure to the corresponding quarter. When no specific implementation quarter is stated, the measure is assigned by default to the first quarter of the fiscal year. The table therefore describes tax policy adjustments as they appear in executive budget proposals rather than the universe of all federal tax legislation.

Table 5: Distribution of New Tax Measures by Category (United States)

Tax Category	# New Measures
Individual Income Tax	371
Corporate Income Tax	361
Consumption Tax	238
Payroll / Social Insurance Tax	114
Other Receipts	89
Wealth Transfer Tax	17
Total	1190

5.2 Document corpus

A final step in constructing the U.S. tax dataset is to assemble a corpus of contemporaneous documents that records policymakers’ stated motivations for legislated tax changes. Sources are selected to reflect the policy rationale articulated at the time of enactment and to remain consistent and comparable

over long horizons.⁴ The broader U.S. corpus relies on two recurring source families: the *Budget of the United States Government* and the *Annual Report of the Secretary of the Treasury*.

For the end-to-end benchmark in this subsection, the relevant document source is the annual Budget itself. These budget volumes are the full-document analogue of the edited RR narratives used above: instead of starting from hand-curated narrative excerpts, the procedure retrieves and summarizes contemporaneous budget text. The application follows the common procedure and parameter choices present in Section 3.

5.3 Variation in narrative support and quarterly tax shocks

The budget application combines the policy-level support measures and tax changes into quarterly series that can be matched to the macroeconomic data. This yields a quarterly tax-shock series and a quarterly narrative-support index (p_t), which are the objects used in the support-indexed moment estimator.

Figure 3 reports the distribution of the resulting quarter-level support index. The 1,078 matched budget measures collapse to 134 event quarters. The distribution is strongly interior: the mean is about 0.46, the median is about 0.44, and no event quarter is assigned exactly $p_t = 0$ or $p_t = 1$. This is the relevant empirical environment for the support-indexed estimator. Budget measures do not separate naturally into fully exogenous and fully endogenous observations.

Figure 4 relates this support variation to the quarterly tax-shock series. The upper panel compares the all-budget tax shock with the original Romer–Romer exogenous shock. The budget series is much denser, with 133 non-zero quarters compared with 45 in the RR series. Its timing is distinct from the RR benchmark: the correlation between the two quarterly shock series is -0.03 . The lower panel plots p_t for the same budget-event quarters. The support index varies substantially across time, including among periods with sizable tax changes, and is nearly uncorrelated with the absolute net budget shock: the correlation is about 0.07.

The support-indexed estimator exploits the variation in (p_t) displayed in Figure 4. The estimates below trace how the controlled outcome–tax moment changes with narrative support and then evaluate the affine benchmark at high support. The resulting responses are therefore identified from support variation within a distinct and denser set of routine budget measures rather than from the timing of the Romer–Romer shock series.

5.4 Aggregate budget endpoint

For the end-to-end document exercise, the budget-measure evidence should use the same affine GMM estimator that is the baseline object in Section 1.4. The main RR validation above re-expresses the RR panel in that support-indexed moment estimator; Appendix F.2.2 separately reports the exact distributed-lag comparison closest to Romer and Romer (2010). I therefore keep the main budget figure focused on the full-quarter GMM endpoint constructed from the budget measures alone.

Figure 5 should therefore be read as the paper’s document-based affine GMM endpoint, not as a direct RR replication exercise. The RR moment-sorting check above is the main validation bench-

⁴Romer and Romer (2010) draw on a broad set of contemporaneous sources, including legislative debates and discussions in congressional chambers. Some of these materials are not systematically used here, primarily because their access and coverage are uneven over time. The analysis therefore focuses on recurring executive and Treasury documents that are consistently available throughout the sample period.

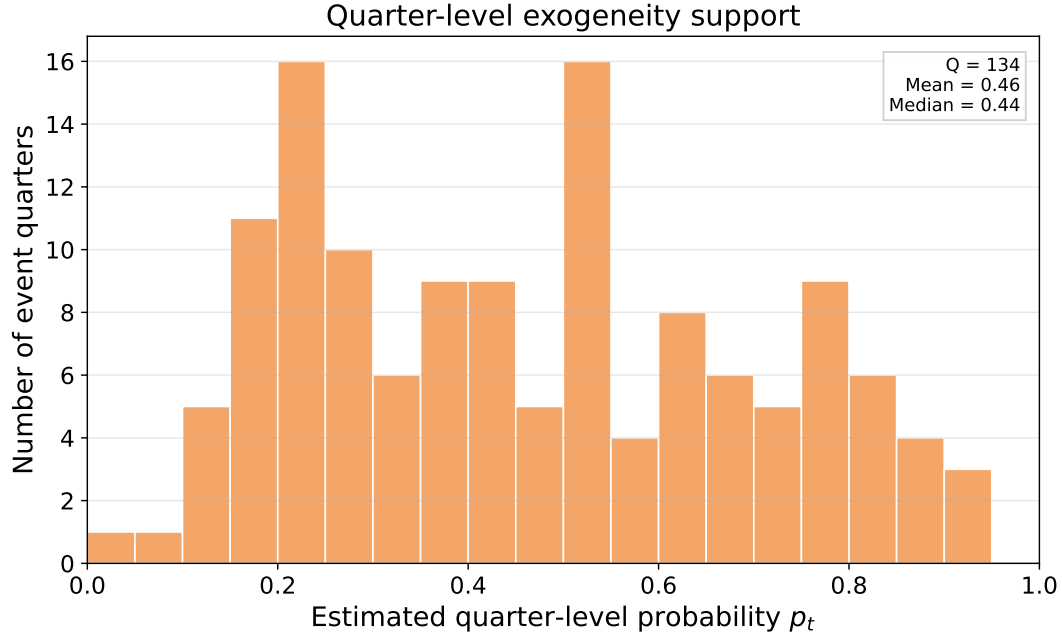


Figure 3: Distribution of Quarter-Level Exogeneity Support for U.S. Budget Tax Measures

Note: The figure uses budget-based U.S. tax measures with complete effective dates, tax amounts, and structural probabilities that can be matched to the quarterly macro sample. Policy-level probabilities are collapsed to the quarter level using absolute tax amounts as weights. The matched sample contains 134 quarters with budget-tax activity.

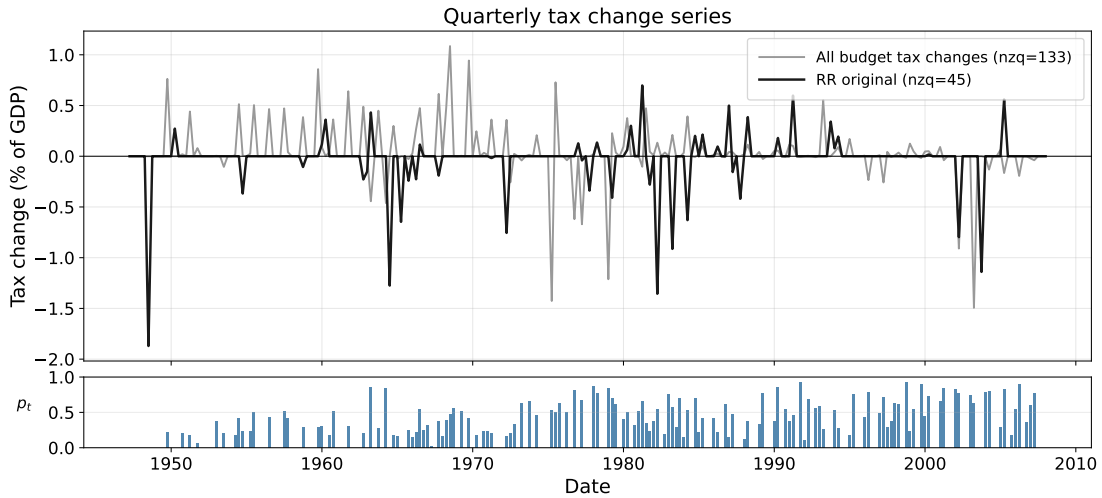


Figure 4: Quarterly Budget Tax Shocks and Narrative Support

Note: The upper panel plots quarterly tax-shock series expressed as a percentage of contemporaneous GDP: all budget tax changes and the original Romer–Romer exogenous tax shock series. The lower panel reports the quarter-level support index p_t for quarters with budget-tax activity. “nzq” denotes the number of non-zero quarters.

mark for the estimator, while the exact distributed-lag replication is kept in the appendix as the closest comparison to the published RR equation. The budget figure asks whether the document-based budget-measure universe produces a plausible high-support endpoint under the paper’s baseline moment estimator.

The document-based budget endpoint is contractionary over the medium run, with its largest GDP decline around quarters 8–12. Inflation also declines after a delay, consistent with the interpretation that the high-support tax increases operate primarily through weaker demand. Appendix F.3 reports the corresponding low-support endpoint as a diagnostic for the affine specification. Because the support index is generated from text, Appendix F.4 also reports two measurement diagnostics for GDP and inflation: an approximate-moment sensitivity exercise and a recomputation of the endpoints using each evaluator family separately.

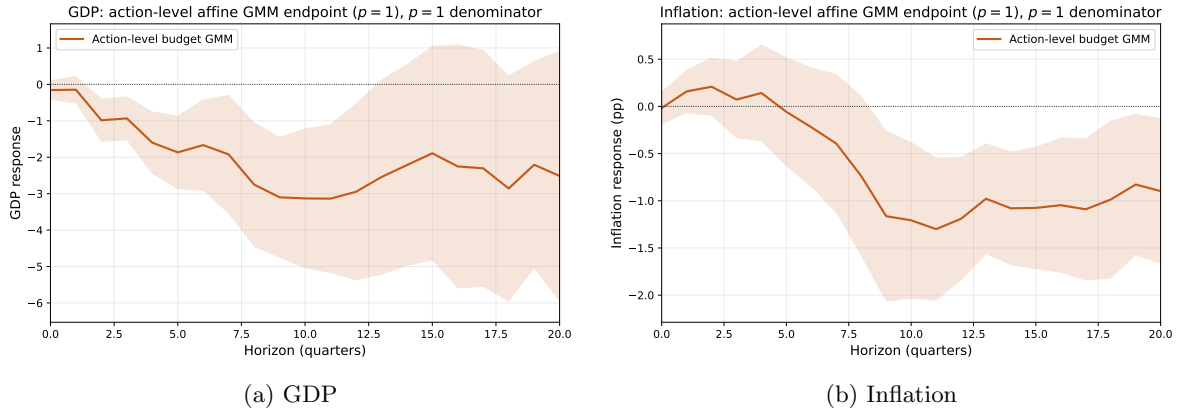


Figure 5: Budget Measures Under the Baseline GMM Estimator

Note: The figure reports affine $p = 1$ endpoint responses estimated with the baseline full-quarter GMM estimator. The estimator residualizes the outcome, the complementary macro control, and source-specific all-tax-shock controls, forms the numerator moment, and reports $\widehat{M}_{A,h}/\widehat{Q}_{A,h}$, where $\widehat{Q}_{A,h}$ is the affine denominator evaluated at $p = 1$. Shaded areas denote 68% HAC confidence bands.

5.5 Tax composition of the budget endpoint

The document-based budget response aggregates a dense set of tax measures across instruments. A natural question is therefore which parts of the tax code account for the aggregate contraction. I answer this question with an additive decomposition of the aggregate support-indexed endpoint, rather than with separate instrument-specific multipliers. For each instrument family k , I form the category-quarter tax shock $\Delta\tau_{k,t}$, use the same aggregate support probability p_t^{all} as in the all-budget exercise, and divide every contribution by the same aggregate endpoint denominator $\widehat{Q}_{A,h}^{all}$. The resulting object,

$$\widehat{\beta}_{k,h}^{add} = \frac{\widehat{M}_{k,h}(p_t^{all} = 1)}{\widehat{Q}_{A,h}^{all}},$$

is a historical contribution to the aggregate high-support budget response. It is not the effect of changing instrument k in isolation. This convention is useful because the components add back exactly

to the aggregate endpoint:

$$\hat{\beta}_h^{all}(1) = \sum_k \hat{\beta}_{k,h}^{add}.$$

Table 6 reports the GDP decomposition at the main horizons together with the composition of the underlying budget measures. Event shares describe the distribution of matched policy rows across tax instruments, shock-size shares describe the distribution of gross tax-shock mass, and mean p reports average narrative support within each instrument family. The medium-run contraction is mostly accounted for by individual income tax changes. This is not because they are uniquely frequent: corporate income tax rows are slightly more numerous. Individual income taxes instead combine a large share of gross shock mass, relatively high narrative support, and the largest negative contribution to the aggregate endpoint. Corporate income taxes also contribute negatively, but less strongly.

The consumption-tax split is important. Clean standard excises contribute little to the aggregate response. The broader nonstandard consumption/excise residual is positive, reflecting heterogeneous excise, energy-related, and non-VAT-like measures rather than a clean consumption-tax experiment. The U.S. consumption/excise rows therefore do not provide a readable indirect-tax estimate. This ambiguity is consistent with the warning in Nguyen et al. (2021) that broad consumption-tax objects can be difficult to interpret when they mix instruments and packages. It also clarifies the role of the VAT application below: VAT provides a sharper indirect-tax endpoint than the U.S. consumption/excise residual.

Table 6: Composition and Additive GDP Contributions to the U.S. Budget Endpoint

Instrument	Event share	Shock-size share	Mean p	GDP $h = 4$	GDP $h = 8$	GDP $h = 12$
Individual income	30.6%	37.0%	0.65	-1.50 (0.64)	-2.46 (1.18)	-2.57 (1.54)
Corporate income	32.2%	22.7%	0.64	-0.82 (0.41)	-1.02 (0.63)	-1.08 (0.60)
Clean standard excise	8.9%	11.2%	0.35	0.05 (0.12)	-0.06 (0.20)	-0.31 (0.31)
Nonstandard consumption/excise	11.1%	15.5%	0.40	0.87 (0.38)	0.99 (0.53)	0.99 (0.64)
Other low-support taxes	17.2%	13.6%	0.32	-0.19 (0.19)	-0.21 (0.31)	0.03 (0.31)

Note: Event shares are shares of matched policy rows. Shock-size shares are shares of gross tax-shock mass across the displayed instrument taxonomy. Mean p is the instrument-level average support. GDP entries are additive contributions to the affine $p = 1$ aggregate budget endpoint under the baseline full-quarter GMM estimator; parentheses report HAC standard errors. Each instrument contribution uses the aggregate all-budget support probability and aggregate all-budget endpoint denominator, so the GDP columns sum to the aggregate endpoint up to rounding. The table should not be read as a set of standalone instrument-specific multipliers.

5.6 Macroeconomic mechanism of the U.S. endpoint

The composition evidence disciplines the interpretation of the U.S. mechanism. Since the high-support endpoint is an income- and corporate-tax-heavy composite, the relevant margins are output, prices, private consumption, and especially investment. Figure 6 reports these responses. GDP, consumption, and investment are measured as percent responses to a tax increase normalized by the tax-shock denominator, rather than as GDP-share contributions.

The mechanism is consistent with the composition of the high-support endpoint. Output falls, inflation weakens, and investment is the most elastic domestic-demand component. Consumption

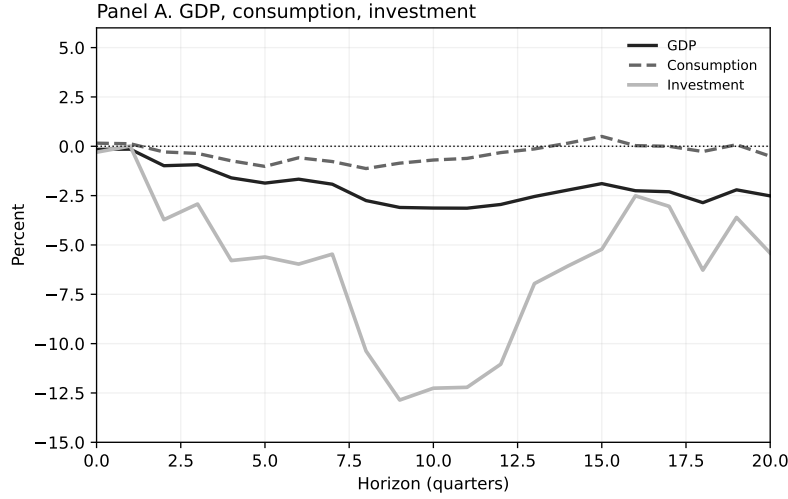


Figure 6: GDP, Consumption, and Investment Under the Baseline GMM Estimator

Note: The figure reports affine $p = 1$ endpoint responses using the same budget-measure probabilities, quarterly tax shocks, endpoint denominator, and HAC inference as Figure 5. Components are log quantity indexes, so responses are percent changes. The figure follows the layout of Panel A in Figure 14 of Romer & Romer (2010); it is not a GDP-contribution accounting exercise.

also declines, but less strongly than investment. The interpretation is therefore not that the budget endpoint estimates a generic tax multiplier or a clean consumption-tax response. It is a high-support composite tax shock whose medium-run contraction is concentrated in income-tax margins and whose expenditure response is carried mainly by investment. Appendix F.5 reports the remaining Figure 14-style component panels and compares them with the Romer–Romer results.

Three strands of prior work motivate this composition reading. Personal and corporate income-tax shocks load on different margins (Mertens & Ravn, 2013). Income and consumption taxes generate different empirical responses, and broad consumption-tax packages can be hard to interpret (Nguyen et al., 2021). Structural tax models similarly distinguish labor, capital, and consumption taxation as separate wedges with different behavioral and revenue implications (Trabandt & Uhlig, 2011). The contribution of the document-based data is to make this heterogeneity visible at the action level. The VAT application below is therefore not a repetition of the U.S. consumption-tax row. It is the clean indirect-tax counterpart to the U.S. aggregate budget endpoint.

Taken together, the U.S. document exercise should be read as the paper’s main out-of-sample extension of the RR identification logic. The exercise does not start from hand-written RR narratives and it does not attempt to reproduce the original episode list. Instead, it asks whether the document-grounded construction can recover economically meaningful identifying variation when applied to a denser universe of routine policy measures and their contemporaneous documentation. The answer from the baseline GMM environment is positive, but nuanced: the document-based budget endpoint is economically large, and its mechanism depends on the tax instruments that account for it.

6 The Indirect-Tax Contrast: VAT in the United Kingdom and France

The U.S. budget endpoint is a composite high-support tax shock. The decomposition above shows that its medium-run contraction is concentrated in income-tax margins, while the U.S. consumption/excise rows are too heterogeneous to provide a clean indirect-tax experiment. I therefore turn to VAT changes in the United Kingdom and France as the indirect-tax contrast. The goal is not to replace the U.S. end-to-end application, but to ask whether the same support-indexed estimator recovers the transmission pattern associated with a specific tax base. Value-added-tax changes are central to modern fiscal policy, but they are hard to study with binary narrative methods. VAT reforms are sparse, often legislated as part of broader tax packages, and frequently justified by several motives at once: revenue needs, price stabilization, purchasing power, simplification, and competitiveness. These are precisely the cases in which forcing a policy action into a binary exogenous/endogenous label discards useful information.

The VAT application asks whether document-based probabilities can turn this mixed narrative evidence into a macroeconomic object. The economic channel is sharp. A VAT increase raises the tax-inclusive price of consumption goods, lowers household purchasing power, and should contract domestic absorption, especially private consumption. It may also generate a short-run inflation response and an external offset through lower imports. The empirical question is therefore not only whether VAT shocks reduce output, but whether the response has the mechanism one would expect from an indirect tax.

The United Kingdom and France provide a natural two-country setting. Both have long postwar VAT or VAT-equivalent histories and rich documentary records, while the United States does not have a federal VAT. The pooled application estimates the response to a high-support VAT increase using the same probability scale and GMM logic as the U.S. application. The section therefore provides the indirect-tax counterpart to the U.S. aggregate budget endpoint: after the U.S. application shows how a broad federal tax mix transmits, the VAT exercise asks whether a specific consumption-price wedge produces the mechanism predicted for an indirect tax.

6.1 VAT policy data and documentary sources

The French VAT data are built from the *Voies et Moyens Tome I*, the revenue annex to the annual draft budget law. This source decomposes the projected change in tax revenues into mechanical growth in the tax base, accounting-scope changes, the continuing effects of previous reforms, and newly proposed measures. The application focuses on discretionary policy actions with direct macroeconomic content, rather than spontaneous revenue movements or scope transfers. In the full French corpus, the budget documents record 140 new VAT measures and 324 previous-measure entries. VAT is also quantitatively important: in the 2024 budget documents it accounts for about 38 percent of gross central-government tax revenues and about 6 percent of GDP.

The annual budget source is converted into a quarterly policy dataset using implementation dates. Measures introduced in the annual budget are assigned to the quarter in which they take effect, usually the first quarter of the fiscal year, unless the legal text or external statute-level information records a different implementation date. Measures adopted outside the annual budget process, such as amended

finance laws, enter in the quarter in which the relevant legislation is implemented. Signed tax amounts are scaled by the country-specific GDP normalization used in the panel application.

The documentary corpus for the French side consists primarily of contemporaneous Senate finance reports. These reports are useful because they discuss the budget articles in detail, record the government’s stated rationale, and often scrutinize the economic and fiscal consequences of the proposed measures. For the VAT corpus, each measure can be linked to the Senate-report discussion of the corresponding budget-law provision. This structure is used to audit retrieval quality in the implementation section: the document retriever locates the relevant discussion for almost all VAT measures among the top retrieved passages. The extracted motivations are then mapped into the same robust support score used in the U.S. application.

For the United Kingdom, I use the action-level tax data underlying the Cloyne application and select VAT rows from tax-type and source labels containing VAT or value-added-tax language. The UK and French probabilities are placed on the same robust hidden-state calibration scale, so the panel uses a common support measure rather than country-specific binary classifications.

Table 7 summarizes the country-level inputs. The main VAT panel is the common UK–France sample from 1955Q1 to 2007Q4. It contains enough VAT timing variation to estimate an instrument-specific response, while preserving the same quarter-level support construction across countries.

Table 7: VAT Policy Data Used in the UK–France Panel

Country	Main documentary source	Rows	nzq	Mean p	p_{75}	p_{90}
France	<i>Voies et Moyens</i> Tome I; Senate finance reports	122	39	0.58	0.73	0.80
United Kingdom	Cloyne action-level data; Budget, Hansard, Treasury record	168	72	0.72	0.82	0.87

Note: “Rows” counts VAT policy rows. “nzq” counts nonzero VAT quarters in the common 1955Q1–2007Q4 macro sample. The support statistics are computed on nonzero VAT quarters after aggregating action-level probabilities with absolute tax-change weights.

Figure 7 reports the resulting quarter-level support distribution. The object is not a country-specific binary label: event-quarter support is interior in both countries, with the UK distribution shifted toward higher support and the French distribution retaining more mass in the middle of the unit interval.

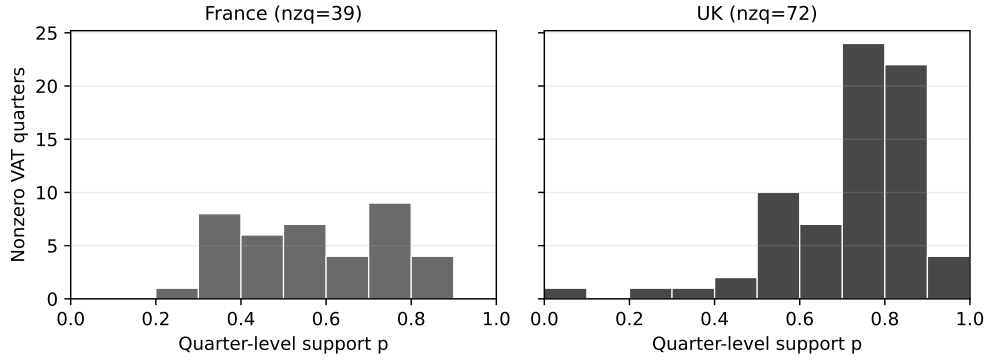


Figure 7: Quarter-Level Exogeneity Support for VAT Changes

Note: The figure plots the distribution of quarter-level support p_{ct}^{VAT} for nonzero VAT quarters. Quarter-level support is the absolute-tax-weighted average of action-level support among VAT measures implemented in the same country-quarter.

The average support is high, especially in the United Kingdom. This is consistent with the nature of the VAT policy record. Many VAT changes are structural reforms of the tax system: introduction or coverage changes, rate unification, simplification, European harmonization, or long-run revenue design. They are therefore often documented as tax-design or fiscal-structure measures rather than as short-run stabilization responses. The probabilities should not be read as saying that every VAT action is an ideal isolated shock. They say that the stated motives often look closer to the exogenous/structural side of the narrative taxonomy than to contemporaneous macroeconomic management. This distinction also matters for Nguyen et al. (2021). The UK episodes they emphasize, including the VAT introduction transition in 1973Q2 and the 1979Q3 rate unification, are not assigned low support in this calibration: their quarter-level support is about 0.83 and 0.73, respectively. Their concern is therefore better viewed as a concern about anticipation, tax packages, and the isolation of the VAT component within broader consumption-tax measures, rather than as a prediction that these specific episodes should have near-zero narrative support.

Figure 8 translates the VAT policy rows into the quarterly shock series used in the estimation. The upper panels plot signed quarterly VAT shocks; the lower panels show the corresponding quarter-level exogeneity support. This is the direct UK–France analogue of the U.S. shock-construction figure, but restricted to the VAT instrument.

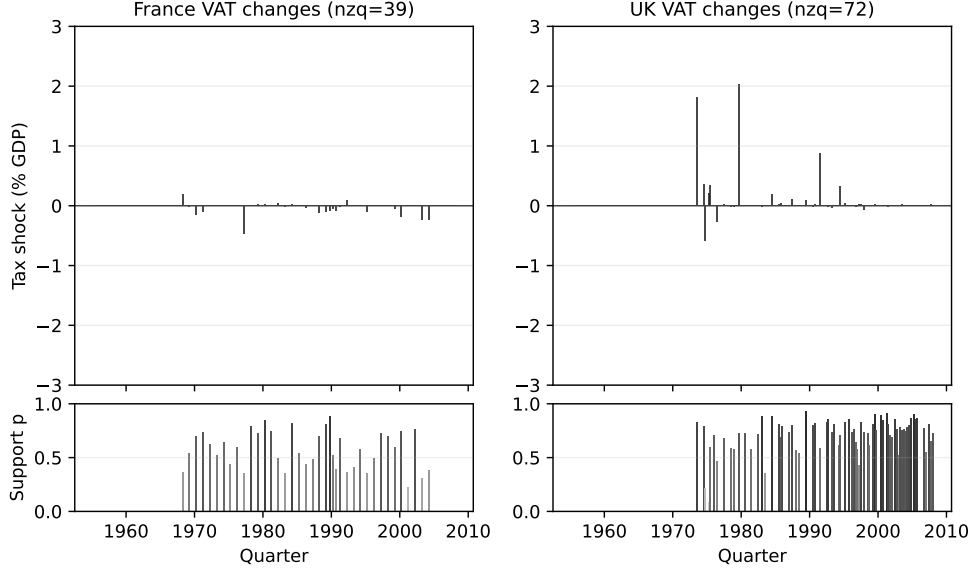


Figure 8: From VAT Measures to Quarterly VAT Shocks in France and the United Kingdom

Note: The upper panels plot VAT policy changes aggregated to country-quarter shocks, expressed as a percent of GDP. The lower panels report the corresponding absolute-tax-weighted quarter-level support. “nzq” denotes the number of nonzero VAT quarters in the common 1955Q1–2007Q4 sample.

6.2 Empirical design

The data are organized at the country-quarter level. In this subsection, τ_{ct} and p_{ct} refer to the VAT shock and VAT support in country c , quarter t . Let i index VAT policy rows. I aggregate signed tax changes and probabilities as

$$\tau_{ct} = \sum_{i \in (c,t)} \tau_{ict}, \quad p_{ct} = \frac{\sum_{i \in (c,t)} |\tau_{ict}| p_{ict}}{\sum_{i \in (c,t)} |\tau_{ict}|}, \quad (10)$$

with $\tau_{ct} = 0$ and $p_{ct} = 0$ in quarters without a VAT event. The probability is therefore a quarter-level measure of narrative support. When several VAT actions occur in the same quarter, actions with larger revenue effects receive more weight.

The estimator is the pooled version of the high-support GMM estimator used in the U.S. application. For each horizon h , I residualize the outcome and the VAT shock using the common macro controls and form the numerator moment $x_{ct,h} = \tilde{y}_{c,t+h} \tilde{\tau}_{ct}$. I then estimate

$$x_{ct,h} = A_h + d_{c,h} (p_{ct} - 1) + u_{ct,h}. \quad (11)$$

The parameter A_h is the common high-support numerator endpoint. The country-specific slopes $d_{c,h}$ allow the raw probability scale to differ across countries, while imposing that a high-support VAT change has the same endpoint effect in the pooled application. The impulse response divides A_h by the same linear-in- p denominator used in the current U.S. baseline, evaluated at $p = 1$. The estimates are therefore responses to a high-support VAT increase of one percent of GDP. Inference uses calendar-

quarter HAC scores aggregated across countries.

6.3 Macroeconomic mechanism

A VAT increase changes the tax-inclusive price of consumption goods and reduces household purchasing power. The relevant margins are therefore output, prices, private consumption, investment, and the external offset through imports and net exports. Figure 9 reports the main response. The top row shows output and prices; the bottom row shows the two domestic-absorption components that are most directly relevant for the VAT mechanism. A high-support VAT increase of one percent of GDP lowers GDP by about 2.5 percentage points of trend GDP after four quarters and by about 4.7 percentage points after eight quarters. Inflation rises initially and then returns toward zero, consistent with a VAT shock that combines a direct tax-inclusive price effect with a contraction in real activity.

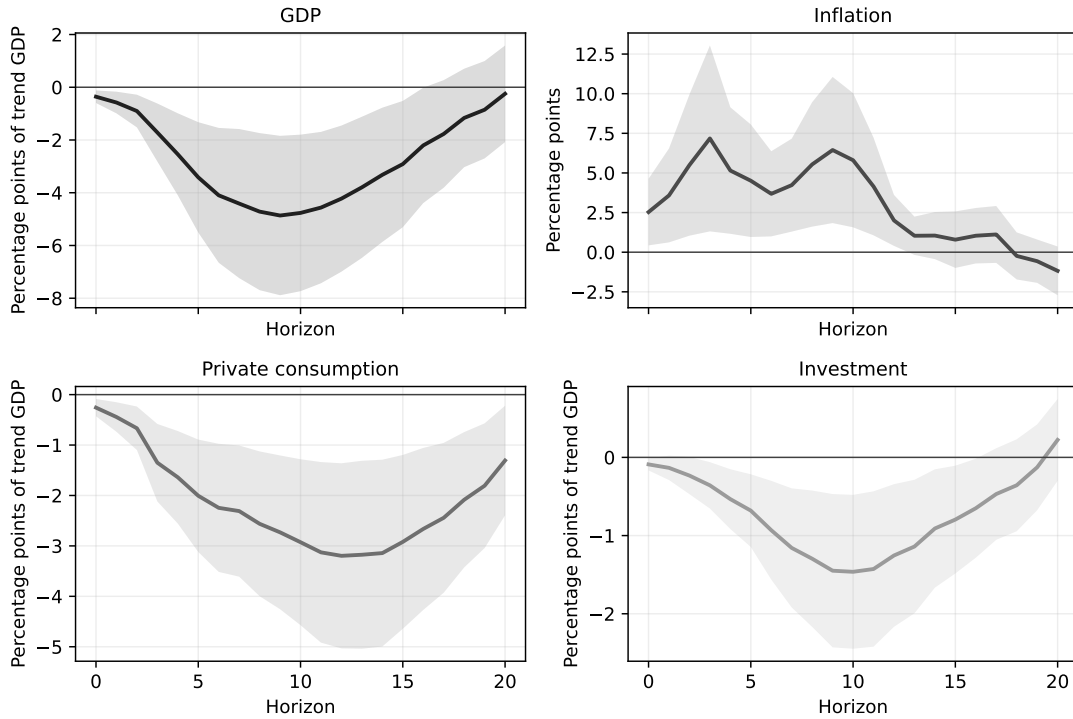


Figure 9: UK-France VAT: Output, Prices, and Domestic Absorption

Note: The figure reports the pooled high-support VAT endpoint. GDP, private consumption, and investment are expressed as $y_{c,t+h} - y_{c,t-1}$ in percentage points of trend GDP. Inflation is in percentage points. Responses are normalized by the linear-in- p denominator evaluated at $p = 1$. Shaded areas are 68 percent HAC confidence bands.

The response is concentrated in domestic absorption. Private consumption falls immediately and continues to decline through the medium run. Investment also falls, but the pooled response is smaller than the consumption response. Table 8 summarizes the same pattern at horizons 4, 8, and 12. At horizon 8, GDP is 4.7 percentage points below its pre-event trend-GDP position. Private consumption accounts for 2.6 percentage points of the decline, while investment accounts for 1.3. The result is therefore not only a negative VAT multiplier. It is a mechanism result: high-support VAT increases

raise prices on impact and then depress real activity primarily through household demand.

Table 8: VAT Main Mechanism at Key Horizons

Outcome	$h = 4$	$h = 8$	$h = 12$
GDP deviation from trend	-2.54	-4.72	-4.22
Inflation	5.15	5.54	2.00
Private consumption response	-1.64	-2.56	-3.20
Investment response	-0.53	-1.30	-1.25

Note: GDP, private consumption, and investment are $y_{c,t+h} - y_{c,t-1}$ in percentage points of trend GDP. Inflation is in percentage points.

The additional expenditure components, reported in Appendix G.1, complete the accounting interpretation. Government consumption does not explain the initial contraction. Net exports move in the opposite direction and partly offset the fall in domestic demand, mostly because imports decline. The VAT application therefore complements the aggregate U.S. evidence by showing a different transmission pattern for a different tax base: the U.S. composite endpoint is investment-heavy and disinflationary, whereas the VAT endpoint combines an output decline with an initial price response and a consumption-centered absorption channel.

A natural concern is that VAT changes may be bundled with other tax instruments. This is the central warning in Nguyen et al. (2021): a broad consumption-tax proxy can attribute package effects to indirect taxes if income-tax and consumption-tax changes are systematically linked. The same concern is relevant here, but the diagnostics point to a VAT-specific result. Same-quarter overlap is common because income-tax measures are frequent in budget quarters, yet the VAT-income correlation is small. Appendix G.2 shows that an explicit VAT/income adjustment leaves the GDP endpoint essentially unchanged and that dropping opposite-sign VAT-income package quarters does not weaken the response. By contrast, broadening the object from VAT to VAT plus duties and excises substantially attenuates the response. The result should therefore be read as a VAT-specific high-support endpoint, not as evidence about the broader average-consumption-tax-rate object studied by Nguyen et al. (2021).

7 Conclusion

This paper revisits narrative identification in macroeconomics and argues that its core logic naturally admits a soft interpretation. Traditional narrative designs rely on binary inclusion rules that treat selected policy actions as fully exogenous while discarding the remainder. This paper generalizes that approach by allowing qualitative documentary evidence to enter continuously, reallocating identifying weight across policy actions according to the strength of narrative support for orthogonality in expectation. Identification is recast at the level of reduced-form moments, exploiting systematic variation in narrative support without requiring the recovery of observation-level structural policy shocks.

To operationalize this perspective, the paper develops a document-grounded procedure that translates policymakers' stated motivations into a continuous index of narrative support for exogeneity. The procedure systematizes the use of historical sources while preserving traceability to the underlying documentary record and incorporating diagnostics that discipline how textual evidence contributes to identification. These features are designed to support, rather than replace, economic judgment, and

to make explicit the evidentiary basis underlying narrative restrictions.

The empirical analysis first validates this object against external benchmarks and then applies it to new U.S. budget measures. Edited Romer–Romer narratives show that the probability-construction stage assigns higher support to episodes classified as exogenous in the canonical U.S. narrative series. Cloyne’s UK tax data then provide a full document-based external-label validation: probabilities constructed from reconstructed policy documents are systematically higher for tax changes that Cloyne classifies as exogenous. The end-to-end U.S. budget exercise applies the full procedure to routine federal tax measures outside the original Romer–Romer sample; under the paper’s baseline affine GMM estimator, the document-based budget endpoint generates a sizable contractionary GDP response. The action-level decomposition shows that this endpoint should not be read as a generic tax multiplier: it is concentrated in income-tax margins, lowers inflation, and operates mainly through investment. The UK–France VAT application then provides the indirect-tax contrast. High-support VAT increases contract output primarily through domestic absorption, especially private consumption, while generating the short-run price response expected from an indirect tax.

More broadly, the framework clarifies how narrative evidence can discipline causal inference without requiring sharp exogeneity claims at the level of individual policy actions. By relaxing the implicit zero-contamination assumption underlying binary classifications, soft narrative identification retains information from a wider range of policy interventions while preserving the interpretive foundations of narrative approaches. This perspective helps address the sparsity of traditional narrative series and provides a principled way to trade off credibility and statistical power.

The analysis also illustrates how large collections of policy documents can be incorporated systematically into empirical macroeconomics when combined with transparent, document-grounded procedures. By lowering the cost of organizing and interpreting qualitative evidence, the framework expands the feasibility of narrative designs in settings characterized by frequent, lower-salience policy actions. Future work can build on this foundation by developing estimators that directly exploit variation in narrative support under weaker orthogonality assumptions and by applying the same validated procedure to other policy domains.

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8 Appendix

A When a Two-Way Contamination Representation Is Valid

This appendix formalizes the scope of the two-way representation used in the main text. The contaminated component need not consist of homogeneous observations. It may pool several latent sources of bias, including sources with opposite signs. What matters for the validity of the two-way decomposition is not sign homogeneity, but whether the *average* contaminated moment remains stable conditional on the narrative support index.

Let S_t denote a latent state taking values in

$$\mathcal{S} = \{A, 1, \dots, K\},$$

where $S_t = A$ denotes the orthogonal state and $S_t = k \in \{1, \dots, K\}$ denotes one of K contaminated states. Let

$$p_t \equiv \Pr(S_t = A \mid \mathcal{D}_t)$$

be the narrative support index, where $\mathcal{D}_t$ is the contemporaneous documentary record. For each p in the support of p_t , define

$$\pi_k(p) \equiv \Pr(S_t = k \mid p_t = p), \quad k = 1, \dots, K,$$

so that

$$p + \sum_{k=1}^K \pi_k(p) = 1.$$

For each horizon $h \geq 0$, define the state-specific cross-moments

$$r_h^A \equiv \mathbb{E}(y_{t+h} \Delta \tau_t \mid S_t = A), \quad r_h^k \equiv \mathbb{E}(y_{t+h} \Delta \tau_t \mid S_t = k).$$

The observable conditional outcome-tax moment is therefore

$$r_h(p) \equiv \mathbb{E}(y_{t+h} \Delta \tau_t \mid p_t = p) = pr_h^A + \sum_{k=1}^K \pi_k(p) r_h^k, \quad (12)$$

which allows the contaminated part to be arbitrarily heterogeneous. The question is when this richer structure can still be summarized by the simpler two-way representation used in the main text.

Proposition .1 (Validity of the two-way representation). *There exists a constant r_h^B such that, for all p in the support of p_t ,*

$$r_h(p) = pr_h^A + (1 - p)r_h^B, \quad (13)$$

if and only if

$$\frac{\sum_{k=1}^K \pi_k(p) r_h^k}{1 - p}$$

is constant in p on that support.

Proof. Starting from (12), define the average contaminated outcome-tax moment conditional on p by

$$\bar{r}_h^B(p) \equiv \frac{\sum_{k=1}^K \pi_k(p) r_h^k}{1-p}, \quad p < 1.$$

Then

$$r_h(p) = pr_h^A + (1-p)\bar{r}_h^B(p).$$

Hence a representation of the form (13) with a constant r_h^B exists if and only if $\bar{r}_h^B(p)$ is constant in p . $\square$

Proposition .1 shows that the two-way decomposition does not require all contaminated observations to have the same sign of bias or to be generated by a single latent source. It requires only that, conditional on the narrative support index, the contaminated observations can be summarized by a stable average moment. In this sense, the two-way representation is a parsimonious reduction of the first moments relevant for identification, not a claim of homogeneity of all endogenous tax changes. The next corollary specializes the result to the case in which the contaminated component combines sources of positive and negative bias.

Corollary .2 (Positive and negative contamination). *Suppose there are two contaminated states, denoted $+$ and $-$, with outcome-tax moments r_h^+ and r_h^- . Let*

$$\omega(p) \equiv \Pr(S_t = + \mid S_t \neq A, p_t = p),$$

so that $1 - \omega(p) = \Pr(S_t = - \mid S_t \neq A, p_t = p)$. Then the average contaminated cross-moment conditional on p is

$$\bar{r}_h^B(p) = \omega(p)r_h^+ + [1 - \omega(p)]r_h^-. \quad (14)$$

If $r_h^+ \neq r_h^-$, the two-way representation

$$r_h(p) = pr_h^A + (1-p)r_h^B$$

holds if and only if $\omega(p)$ is constant in p .

Proof. The result follows immediately from (14). If $r_h^+ \neq r_h^-$, the average contaminated moment $\bar{r}_h^B(p)$ is constant in p if and only if the composition weight $\omega(p)$ is constant in p . $\square$

Corollary .2 makes precise the sense in which a changing composition of bias can invalidate the two-way model. Opposite-sign sources of contamination are not a problem by themselves. They become a problem only when their relative incidence varies systematically with p_t , so that the average contaminated moment is no longer stable. By contrast, if positive and negative contamination are balanced in a way that does not vary with p_t , then the two-way representation remains valid for the purpose of recovering the orthogonal moment.

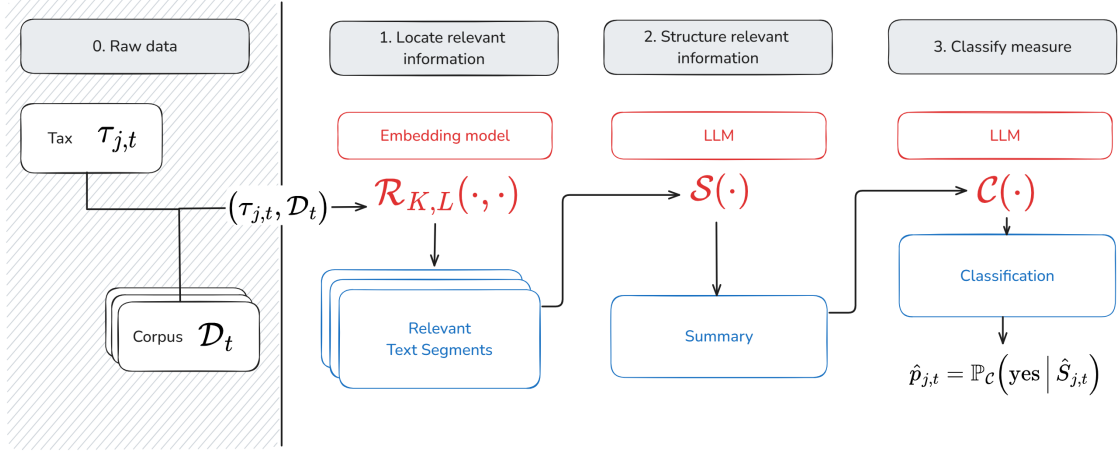


Figure 10: Mapping from policy descriptions and contemporaneous documents to narrative classifications. The figure illustrates the sequence $(q_i, \mathcal{D}_i) \rightarrow \hat{R}_i \rightarrow \hat{S}_i \rightarrow \hat{c}_i$ defined in Section 3.

B A Document-Grounded Implementation of Soft Narrative Identification

B.1 Illustration of the mapping

B.2 Embedding Model and Distance Metric

Semantic retrieval relies on vector representations of text segments and policy descriptions generated by a transformer-based embedding model. In the baseline specification, embeddings are obtained using OpenAI’s `text-embedding-3-large` model, which maps each text input to a 3,072-dimensional dense vector in $\mathbb{R}^{3072}$. The model is trained to capture semantic relationships between texts such that semantically similar passages are represented by vectors with small angular separation.

For each policy description q_i and document segment $d_{i,k} \in \mathcal{D}_i$, the corresponding embedding vectors e_i and $v_{i,k}$ are normalized to unit length. Semantic proximity is then evaluated using the cosine distance,

$$s_{i,k} = 1 - \frac{e_i \cdot v_{i,k}}{\|e_i\| \|v_{i,k}\|},$$

so that smaller values of $s_{i,k}$ indicate greater semantic similarity. Semantic proximity between text embeddings is measured using the cosine distance. Because all embeddings are normalized to unit length, cosine and Euclidean distances are mathematically equivalent in this setting, yielding identical retrieval rankings under either metric. Distances are computed through standard nearest-neighbor search routines provided by the underlying vector store, which efficiently identifies the text segments closest to each policy query in the embedding space. This normalization ensures that retrieval reflects semantic similarity rather than differences in vector magnitude. The retrieval operator defined in Equation (8) in the main text therefore selects, for each policy event i , the K document segments with the smallest cosine distances to the corresponding policy query q_i .

B.3 Example details

Applying the retrieval operator $\mathcal{R}_{K,L}$ to the query q_i and corpus $\mathcal{D}_i$, I obtain the top- K most relevant passages. Here are the top-5 passages:

- **Chunk 1.** On December 31, 1976, to mitigate, by a reduction of the weight of taxes bearing on consumption, the upward movement likely to occur at the end of this phase of price freeze; — it constitutes the last stage of a reform undertaken since 1969 — the date on which the standard rate of VAT was 23.456% — which results, ultimately, in a reduction, in less than seven years, of one quarter of the standard rate and in the merging of the latter with the intermediate rate.
- **Chunk 2.** An exceptional contribution would be requested in 1977 from the wealthiest French; it would affect all of them and only them, contrary to what had been provided for in the deleted article 5. This taxation would have as its base the total of the bases corresponding to the elements of lifestyle enumerated in article 168 of the General Tax Code, with the exception of main residences and cars of a power equal to or less than 16 horsepower. The income-equivalent threshold, calculated according to the norms of the aforementioned article 168, above which the exceptional contribution applies, is set at 60,000 F and the number of accumulated elements provoking taxation is three. Your Finance Committee proposes that you adopt the present article. VAT (Value Added Tax) – Article 6. Reduction of the standard rate and unification of the standard rate and the intermediate rate. Text. — I. — The standard rate of value-added tax is set at 17.60%. In the departments of Guadeloupe, Martinique and Réunion, it is set at 7.50II. — The upper limit of the special rebate for artisans fixed in paragraph 3 of article 282 of the General Tax Code is raised to 20,000 F. III. — The provisions of the present article enter into force on January 1, 1977. Comments. — It is proposed in the present article to lower by 2.4 points, starting from January 1, 1977, the standard rate of value-added tax which is currently 20% and, thereby, to bring it down to the level of the intermediate rate, namely 17.60%. This measure fits within a double perspective: — it is part of the program of combating inflation and is of a nature to permit, after the price freeze applied during the period from September 15 to December 31, 1976, to mitigate, by a reduction of the weight of taxes bearing on consumption, the upward movement likely to occur at the end of this phase of price freeze;
- **Chunk 3.** The number of rates applied in the matter of value-added tax, starting from next January 1, will be reduced to three instead of four; there would remain only: — the increased rate (33 %: cars, motorcycles, photography and cinematography equipment, sound and image recording devices, precious stones and precious metals, etc.); — the standard rate (17.6%: energy products, beverages, services and all transactions hitherto subject to the 20% rate, that is to say all those which were not taxed at the increased rate, nor at the intermediate rate, nor at the reduced rate); — the reduced rate (7%: essentially food products, pharmaceutical products, water and fertilizers). At the same time, it is planned to carry out a reduction of 2.5 points of the standard VAT rate and to align it with the intermediate rate in the departments of Guadeloupe, Martinique and Réunion, French Guiana not being concerned since the value-added tax regime is not applied there. In the three departments concerned, where, due to constraints inherent in the economies of developing countries, VAT rates are significantly lower than those of Metropolitan France, starting from January 1, 1977, one will record: — the increased rate: 14%; — the standard rate: 7.5%; — the reduced rate: 3.5%. In order to support the artisanal professions, it is proposed in paragraph II of the present article to raise, starting from January 1, 1977, the upper limit of the special rebate from which they benefit, raising it from 13,500 F to 20,000 F. Let us recall that, in order to lighten the burdens of small taxpayers, and in particular artisans,

article 282 of the General Tax Code has provided, in their favor, special provisions resulting in a waiver or a reduction of the VAT normally incumbent upon them.

- **Chunk 4.** Standard of 6.5% per year. b) The elimination of the levy. As had been provided by the Finance Law for 1976, in the initial text presented by the Government, the levy was to cease to apply when the progression of prices was below a set norm: this one, in the governmental project, was fixed at 2.50% for the consumer price index of the group of manufactured products during a period of six consecutive months.
- **Chunk 5.** Article: — in paragraph I, the objective is to encourage business leaders to convert into capital the advances they are led to grant in the form of deposits in current accounts to the companies they manage. Up to now, in such cases, the ordinary contribution right was a proportional duty of 1%; it is planned to substitute for it a fixed duty of 220 F on the condition, however, that the sums thus incorporated into the capital have been placed at the constant disposal of the company, for a minimum period of twelve months.

C Grounding Diagnostics: Marginal Narrative Contributions

C.1 Robustness of the Linear Surrogate Estimator

Because the surrogate model is estimated with $n = 64$ ablations and up to $K \approx 30$ retrieved passages, I assess whether the resulting coefficients are sensitive to the choice of regularisation. Ridge regression yields estimates that are nearly identical to the OLS baseline and improves numerical stability when inclusion patterns exhibit collinearity. A LASSO with $\lambda = 0.01$ selects the same high-weight passages as OLS and produces coefficients of similar magnitude. These results indicate that the dependence patterns reported in the main text are not driven by estimator choice or overfitting. For settings with substantially larger K , penalised estimators would provide a natural extension.

C.2 Global Placebo Test for Summarisation Grounding

The coefficient-level placebo test in the main text evaluates, for each passage individually, whether its positive marginal contribution is unusually large relative to the distribution of positive placebo contributions. Because this diagnostic is applied to multiple passages within each event, some significant cells may be mechanically expected under the null of no grounding. This appendix reports an omnibus test that aggregates evidence across passages and verifies that the overall pattern of significant contributions cannot be attributed to chance.

Placebo mass. For each event, let $w_k^+ = \max\{\hat{w}_k, 0\}$ denote the positive marginal contribution of passage k , and let R and P denote the sets of relevant and placebo passages, respectively. The share of positive weight falling on placebo passages is

$$\phi = \frac{\sum_{k \in P} w_k^+}{\sum_{k \in R \cup P} w_k^+} \in [0, 1],$$

which summarises, at the event level, how much of the model’s total positive dependence is placed on semantically irrelevant text. Grounded behaviour is characterised by low ϕ , while ungrounded dependence yields large values.

Randomisation inference. To assess whether the observed value ϕ_{obs} is unusually large, I test the sharp null that placebo labels are irrelevant to the pattern of marginal contributions. Let K^+ denote the number of passages with positive weights. Under the null, any assignment of K^+ placebo labels across passages is equally likely. I therefore compute the randomisation distribution by permuting placebo labels across the K^+ positions, recomputing placebo mass ϕ_b for each permutation, and forming the right-tailed p -value:

$$p = \frac{1 + \sum_{b=1}^B \mathbb{1}\{\phi_b \geq \phi_{\text{obs}}\}}{1 + B}.$$

A small p -value indicates that the model places significantly more positive weight on placebo passages than expected under random assignment, implying ungrounded behaviour. Conversely, when ϕ_{obs} falls well within the reference distribution, the observed coefficient-level significance pattern is unlikely to arise from multiple testing alone.

Interpretation. This global test complements the passage-level diagnostic in the main text. While the main procedure identifies *which* passages provide statistical support for each motivation, the global placebo mass test assesses *whether the overall pattern of support is itself meaningful*. Across events, the observed placebo masses cluster well below their randomisation-based critical values, confirming that the significant passages identified in the main text do not arise from chance alone.

D Application of the Document-Grounded Construction

D.1 Retrieval Performance Metrics and Parameter Selection [TO BE ADDED]

This Appendix provides additional detail on the evaluation of the retrieval operator $\mathcal{R}_{K,L}$, the calibration of its parameters (K, L) , and the performance of alternative segmentation strategies. The objective is to document the robustness of retrieval accuracy and to clarify how the parameter settings used in the main text were chosen.

Benchmark Setup Each VAT measure i is manually linked to its corresponding Senate report article, denoted the *gold article*. For every candidate pair of parameters (K, L) , I perform retrieval and record all text segments d_k whose span overlaps the gold article. These are counted as *hits*. Because each article can extend across multiple segments, a single event may have several true positives.

D.2 Classification Prompts and Category Scheme

This appendix reports the prompt templates that define the eight narrative categories used in the tax applications. Each category corresponds to one of the subtypes summarized in Figure 1, following the narrative taxonomy of Cloyne (2013). The prompts should be read as the semantic dictionary of the measurement problem, not as the main probability estimator. In the baseline construction, these definitions guide the generation of labeled synthetic motivations, and hidden-state features are calibrated on that synthetic corpus. In the alternative signal in Appendix D.3, the same prompts are used directly to compute the model’s probability of answering *Yes*.

Prompt templates

Each prompt is presented in a standardized format and corresponds to one category listed in Figure 1. The prompts are organized by group: *endogenous* (Demand Management 1, Supply Stimulus 2, Deficit Reduction 3, Spending-driven 4) and *exogenous* (Long-term 5, Ideological 6, External 7, Deficit Consolidation 8). Each prompt provides the task definition, inclusion and exclusion criteria, and output format. Together, they make the category definitions reproducible and provide the pivot from qualitative narrative taxonomy to synthetic calibration data.

English translation of the classification prompt (Demand Management / DM):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label “Demand Management (DM)”?

Definition of Demand Management (DM):

A passage is classified as Demand Management if the measure is motivated by the need to influence, in the short term, the level or composition of aggregate demand in the economy—either to curb inflationary pressures and cool overheated sectors, or to stimulate demand during a slowdown or recession.

This includes:

- Tax increases or restrictions explicitly justified as reducing consumption, restraining demand, or mitigating inflationary pressures.
- Tax cuts or reliefs explicitly presented as stimulating demand, consumption, investment, or economic activity.
- Measures related to concerns over the balance of payments, inflation control, or unemployment, where the mechanism operates through aggregate demand.
- Temporary or reversible measures (e.g., surcharges, freezes, deferrals) justified by immediate economic circumstances, particularly short-term demand conditions.

*Do **not** classify as Demand Management if:*

- The motivation is solely deficit reduction or fiscal consolidation.
- The measure is driven exclusively by environmental, health, or equity objectives without reference to aggregate demand.
- The change is a routine indexation or neutral adjustment without intent to influence demand.
- The measure is presented as a long-term structural reform (improving incentives or competitiveness) rather than targeting short-term demand effects.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
 - Yes — [short rationale]
 - No — [short rationale]

Prompt 1: Classification Prompt for the “Demand Management (DM)” Category

English translation of the classification prompt (Supply stimulus / SS):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label “Supply stimulus (SS)”?

Definition of Supply stimulus (SS):

A passage is classified as Supply stimulus if the measure is motivated by the need to stimulate, in the short term, production, investment, competitiveness, or employment on the supply side of the economy. Such measures typically involve tax cuts, reliefs, or structural changes aimed at improving incentives, profitability, liquidity, or business conditions to increase output and growth potential.

This includes:

- Reductions in income tax rates, allowances, or thresholds explicitly presented as improving work incentives or effort.
- Corporate or business tax cuts justified as encouraging investment, profitability, or competitiveness.
- Investment allowances or reductions in social contributions intended to lower hiring costs or promote employment.
- Measures described as supporting recovery or short-term growth through improvements on the supply side.
- Cyclical adjustments designed to maintain the immediate competitiveness of a sector.

*Do **not** classify as Supply stimulus if:*

- The justification emphasizes long-term structural, productivity, or competitiveness effects.
- The measure is presented as deficit reduction or fiscal consolidation.
- The measure targets aggregate demand management (short-term demand stimulation or restraint).
- The justification is primarily environmental, health-related, or social without a supply-side argument.
- The change is a neutral indexation, a temporary demand stimulus, or a routine adjustment without an incentive-based rationale.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
 - Yes — [short rationale]
 - No — [short rationale]

Prompt 2: Classification Prompt for the “Supply stimulus (SS)” Category

English translation of the classification prompt (Deficit Reduction / DR):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label “Deficit Reduction (DR)”?

Definition of Deficit Reduction (DR):

A measure is classified as Deficit Reduction if its objective is to generate immediate revenue or prevent a loss of revenue in order to contain the short-term budget deficit.

This includes:

- The freezing or restriction of tax allowances or reliefs to raise funds for the national treasury “this year” or “next year.”
- Increases in taxes or duties explicitly presented as necessary to raise additional revenue (e.g., “I need to raise X in additional receipts,” “to finance this,” “given the need to generate extra revenue”).
- The elimination or reduction of costly reliefs whose expense is deemed unsustainable or unjustified.
- Adjustments to consumption taxes—such as fuel, alcohol, tobacco, or VAT—to “protect real yields” or prevent erosion of the tax base.
- Temporary or exceptional levies introduced to cover revenue needs within the current fiscal year.

*Do **not** classify as Deficit Reduction if:*

- The measure is primarily motivated by medium- or long-term fiscal consolidation (this belongs to another category).
- The measure is justified mainly by non-fiscal motives (environmental, health, or social equity) without clear reference to deficit control.
- The measure reflects routine indexation or neutral adjustments that do not raise revenue.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
 - Yes — [short rationale]
 - No — [short rationale]

Prompt 3: Classification Prompt for the “Deficit Reduction (DR)” Category

English translation of the classification prompt (Spending-driven / SD):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label “Spending-driven (SD)”?

Definition of Spending-driven (SD):

A passage is classified as Spending-driven if the measure is explicitly motivated by the need to finance, offset, or enable increases in public spending or the introduction of new expenditure programs.

This includes:

- Tax increases or restrictions on reliefs introduced to cover the cost of new spending or the expansion of existing programs (e.g., food subsidies, NHS funding, social reforms).
- Statements indicating that additional revenue is needed to “finance,” “help finance,” or “enable” proposed spending measures.
- Exceptional or one-off taxes whose purpose is to fund a specific expenditure program (e.g., employment initiatives, child poverty measures).
- Reallocation or withdrawal of tax reliefs to free resources for targeted spending programs.

*Do **not** classify as Spending-driven if:*

- The motivation is deficit reduction or fiscal consolidation for long-term sustainability.
- The measure is justified by demand management (inflation control, balance-of-payments adjustment, or demand stimulus).
- The change is primarily motivated by environmental, health, or equity considerations without explicit reference to spending needs.
- The measure is a routine indexation or neutral adjustment with no link to expenditure financing.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
 - Yes — [short rationale]
 - No — [short rationale]

Prompt 4: Classification Prompt for the “Spending-driven (SD)” Category

English translation of the classification prompt (Long-term / LR):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label “Long-term (LR)”?

Definition of Long-term (LR):

A passage is classified as Long-term if the measure is motivated by structural, supply-side, or long-term incentive effects on the economy—aimed at improving productivity, competitiveness, investment, saving, or economic efficiency over time.

This includes:

- Tax cuts or reforms justified by improving work effort, incentives, or morale.
- Measures designed to stimulate business investment, capital formation, or technological modernization.
- Reforms targeting international competitiveness, efficiency, or simplification of the tax system.
- Policies encouraging entrepreneurship, risk-taking, or saving to support future growth.
- Changes presented as necessary for the transition to a new economic regime (e.g., from wartime to peacetime).

*Do **not** classify as Long-term if:*

- The main motivation is short-term revenue needs or deficit consolidation.
- The measure concerns only short-term demand management (stimulus or demand restraint).
- The justification is purely environmental, health-related, or equity-driven without reference to long-term incentives or growth.
- The change is a routine indexation or minor administrative adjustment with no structural intent.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
 - Yes — [short rationale]
 - No — [short rationale]

Prompt 5: Classification Prompt for the “Long-term (LR)” Category

English translation of the classification prompt (Ideological / IL):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label “Ideological (IL)”?

Definition of Ideological (IL):

A passage is classified as Ideological if the measure is primarily motivated by political values, social philosophy, principles of fairness, or a broader vision of society and the tax system—rather than by short-term macroeconomic management (demand control) or fiscal balance (deficit reduction or consolidation).

This includes:

- References to fairness, justice, tradition, or “social values,” including environmental or ethical considerations.
- Tax reliefs or exemptions targeted at “deserving” groups (e.g., retirees, widows, children, families, vulnerable populations).
- Explicitly moral, social, or political justifications (e.g., “people hate income tax more than any other tax,” “to reward work,” “to encourage saving,” “to support family unity”).
- Long-term reforms aimed at making the tax system simpler, fairer, or more logical, without references to deficit or demand management.
- Measures presented as fulfilling electoral promises or party commitments (e.g., VAT cuts on essential goods, new redistributive taxes).
- Reforms introduced as part of a broader vision of taxation reflecting social or political values (e.g., taxing capital gains “as income,” inheritance thresholds for family fairness).

Do not classify as Ideological if:

- The measure is primarily motivated by deficit reduction, fiscal sustainability, or budget consolidation.
- The change is framed in terms of short-term demand management (stimulus, inflation control, balance of payments).
- The measure is a routine indexation or a neutral technical adjustment.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
 - Yes — [short rationale]
 - No — [short rationale]

Prompt 6: Classification Prompt for the “Ideological (IL)” Category

English translation of the classification prompt (External / EXT):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label “External (EXT)”?

Definition of External (EXT):

A passage is classified as External if the measure is motivated by external economic pressures, trade obligations, or international agreements.

This includes:

- Measures adopted in response to balance-of-payments problems, exchange rate pressures, or the need to protect reserves.
- Tax modifications explicitly linked to supporting exports, foreign trade operations, or competitiveness in international markets.
- Adjustments mandated by commitments to international organizations or treaties (e.g., WTO, EU, EFTA).
- Policy changes justified by concerns about investor confidence or market reactions to the government’s financing needs.

*Do **not** classify as External if:*

- The motivation is purely domestic (e.g., deficit reduction, consolidation, demand management, social equity, environment).
- The change is a routine adjustment unrelated to trade, finance, or international agreements.
- The measure concerns only domestic consumption, revenue, or distribution without an external framing.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
 - Yes — [short rationale]
 - No — [short rationale]

Prompt 7: Classification Prompt for the “External (EXT)” Category

English translation of the classification prompt (Deficit Consolidation / DC):

Task: Classify a fiscal measure based on the stated motivations. Does this measure correspond to the label “Deficit Consolidation (DC)”?

Definition of Deficit Consolidation (DC):

A passage is classified as Deficit Consolidation if the measure is motivated by a structural or medium- to long-term strategy aimed at strengthening public finances or reducing deficits, borrowing, or public debt.

This includes:

- Measures presented as part of a sustained fiscal plan to ensure the soundness of public finances.
- Structural revenue generators (e.g., base broadening, duty escalators, restrictions on reliefs) intended to improve the budget balance over several years.
- Policies explicitly linked to fiscal consolidation, fiscal sustainability, or medium-term deficit reduction, even when other justifications are also mentioned.
- Measures whose initial intent was deficit consolidation, even if the rhetoric later evolved.

*Do **not** classify as Deficit Consolidation if:*

- The motivation concerns short-term revenue needs (“to raise additional revenue this year,” “to finance support for the economy”)—this falls under Deficit Reduction (DR), not Consolidation.
- The change is justified mainly by environmental, health, or social equity objectives without a fiscal framing.
- The measure reflects routine indexation or neutral adjustments without a consolidation goal.
- The policy reduces taxes or increases spending without compensating measures.

Output format:

- Respond with one option only for the entire set of motivations (do not provide multiple answers):
 - Yes — [short rationale]
 - No — [short rationale]

Prompt 8: Classification Prompt for the “Deficit Consolidation (DC)” Category

D.3 Alternative LLM Signal: Incremental Log Probabilities

The main text uses hidden-state calibration as the baseline way to extract category probabilities from language models. A simpler alternative is to use the same semantic prompts more directly. For motivation $M_{i\ell}$ and category $c \in \mathcal{G}$, let X_c be the category-specific prompt reported in Appendix D.2. Define the incremental log-probability signal

$$s_{i\ell,c}^{\log} := \log \Pr_{\text{LM}}(\text{Yes} \mid X_c \cup M_{i\ell}) - \log \Pr_{\text{LM}}(\text{Yes} \mid X_c). \quad (15)$$

The first term measures the model’s support for category c after observing the motivation; the second removes the prompt-specific baseline tendency to answer “Yes.” Collect these signals in

$$Z_{i\ell}^{\log} = \left(s_{i\ell,c}^{\log} \right)_{c \in \mathcal{G}}.$$

Because token-probability signals generated by different prompts need not be on a common probability scale, they should be treated as features to be calibrated rather than as final category probabilities. The same synthetic motivations used in the baseline calibration can provide the training sample: each synthetic motivation is generated from the category definitions in Appendix D.2, so its category label is known by construction. One can therefore estimate, on this calibration sample, a multinomial mapping

$$\Pr(C_{i\ell} = c \mid Z_{i\ell}^{\log}) := \frac{\exp(\alpha_c + Z_{i\ell}^{\log'} \beta_c)}{\sum_{c' \in \mathcal{G}} \exp(\alpha_{c'} + Z_{i\ell}^{\log'} \beta_{c'})}, \quad c \in \mathcal{G}. \quad (16)$$

The resulting fitted probabilities can then be aggregated exactly as in Section 2.2: sum the calibrated category probabilities over $\mathcal{G}_{\text{exo}}$ to obtain motivation-level support and average motivation-level support within the policy action to obtain p_i . This construction is useful as an alternative signal extraction method that keeps the prompt templates directly in the measurement equation. The main application instead uses hidden-state calibrated probabilities, reported in Table 1, because the hidden-state route avoids relying on the model’s final Yes/No token probabilities as the primary signal.

E Calibration Diagnostics

This appendix reports diagnostics for the synthetic calibration exercise used to construct the narrative-support probabilities. The objective of these diagnostics is not to validate the economic interpretation of the support score. That role is played by the external Romer–Romer and Cloyne exercises in Section 4. Instead, the diagnostics reported here assess whether the probability scale is stable across alternative calibration datasets and language-model representations, and whether the synthetic category definitions are internally learnable.

E.1 Model Replication Details

The exact language-model identity is pinned by the pair consisting of the Hugging Face repository identifier and the resolved revision SHA. Hidden-state signals were extracted with the following models: Runs used Python 3.11.5, PyTorch 2.10.0+cu128, and a single NVIDIA L40S GPU.

Table 9: Language-model identifiers used for hidden-state extraction

Short name	Hugging Face repository	Resolved revision
Gemma	<code>google/gemma-4-E4B-it</code>	<code>fee6332c1abaafb77f6f9624236c63aa2f1d0187</code>
Mistral	<code>mistralai/Mistral-7B-Instruct-v0.2</code>	<code>63a8b081895390a26e140280378bc85ec8bce07a</code>
Qwen	<code>Qwen/Qwen3.5-9B</code>	<code>c202236235762e1c871ad0ccb60c8ee5ba337b9a</code>

E.2 Synthetic construction diagnostics

I audit the synthetic calibration corpus before using it to train the hidden-state classifier. These checks are construction diagnostics: they are designed to detect mechanical artifacts in the synthetic examples, not to validate the empirical identifying assumptions.

First, I verify exact balance. Each independently generated calibration set must contain the same eight motivation categories and exactly 60 examples per category. This preserves the row-block structure used by the hidden-state extraction workflow and prevents category imbalance from mechanically affecting the multinomial decoder.

Second, I scan each motivation for prompt artifacts. The audit searches for banned prompt residues and template phrases such as “demand management motive”, “program funding motive”, “recovery through incentives”, “values-based reform”, “structural reform motive”, and related heading-like expressions. These checks are intentionally narrower than a general search for economically meaningful words: terms such as “long-term” or “deficit” may be legitimate content, while prompt headings or classifier language would indicate leakage from the generation prompt.

Third, I check for duplicate and near-duplicate examples. Exact duplicates are computed after lowercasing and normalizing punctuation and whitespace. Near duplicates are computed within each set and category using Jaccard similarity over normalized token sets, with the configured threshold equal to 0.72. This check is stricter than simply looking for identical strings and is meant to ensure that a category block is not populated by lightly reworded copies of the same example.

Fourth, I check for repeated openings. For each motivation, the audit records the first seven words and counts repeated openings within each set-category cell. The construction gate flags openings that occur more than three times within the same category and set. This prevents the classifier from exploiting formulaic first clauses rather than the substantive motivation.

Fifth, I estimate simple lexical shortcut diagnostics. For each synthetic set, I train cross-validated TF-IDF multinomial classifiers to predict the eight categories. One classifier uses the full motivation text; the other uses only the opening phrase. High full-text separability is expected in a synthetic category-labeled corpus and is reported only as a diagnostic. The more important shortcut check is the opening-only classifier: if the first few words alone predict category too well, the set may encode category through templates rather than motivation content.

Finally, I compare the synthetic texts to real extracted motivations from the extension and Romer–Romer/Cloyne replication files. This is a coverage diagnostic, not a training step. The audit computes TF-IDF nearest-neighbor similarities between real motivations and synthetic examples, reports average and lower-tail similarities, and stores the least-covered real motivations for inspection. Real labeled observations are not added to the synthetic training set in this version of the calibration exercise.

For the main robust synthetic corpus, all twelve generated sets are exactly balanced. The audit finds zero banned prompt-artifact rows, zero repeated-opening violations, zero exact duplicates, and

zero near-duplicate pairs above the configured threshold. The opening-only classifier remains much less predictive than the full-text classifier, although a small number of generated sets are flagged as mild shortcut warnings. These warnings are retained as diagnostics and motivate the stability analysis across independently generated calibration sets.

E.3 Synthetic Holdout Performance

For each synthetic calibration set and each language model, I train the multinomial hidden-state decoder using five-fold stratified cross-validation within the synthetic corpus. The resulting diagnostics evaluate whether the category distinctions embedded in the synthetic taxonomy can be recovered from the hidden-state representations.

Table 10 reports average holdout performance across the twelve independently generated calibration sets. Performance is uniformly strong across models. Brier scores and log-loss values are low, classification accuracy is high, and the binary exogeneity classification implied by the taxonomy is almost perfectly separated in the synthetic environment.

Table 10: Synthetic holdout performance

Model	Brier	Log-loss	Accuracy	AUC	8-category log-loss
Gemma	0.013	0.050	0.985	0.999	0.085
Mistral	0.004	0.015	0.996	1.000	0.020
Qwen	0.003	0.014	0.996	1.000	0.018

These results should be interpreted with caution. Synthetic examples are necessarily cleaner and more category-separated than real policy language. Strong synthetic performance therefore demonstrates that the category definitions are internally coherent and recoverable from hidden-state representations, but does not by itself establish external validity.

E.4 Stability Across Calibration Sets

The calibration design deliberately uses twelve independently generated datasets. Each dataset is used to train a separate decoder, producing twelve independent probability estimates for every policy action. This design makes it possible to assess how sensitive the final support score is to the particular synthetic examples used during calibration.

Table 11 reports agreement across the twelve set-specific probabilities. Correlations are generally high, although meaningful variation remains in some samples. The median standard deviation across set-trained decoders ranges from roughly 0.09 to 0.12, indicating that calibration uncertainty is non-negligible but substantially smaller than the overall variation in support probabilities. The 90th percentile of the set-level standard deviation captures the uncertainty in harder-to-classify observations.

The baseline support probability averages over these twelve calibration datasets. The remaining dispersion is retained as a calibration-uncertainty measure and reported alongside the final support scores.

Table 11: Agreement across calibration sets

Sample	Mean Pearson	Mean Spearman	Mean abs. diff.	Median set s.d.	P90 set s.d.
U.S. RR replication	0.898	0.878	0.099	0.086	0.129
U.S. extension	0.820	0.816	0.143	0.117	0.188
France VAT	0.784	0.777	0.114	0.096	0.139
France non-VAT	0.811	0.798	0.104	0.087	0.138

E.5 Stability Across Language Models

The baseline specification also averages probabilities across three open-weight language models: Gemma, Mistral, and Qwen. This averaging is intended to reduce dependence on any particular hidden-state geometry.

Table 12 reports agreement across the model-specific probabilities after averaging over calibration sets. Pairwise correlations are generally high across applications, although the French samples exhibit somewhat greater dispersion than the U.S. samples. Median within-observation model standard deviations remain modest relative to the full support distribution.

Table 12: Agreement across language models

Sample	N	Mean Pearson	Mean Spearman	Median model s.d.
U.S. RR replication	50	0.82	0.80	0.09
U.S. extension	1188	0.83	0.84	0.11
France VAT	140	0.72	0.72	0.10
France non-VAT	1020	0.72	0.74	0.11

These results indicate useful agreement across model representations while leaving enough variation to justify averaging across models rather than conditioning the final support score on a single language model.

F Replication of Romer and Romer

F.1 Probability Benchmark Settings

F.1.1 Narrative extraction

This appendix documents the benchmark procedure behind Section 4.1. The edited RR narratives are used to validate the probability construction rather than a direct binary relabeling exercise. Each example below shows the portion removed before the narrative is passed to the summarization stage. As in the main text, the deleted block includes the explicit RR classification sentence and the two immediately preceding sentences. Removing these passages prevents the model from seeing the authors’ bottom-line judgment while leaving enough narrative detail for the summarizer to recover multiple stated motivations from the remaining text.

Example 1: an endogenous measure

The motivation for the Revenue Act of 1951 was again the increase in spending related to the Korean War. The *1951 Economic Report* stated: “These new taxes are required to finance the defense effort; and to help keep total spending within the capacity of current production, so that inflation does not reduce the purchasing power” (p. 17). This same sentiment was echoed in a number of presidential speeches. On February 2, 1951, Truman said, “If we do not tax ourselves enough to pay for defense expenditures, the Government will . . . add to total purchasing power and inflationary pressures” (*Special Message to the Congress Recommending a “Pay as We Go” Tax Program*, p. 2). These quotations make it clear that the tax increase was designed to keep growth normal.

It stated: “The military action in Korea, coupled with the general threat to world peace, has made it necessary to provide extraordinary increases in revenues to meet essential national defense expenditures” (82d Congress, 1st Session, House of Representatives Report No. 586, 6/18/51, p. 1). It also stressed the temporary nature of the tax, saying: “The 12½-percent flat across-the-board increase [in individual income tax rates] was selected as the form of increase, since it is not intended that this increase will be permanent, and therefore, it was desired to provide an increase which would not become an integral part of the rate structure” (p. 8). Because the tax increase was designed to counteract the effects of higher military spending and thereby keep growth normal, I classify it as an endogenous, spending-driven action.

Truman’s speech upon signing the bill stated that it would raise about \$5.5 billion of additional revenues at an annual rate (*Statement by the President Upon Signing the Revenue Act of 1951*, 10/20/51, p. 1). Both the *1952 Economic Report* (p. 135) and the *1951 Treasury Annual Report* (p. 44) gave the figure as \$5.4 billion. I follow my usual practice and use the number from the *Economic Report*.

The increase in corporate taxes was retroactive to April 1, 1951, while the remainder took effect on November 1, 1951 (*1951 Treasury Annual Report*, p. 51). Because the action took place before the middle of the quarter, I date the tax increase in 1951Q4. Approximately \$2.3 billion of the tax increase was from corporate taxes (p. 501). Therefore, the retroactive portion of the bill increased taxes at an annual rate of $\$2.3 \text{ billion} \cdot 2 \text{ extra quarters}$, or \$4.6 billion. Adding this to the steady-state effect of \$5.4 billion yields a tax increase of \$10 billion in 1951Q4. The return to the steady-state level of tax increase implies a tax cut of \$4.6 billion in 1952Q1. If one ignored the retroactive part of the legislation, there would be just a tax increase of \$5.4 billion in 1951Q4.

The tax increase largely took the form of an increase in marginal rates. The act also raised the capital gains tax, the tax on corporate profits, and some excise taxes (*1951 Treasury Annual Report*, pp. 50–52).

Most of the changes were explicitly temporary. The individual income tax increases were legislated to expire on January 1, 1954; the corporate and excise tax increases were to continue until March 31, 1954 (*1951 Treasury Annual Report*, pp. 51–52).

Example 2: an exogenous measure

This bill was a significant change of the income tax system. It was the first comprehensive revision of the Internal Revenue Code in 75 years (*Statement by the President Upon Signing Bill Revising the Internal Revenue Code*, 8/16/54, p. 1).

This bill was passed primarily to increase the fairness and efficiency of the tax system. It had been under discussion since 1952 (Holmans, 1961, p. 219), well before the 1953–54 recession. At the signing ceremony, Eisenhower stressed: “It is a law which will help millions of Americans by giving them fairer tax treatment than they now receive.” He also said, “economic growth will be fostered by such provisions as more flexible depreciation and better tax treatment of research and development costs, thus encouraging all business—large and small—to modernize and expand” (*Statement by the President Upon Signing Bill Revising the Internal Revenue Code*, 8/16/54, p. 1). The Report of the Committee on Ways and Means to Accompany H.R. 8300 (Internal Revenue Code of 1954) repeated this motivation. It stated: “In general, the purpose of these changes has been to remove inequities, to end harassment of the taxpayer and to reduce tax barriers to future expansion of production and employment” (83d Congress, 2d Session, House of Representatives Report No. 1337, 3/9/54, p. 1).

From this perspective, the reduction in revenues at a given level of income was the result of a belief that the bill would stimulate growth and of a desire to make the reform roughly revenue neutral after accounting for the growth effects. The *1954 Economic Report* stated, “A number of additional tax measures should now be taken in order to strengthen the forces of growth in employment and production. These measures, which involve some immediate sacrifices of revenue, contain the seeds of important future revenue gains to be reaped from the economic growth they will stimulate” (p. 79). The *1955 Economic Report* described the motivations for the bill as “to stimulate competitive enterprise, to strengthen the floor of security for the individual, and to curb tendencies toward either depression or inflation,” and referred to “the loss of 1.4 billion dollars as a result of enacting structural tax changes” as almost a side effect of the reform (pp. 19–20).

The possible countercyclical benefits of the bill were certainly mentioned frequently. As discussed above, the administration favored some additional tax relief in mid-1954 because of further declines in defense spending. And, much of the Congressional debate focused on the need for additional stimulus. Indeed, several amendments were offered that involved increasing the personal exemption and cutting tax rates (Holmans, 1961, pp. 229–233). However, once the Excise Tax Reduction Act of 1954 was passed, the Administration argued strenuously against further tax cuts, especially the increase in the personal exemption.

[1] On March 15, 1954, Eisenhower said: “at this time economic conditions do not call for an emergency program that would justify larger Federal deficits and further inflation through large additional tax reductions” (*Radio and Television Address to the American People on the Tax Program*, 3/15/54, p. 4; see also *Annual Budget Message to the Congress: Fiscal Year 1955*, 1/21/54, p. 8). Also, the House report on the bill explicitly downplayed the countercyclical aspects, saying: “This bill is a long overdue reform measure which is vitally necessary regardless of the momentary economic conditions and should not be confused with other measures which may be, or might become, appropriate in the light of a particular

short-run situation” (House Report No. 1337, pp. 1–2). Thus, it seems clear that at least some of the tax cuts contained in the Excise Tax Reduction Act of 1954 and the Internal Revenue Code of 1954 should be classified as exogenous.

[2] Thus, it seems clear that at least some of the tax cuts contained in the Excise Tax Reduction Act of 1954 and the Internal Revenue Code of 1954 should be classified as exogenous. There was a cut beyond what the administration thought was justified by the decline in spending and the state of the economy. Because the excise tax reduction came first and had no obvious other motivation, I choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action.

[3] There was a cut beyond what the administration thought was justified by the decline in spending and the state of the economy. Because the excise tax reduction came first and had no obvious other motivation, we choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action. We feel this makes sense given that the Internal Revenue Code of 1954 clearly had a substantial exogenous component and indeed was passed after it was widely understood that the trough of the recession (May 30, 1954) had been turned (Holmans, 1961, p. 233).

[4] Because the excise tax reduction came first and had no obvious other motivation, we choose to classify it as endogenous and the Internal Revenue Code of 1954 as an exogenous, long-run action. We feel this makes sense given that the Internal Revenue Code of 1954 clearly had a substantial exogenous component and indeed was passed after it was widely understood that the trough of the recession (May 30, 1954) had been turned (Holmans, 1961, p. 233). But, at some level, which of the two tax cuts is classified as endogenous and which as exogenous is somewhat arbitrary.

The act was signed in mid-August 1954, so I date the initial tax change in 1954Q3. My sources give a figure of −\$1.4 billion for the revenue effects at an annual rate (for example, *1955 Economic Report*, p. 20; *1954 Treasury Annual Report*, p. 44). The bill was retroactive to January 1, 1954 (*1954 Treasury Annual Report*, p. 286). It thus reduced taxes by $3 \cdot \$1.4$ billion, or \$4.2 billion at an annual rate, in 1954Q3. The return to the steady-state reduction of \$1.4 billion in 1954Q4 implies a tax increase of \$2.8 billion in 1954Q4. If one chose to neglect the retroactive element, there would be just a tax cut of \$1.4 billion in 1954Q3.

The changes consisted largely of the removal of obvious inequities, complications, ambiguities, and opportunities for tax avoidance, and of provisions designed to stimulate investment. All the changes were legislated to be permanent.

F.1.2 Overlap Between the RR Series and Budget-Based Measures

The budget-based U.S. data are designed to expand coverage beyond the canonical RR tax series, but the two sources are not disjoint by construction. RR classify enacted federal tax changes. The budget data record tax measures as they appear in annual budget documents. These are different sampling rules. They nevertheless intersect when an RR law is itself part of the budget or reconciliation process, or when a standalone RR law later reappears in budget tables through extensions, accelerations, scheduled expirations, or continuing fiscal effects.

This distinction matters for combining the two sources. A match by law title is insufficient. Some budget rows mention an RR statute only because they report later effects of that statute, not because they correspond to the original RR shock. Conversely, a single RR episode may be split across several budget rows by instrument, implementation date, and fiscal year. The relevant object for a crosswalk is therefore the policy provision: the law, tax instrument, implementation timing, and public-finance amount must line up jointly.

I conducted a preliminary provision-level overlap audit between the 50 RR episodes and the budget-

based action rows. The audit finds a small number of clean cases and a larger set of partial or ambiguous cases. The clean cases are useful as proof that the two datasets sometimes cover the same historical tax action. They also show why a mechanical merge is not appropriate.

Table 13: Clean Examples of RR–Budget Overlap

RR episode	Budget-based evidence	Interpretation
Revenue and Expenditure Control Act of 1968	The budget rows include the temporary 10 percent surcharge on individual and corporate income taxes, together with related excise-tax extensions.	Clean overlap. The same high-level tax action appears in both RR and the budget documents, although the budget data split it into separate instrument rows.
Omnibus Budget Reconciliation Act of 1993	The budget rows include the high-income individual tax increase reported in the fiscal-year 1994 revenue proposals.	Clean overlap for the individual-income component. Other OBRA-93 components require additional reconciliation.

Other cases are informative but not yet clean enough to combine without further manual work. For example, the Economic Growth and Tax Relief Reconciliation Act of 2001 appears in the budget data through later accelerations of EGTRRA provisions, so the match is related to the same policy family but not necessarily to the original RR timing. Similarly, the Economic Recovery Tax Act of 1981, the Tax Equity and Fiscal Responsibility Act of 1982, the Revenue Act of 1978, and OBRA 1990 are visible in the budget universe, but preliminary automatic matches often attach the wrong budget provision or only a small subset of the RR amount. The Revenue Acts of 1950 and 1951 illustrate the opposite problem: the statutes are central RR episodes, but the candidate budget rows currently selected by automatic matching are not sufficiently close to the original enacted provisions.

The practical implication is that the budget application should be interpreted as a new document-based sample, not as a simple extension obtained by appending rows to RR. The overlap cases provide validation and a starting point for future integration. A combined RR–budget dataset would require an explicit provision-level crosswalk that records, for each RR action, whether the corresponding fiscal provisions enter the budget data directly, enter later through budget treatment of an existing law, or are absent from the budget-based source universe.

F.2 Romer and Romer replication Robustness checks

F.2.1 Probabilistic validation against the RR classification

As a complementary check, I estimate a simple probabilistic model for whether the original Romer–Romer label is exogenous. Let

$$R_i = \mathbf{1}\{\text{episode } i \text{ is classified as exogenous by Romer–Romer}\}.$$

I fit the one-variable calibration model

$$\Pr(R_i = 1 \mid p_i) = \Lambda\left(a + b \log \frac{p_i}{1 - p_i}\right),$$

where $\Lambda(\cdot)$ is the logistic cdf. The null $b = 0$ corresponds to no monotone predictive content in the policy-level probability. This exercise is deliberately exploratory: the RR classification is used as an external benchmark, not as a definitive latent truth, and the sample contains only 50 policy episodes.

Table 14: Exploratory Logit Validation Against the Original RR Classification

Statistic	Estimate
Number of RR episodes	50
Coefficient on $\log(p/(1 - p))$	0.936
Standard error	0.331
Wald p -value for $b = 0$	0.005
Likelihood-ratio p -value	< 0.001
McFadden pseudo- R^2	0.181
In-sample AUC	0.761
Leave-one-out AUC	0.736

Note: The dependent variable equals one for episodes classified as exogenous in the original RR classification. p is the robust policy-level probability used in Table 3. The leave-one-out AUC refits the one-variable logit after dropping each episode in turn. The exercise is intended as a monotonicity and calibration check, not as a standalone classifier.

The positive and statistically significant slope indicates that higher policy-level probabilities are systematically associated with the original RR exogenous classification. At the same time, the AUC remains far below one. This is consistent with the interpretation in the main text: the probability contains meaningful narrative information, but ambiguous policy episodes are not perfectly separable by a single continuous score.

F.2.2 Exact Romer–Romer Distributed-Lag Macro Check

As a complementary benchmark, Figure 11 reports the exercise closest to the published Romer–Romer dynamic specification. Quarterly real GDP growth is regressed on the contemporaneous tax shock and twelve lags of that shock, together with eleven lags of GDP growth, over the 1950Q1–2007Q4 sample. The figure reports the simulated cumulative GDP response implied by this distributed-lag equation after a one-percent-of-GDP tax innovation. The inflation panel uses the same original-versus-high-support comparison in the baseline inflation projection. This appendix check is useful because it asks whether the probabilities preserve the canonical RR macro content when the macro equation is held fixed, but it is not the paper’s main validation exercise because the main applications use the

support-indexed moment estimator.

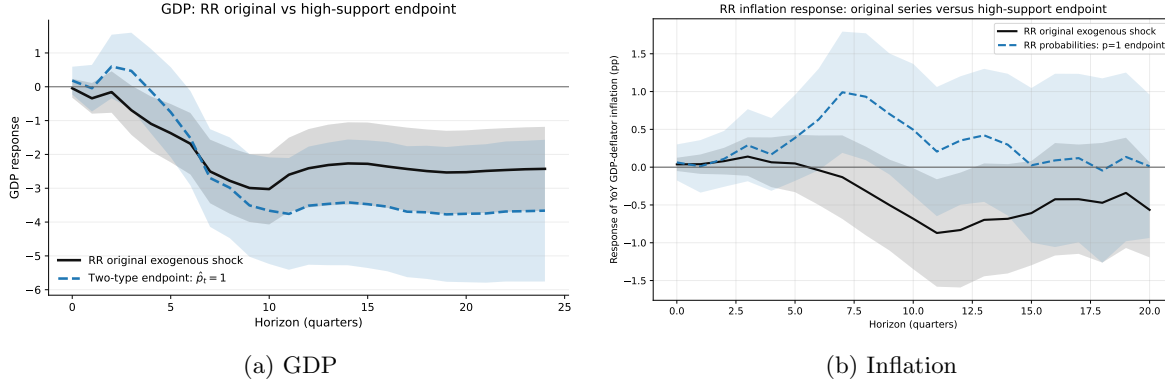


Figure 11: Romer–Romer Responses: Original Series versus High-Support Endpoint

Note: Panel (a) reports the exact Romer–Romer GDP specification. Panel (b) reports the corresponding inflation projection using year-over-year GDP-deflator inflation. The black line is the original RR exogenous shock series. The colored line is the affine two-type estimator evaluated at $p = 1$. Shaded areas denote 68% confidence bands.

The original RR series reaches a peak contraction of roughly 3 log points around quarter 10, in line with the original Romer–Romer result. The two-type $p = 1$ endpoint uses the probability ranking to estimate the high-support endpoint of an affine mixture, rather than converting the continuous score back into a binary rule. The resulting response remains close to the canonical RR response and much farther from the attenuated response obtained by treating all tax changes as equally exogenous. Figure 13 reports the corresponding low- and high-support endpoint diagnostic.

F.2.3 Threshold Sensitivity in the Support-Indexed Moment

Figure 12 reports a binary-style version of the support-indexed moment validation. Instead of estimating the affine endpoint in continuous p_i , the exercise keeps only episodes whose policy-level support exceeds a cutoff $\bar{p}$. This is useful as an interpretive diagnostic because it asks what happens if the continuous support uncertainty is resolved into a hard included/excluded rule.

The hard-threshold version is informative but fragile. Cutoffs $\bar{p} = 0.5$ and $\bar{p} = 0.6$ preserve an RR-like medium-run contraction in the moment estimator: at horizon 8, the estimates are -1.69 and -2.52 , compared with -2.04 for the original RR exogenous indicator in the same moment specification. Raising the cutoff to $\bar{p} = 0.7$ removes additional pivotal episodes, reduces support to about 22 event quarters at short horizons, and weakens or reverses parts of the response. Appendix F.2.5 documents the narrative content of the relevant medium-support episodes. The sensitivity is the reason the main text does not choose a single threshold as the headline validation object.

F.2.4 Two-type endpoint diagnostic

Figure 13 unpacks the two-type estimator used in the exact-RR distributed-lag exercise. The left panel reports the two endpoint responses, $p = 0$ and $p = 1$. The right panel shows the implied linear mapping at horizon $h = 10$, where the exact-RR benchmark reaches its peak contraction in the baseline specification. Figure 14 reports the corresponding diagnostic for inflation using the baseline inflation projection.

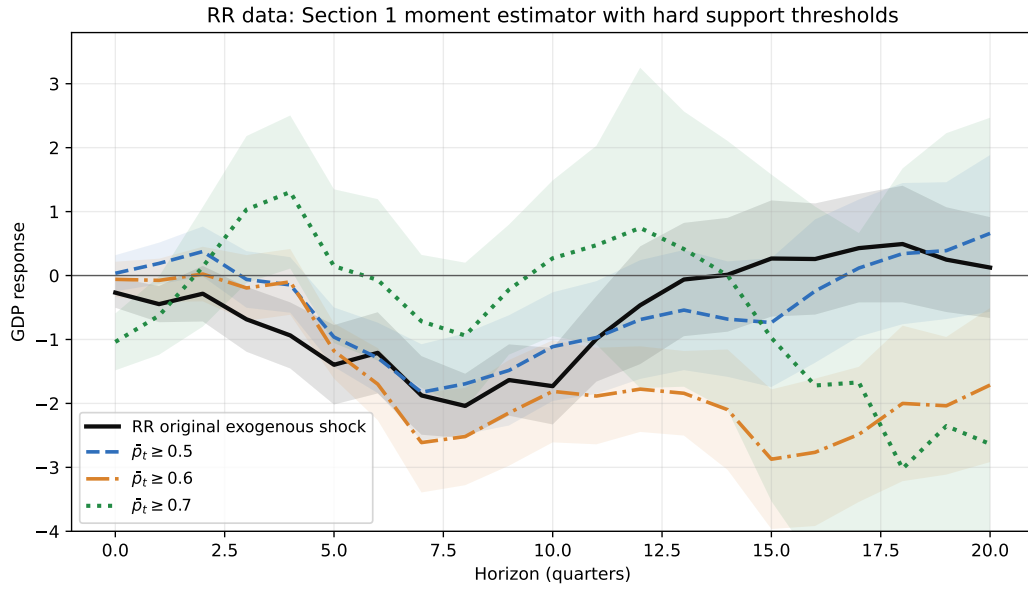


Figure 12: Romer–Romer Moment Validation Under Hard Support Thresholds

Note: The figure reports the support-indexed moment estimator after replacing continuous support with hard threshold rules. The benchmark line uses the original RR exogenous indicator in the same moment estimator. Threshold rules keep episodes with quarter-level support at least $\bar{p}$. Shaded areas denote 68% HAC confidence bands.

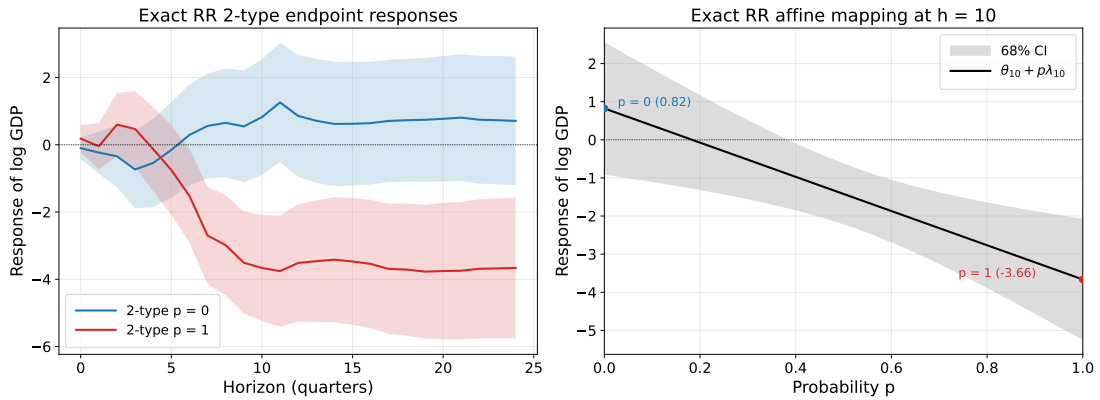


Figure 13: Two-Type RR Benchmark: Endpoint IRFs and the Mapping at $h = 10$

Note: The left panel reports the exact-RR two-type endpoint responses for $p = 0$ and $p = 1$, each with 68% confidence bands. The right panel plots the affine response mapping $\theta_h + p\lambda_h$ at horizon $h = 10$, together with its 68% confidence band.

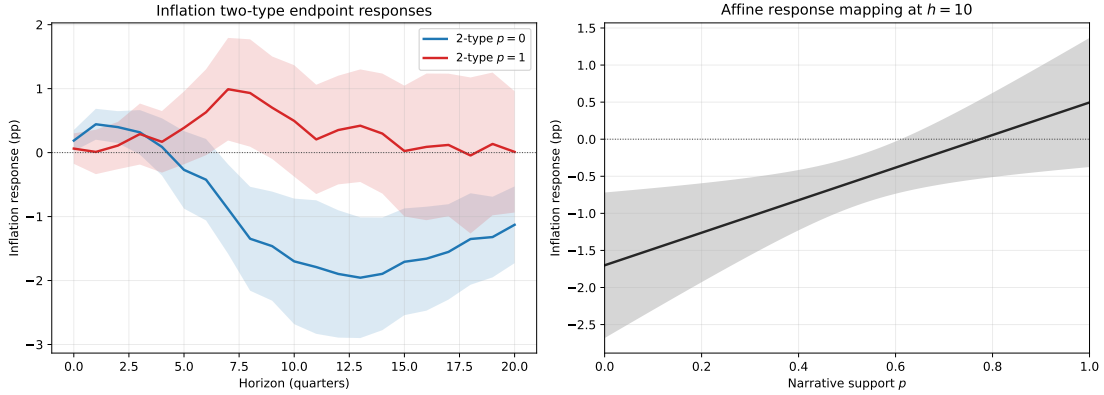


Figure 14: Inflation Two-Type RR Benchmark: Endpoint IRFs and the Mapping at $h = 10$

Note: The left panel reports the inflation two-type endpoint responses for $p = 0$ and $p = 1$, each with 68% confidence bands. The right panel plots the affine response mapping $\theta_h + p\lambda_h$ at horizon $h = 10$, together with its 68% confidence band. This is an inflation projection analogue of the exact-RR GDP diagnostic, not the published Romer–Romer GDP distributed-lag specification.

F.2.5 Narrative Ambiguity in the Romer–Romer Probability Benchmark

This appendix provides a qualitative audit of the policy-level probabilities used in the Romer–Romer benchmark. The exercise uses the edited narrative texts described in Appendix F.1.1: the model observes the policy narrative after the explicit Romer–Romer classification statement and the immediately adjacent classification cues have been removed. The purpose is not to mechanically reproduce the original binary label in Romer and Romer (2010). Instead, the diagnostic asks whether intermediate values of $\hat{p}_i$ correspond to genuine ambiguity in the narrative record, or whether they reveal cases where the estimated support appears too high or too low relative to the semantic object used in the narrative-tax literature.

The key distinction is that the Romer–Romer label combines several judgment margins. Some episodes are excluded because the tax change responds to current or prospective macroeconomic conditions. Others are excluded because the tax change is tied to spending or benefit changes. Conversely, some deficit-driven actions are retained as exogenous under the Romer–Romer convention, even though the narrative is explicitly motivated by a fiscal state variable. A continuous support measure need not collapse these margins into the same binary decision. Table 15 summarizes the main cases, and the discussion below separates clean alignments from boundary cases and potentially informative departures from the hard classification.

Clean alignment. Several cases show that the probability object behaves in the expected direction when the narrative evidence is relatively clear. The Tax Reform Act of 1986 receives very high support, $\hat{p}_i = 0.964$. This is consistent with the Romer–Romer narrative, where the motivation is tax reform, fairness, simplicity, and long-run incentives. The narrative does mention economic growth, but the relevant growth motive is long-run reform rather than contemporaneous stabilization. This is a useful high-support benchmark because it is an RR-exogenous episode that also looks clean under the softer probability object.

At the other end, the expiration of the Korean-War temporary tax increases receives very low

Table 15: Romer–Romer Policy Narratives and Estimated Support

Policy	RR category	$\hat{p}_i$	Interpretation
Expiration of Korean-War temporary taxes	Endogenous, spending-driven	0.217	The low support is consistent with the original narrative: the tax reductions followed the end of Korean-War spending and a changed postwar macro-fiscal environment.
Social Security Amendments of 1958	Endogenous, spending-driven	0.667	The RR exclusion is about benefit financing and actuarial balance, not a direct response to output or inflation. The medium-high support is sensible if $\hat{p}_i$ primarily captures macro exogeneity rather than all package contamination.
Revenue Act of 1964	Exogenous, long-run	0.611	A boundary case. RR emphasize long-run growth and explicitly reject a quick recession-prevention tax cut. The same text, however, repeatedly invokes unemployment, below-capacity production, and fiscal drag.
Revenue and Expenditure Control Act of 1968	Endogenous, countercyclical	0.526	The narrative is directly macro-stabilization oriented: overheating, inflation pressure, fiscal restraint, and external pressure on the dollar. Support remains only moderate.
Tax Reform Act of 1969	Endogenous, countercyclical	0.659	The probability may be high relative to a strict RR rule. The narrative contains inflation-control language, but also fairness, equity, poverty relief, and tax-reform motives.
Crude Oil Windfall Profit Tax Act of 1980	Exogenous, long-run	0.643	RR treat the tax as unrelated to macro stabilization. The medium support reflects a different ambiguity: the policy is tied to oil-price decontrol, energy policy, and consumer-assistance uses of revenue.
Social Security Amendments of 1983	Exogenous, deficit-driven	0.537	The probability is low relative to the RR deficit-driven convention because the narrative is strongly tied to trust-fund solvency, Social Security financing, and deficit reduction.
Tax Reform Act of 1986	Exogenous, long-run	0.964	A clean alignment case: tax reform, fairness, simplicity, and long-run incentives dominate the narrative, with little role for current macro conditions.
OBRA 1990	Exogenous, deficit-driven	0.528	RR classify the act as deficit-driven, but the narrative acknowledges recession timing and weak expected growth. The medium-low support captures this macro-timing discomfort.
OBRA 1993	Exogenous, deficit-driven	0.577	The act is not a short-run stabilization measure, but deficit reduction is the central state-contingent motive. The probability is lower than a strict RR binary label would suggest.
Taxpayer Relief and Balanced Budget Acts of 1997	Endogenous, spending-driven	0.659	The RR exclusion reflects budget-package contamination. The tax-side rationale is less clearly cyclical, so the estimated support is relatively high.
EGTRRA 2001	Endogenous, countercyclical	0.527	The medium-low support is consistent with mixed motives: surplus return and long-run incentives on one side, explicit slowdown and short-run stimulus language on the other.

Note: RR category denotes the original Romer–Romer classification. The probability is the robust policy-level support measure used in the Romer–Romer validation exercise. The table focuses on cases that are useful for interpreting intermediate support values, especially policies entering quarters with $0.5 \leq \bar{p}_t < 0.7$.

support, $\hat{p}_i = 0.217$. Romer–Romer treat the expiration as a deliberate policy choice after the wartime fiscal environment changed. The motivation is tied to the end of Korean-War spending and to the macro-fiscal context in which temporary taxes were allowed to lapse. The low probability is therefore not a failure to recover a label; it reflects the same identifying concern that motivates the original endogenous, spending-driven classification.

The 2001 Economic Growth and Tax Relief Reconciliation Act is also informative. The probability, $\hat{p}_i = 0.527$, is neither close to zero nor close to one. This matches the narrative structure. The act combines an argument about returning projected budget surpluses and improving long-run incentives with an explicit short-run stimulus motive during a slowdown. The intermediate value is therefore more informative than a hard label: it records that the narrative contains both exogeneity-supporting and contamination-supporting elements.

Boundary exogenous cases. The Revenue Act of 1964 is the central boundary case. Romer–Romer classify it as exogenous and long-run because the stated objective was to raise the economy’s longer-run growth path rather than to prevent an immediate recession. In their narrative, the administration explicitly distinguished the proposal from a “quick, temporary tax cut” and emphasized longer-run economic challenges. At the same time, the surrounding text contains substantial language about unemployment, below-capacity production, and the removal of fiscal drag. The resulting probability, $\hat{p}_i = 0.611$, is therefore economically meaningful. It does not say that Romer–Romer are wrong to classify the episode as exogenous. It says that the documentary support for a clean orthogonal interpretation is partial rather than complete.

This case is also empirically important because it helps explain the threshold diagnostic. Moving from a 0.6 threshold to a 0.7 threshold removes the 1964 tax-cut episode, even though that episode is central to reproducing the Romer–Romer-style contractionary response. The high threshold is therefore not merely a stricter version of the same sample. It drops an influential historical episode in which the original narrative classification is defensible, but not unambiguous.

The Crude Oil Windfall Profit Tax Act of 1980 illustrates a different kind of boundary case. Romer–Romer classify the act as exogenous because the tax was motivated by capturing windfall profits from oil-price decontrol rather than by managing output or inflation. The probability is nevertheless only 0.643. This does not necessarily imply that the estimate is biased downward. Rather, the policy is tied to oil-price regulation, energy policy, consumer assistance, and the use of earmarked revenues. Those features are not the same as current-macro endogeneity, but they make the tax change less like an isolated orthogonal shock. The intermediate support is therefore interpretable once the object is broader than a binary replication of the RR label.

Deficit and Social Security cases. Deficit-driven and Social Security cases reveal the main semantic tension between the Romer–Romer taxonomy and a probability of narrative support. Romer–Romer retain some deficit-driven tax increases as exogenous when the stated motivation is not to offset current cyclical conditions. In contrast, a text-based probability may assign less than full support because the tax action is explicitly motivated by a fiscal state variable.

The Social Security Amendments of 1983 make this tension clear. Romer–Romer classify the episode as exogenous, deficit-driven, but the narrative is organized around Social Security solvency, trust-fund financing, and deficit reduction. The estimated support is $\hat{p}_i = 0.537$. This is low relative to

the original binary classification, but it is not arbitrary: the tax change is not presented as an isolated tax-policy experiment. It is a response to a program-financing problem. If the relevant semantic object is "not countercyclical," the RR label is natural. If the object is "high support for orthogonality," the lower probability is also natural.

The 1990 and 1993 budget acts show the same issue. OBRA 1990 receives $\hat{p}_i = 0.528$. The original classification is deficit-driven, but the narrative also notes recession timing and weak expected growth. OBRA 1993 receives $\hat{p}_i = 0.577$. The motive is deficit reduction to improve long-run macroeconomic performance, not short-run demand management. Still, deficit reduction is a state-contingent fiscal motive. The medium probabilities therefore reflect the fact that RR's deficit-driven convention is not the same object as full narrative support for an orthogonal tax shock.

The Social Security Amendments of 1958 point in the opposite direction. They are classified as endogenous, spending-driven, because higher payroll taxes financed higher benefits and improved actuarial balance. The estimated support is 0.667, which is high relative to the hard RR label. This is a plausible departure if the probability is more sensitive to current or prospective macroeconomic conditions than to program-financing bundles. The narrative is clearly contaminated by benefit financing, but it is not primarily a story about output, inflation, or stabilization. This is a case where the probability may be too high if the intended object is the full RR exclusion rule, but reasonable if the intended object is support for macro exogeneity.

Cases with potentially high or low support relative to expectations. The Tax Reform Act of 1969 is a case where the probability may be high relative to a strict reading of the RR taxonomy. Romer–Romer classify the act as endogenous, countercyclical, partly because the administration accepted a period of slower growth to reduce inflation. The extracted motivations also include fairness, poverty relief, and political tax relief. The probability, $\hat{p}_i = 0.659$, appears to weight the non-cyclical components substantially. This is informative, but it is also a case to watch: if the target is strict exclusion of policies motivated by current or prospective macro conditions, the inflation-control motive suggests that support should perhaps be lower.

The 1997 Taxpayer Relief Act and Balanced Budget Act provide a complementary example. Romer–Romer classify the tax cuts as endogenous, spending-driven, because they are tied to a broader budget package in which spending reductions made the tax cuts feasible. The estimated support, $\hat{p}_i = 0.659$, is high relative to that hard label. The reason is understandable: the tax-side motive is not clearly a short-run stabilization response. However, for the RR exclusion rule, the relevant contamination is the package structure itself. The case therefore suggests that a probability focused on macro exogeneity may under-penalize spending-package dependence unless that distinction is made explicit in the measurement target.

Conversely, the Social Security Amendments of 1983 and the deficit-driven budget acts may receive probabilities that are low relative to the original RR convention. This does not mean the measure is biased in a mechanical sense. It indicates that the probability object is stricter than the RR binary label whenever deficit reduction or program solvency is treated as a policy-state motive rather than as a clean exogenous reason for a tax change.

Takeaway. The case readings support the main interpretation of the probability benchmark. The estimated support is not simply a noisy approximation to the Romer–Romer hard label. It often

captures economically meaningful variation inside the hard classes. High-support cases correspond to narratives dominated by tax reform, long-run incentives, or clear orthogonal motives. Low-support cases correspond to war-spending, macro-stabilization, or explicit short-run stimulus motives. Intermediate cases are especially useful: they reveal where RR’s binary decision compresses several forms of ambiguity into a single label. This is precisely the margin on which soft narrative support is meant to operate. The probability should therefore be interpreted as a graded measure of documentary support for orthogonality, not as an automated relabeling of the original Romer–Romer classification.

F.2.6 Probabilistic validation against the Cloyne classification

I apply the same one-variable probabilistic check to the Cloyne validation sample. Let

$$C_i = \mathbf{1}\{\text{policy row } i \text{ is classified as exogenous by Cloyne}\}.$$

I estimate

$$\Pr(C_i = 1 \mid p_i) = \Lambda\left(a + b \log \frac{p_i}{1 - p_i}\right),$$

where the sample consists of Cloyne policy rows with usable canonical labels and complete probability inputs. Because many policy rows belong to the same narrative event, I report both row-level and event-clustered uncertainty, and I also evaluate prediction after holding out entire narrative events.

Table 16: Exploratory Logit Validation Against the Cloyne Classification

Statistic	Estimate
Number of Cloyne policy rows	2,101
Coefficient on $\log(p/(1 - p))$	0.551
Row-level standard error	0.054
Row-level Wald p -value for $b = 0$	< 0.001
Event-clustered standard error	0.100
Event-clustered Wald p -value for $b = 0$	< 0.001
Likelihood-ratio p -value	< 0.001
McFadden pseudo- R^2	0.042
In-sample AUC	0.629
Leave-one-event-out AUC	0.612
Event-grouped 5-fold AUC	0.621
Accuracy from raw $p \geq 0.5$	0.640

Note: The dependent variable equals one for policy rows classified as exogenous by Cloyne. Event-clustered inference clusters rows by narrative event. The leave-one-event-out AUC refits the one-variable logit after dropping each event group in turn; the event-grouped 5-fold AUC holds out folds of entire event groups. The exercise is intended as a monotonicity check against an external expert label, not as a standalone classifier.

The Cloyne probability is therefore statistically predictive of the external expert label: the slope is positive and remains precisely estimated after clustering by narrative event. The predictive strength is still modest, however. The AUC is about 0.63 in sample and around 0.61–0.62 under event-held-out evaluation. This is consistent with the main-text table: the probability ranks Cloyne-exogenous rows above Cloyne-endogenous rows on average, but the two distributions overlap substantially. I interpret this as evidence of monotone external-label validity, not as evidence that the procedure mechanically reproduces Cloyne’s binary taxonomy.

F.2.7 Training-data leakage robustness check

To assess whether the language model has memorized specific source material, I implement a sentence-reconstruction test taken from Golchin and Surdeanu’s (2024) contamination detection framework. The intuition is the following: if a model has encountered a document during training, it may be able to reproduce missing content from that document with unusual precision—especially when provided with cues pointing to the original source. To implement this test, I systematically remove sentences from the Romer and Romer narratives and asks the model to reconstruct them under two conditions: one relying solely on the surrounding context, and the other explicitly guiding the model by referencing the source paper and section. If the model performs significantly better under guided prompting, this may suggest prior exposure to the original text. To quantify and evaluate the reconstruction quality, I use both semantic (embedding cosine similarity) and lexical (ROUGE-L) metrics, followed by a bootstrap-based significance test to flag cases of potential memorization.

Using both metrics, and evaluating with `gpt-4o-mini` and `gemini-2.0-flash`, no narrative is flagged as a potential case of memorization. This result strengthens confidence that the model was not trained on Romer and Romer (2010), and supports the validity of the replication exercise by reducing concerns about training-data leakage as a source of bias in the model’s classifications.

Procedure.

1. **Multiple draws & prompts.** For each narrative d I perform $T = 10$ draws. In draw t , one sentence $s_{d,t}$ is chosen uniformly at random, removed, and replaced by the token [MISSING SENTENCE]. I then issue two prompts a general prompt, and a guided prompt (Prompts 9).

General prompt:

You are provided with a passage in which one sentence has been removed and replaced with [MISSING SENTENCE]. Based **ONLY** on the context provided, generate a plausible sentence that could fill the gap. You do not need to match any specific source text — just ensure the result is coherent and contextually appropriate.

{narrative text from Romer and Romer}

Reconstructed sentence:

Guided prompt:

You are provided with a passage from the scientific paper *A Narrative Analysis of Postwar Tax Changes* by Christina D. Romer and David H. Romer, published in 2010. The passage comes from the section titled “{title}” in the main body of the paper.

One sentence has been removed and replaced with [MISSING SENTENCE]. Based **only** on the context provided and your knowledge of this specific paper, reconstruct the missing sentence exactly as it appeared.

{narrative text from Romer and Romer}

Reconstructed sentence:

Prompt 9: General and Guided Prompt

The model returns two reconstructions $\hat{s}_{d,t}^{\text{gen}}$ and $\hat{s}_{d,t}^{\text{guide}}$.

2. **Similarity metrics.** For each original $s_{d,t}$ and each reconstruction, compute:

- *Embedding cosine similarity* $\rho_{d,t}^{\text{gen}} = \cos(s_{d,t}, \hat{s}_{d,t}^{\text{gen}})$, $\rho_{d,t}^{\text{guide}} = \cos(s_{d,t}, \hat{s}_{d,t}^{\text{guide}})$, using `text-embedding-3-large`.
- *ROUGE-L F1 score* $r_{d,t}^{\text{gen}}$ and $r_{d,t}^{\text{guide}}$ between the original and each reconstruction.

3. **Paired bootstrap test.** For each narrative d , define the mean differences

$$\Delta\rho_d = \frac{1}{T} \sum_{t=1}^T (\rho_{d,t}^{\text{guide}} - \rho_{d,t}^{\text{gen}}), \quad \Delta r_d = \frac{1}{T} \sum_{t=1}^T (r_{d,t}^{\text{guide}} - r_{d,t}^{\text{gen}}).$$

We perform a nonparametric paired bootstrap with $B = 1,000$ resamples of the T pairs $\{(\rho_{d,t}^{\text{guide}}, \rho_{d,t}^{\text{gen}})\}$ (and similarly for $\{(r_{d,t}^{\text{guide}}, r_{d,t}^{\text{gen}})\}$), yielding bootstrap distributions of $\Delta\rho_d$ and Δr_d . Each indicator $\mathbf{1}\{\Delta\rho_d^{*(b)} \leq 0\}$ (or $\mathbf{1}\{\Delta r_d^{*(b)} \leq 0\}$) counts a bootstrap sample in which the similarity under the guided prompt is no better than—or worse than—that under the general prompt; that is, the model failed to improve reconstruction quality despite being explicitly cued about the source. The empirical p -values are

$$p_d^\rho = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\Delta\rho_d^{*(b)} \leq 0\}, \quad p_d^r = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\Delta r_d^{*(b)} \leq 0\}.$$

Decision rule. A narrative d is flagged as potentially memorized if its bootstrap p -value for either similarity metric falls below 0.05, i.e. if guided prompting produces significantly closer reconstructions than general prompting.

F.2.8 Manually Investigating Legacy Binary Misclassifications

The misclassification pattern, mostly exogenous policies misclassified as endogenous, can possibly be attributed to an asymmetry inherent in macroeconomic narrative analysis: while it is relatively straightforward to construct narratives supporting the endogeneity of a policy intervention, establishing its exogeneity typically requires more stringent evidence. In the absence of sufficiently strong signals of such independence LLMs may default to a conservative interpretation, erring on the side of endogeneity.

Steering growth above normal levels. A recurring source of systematic misclassification involves tax cuts enacted with the aim of boosting growth above normal levels. In at least three instances, the LLM interprets these as countercyclical, and thus endogenous, due to the presence of growth-oriented motivation. However, according to Romer and Romer’s decision rule, such motivations qualify as exogenous, as they are not designed to return output to trend but to exceed it. This kind of mislabeling can be mitigated by explicitly including this rule in the prompt design or few-shot examples, highlighting that aspirational growth policies, unless tied to negative output gaps, are to be treated as exogenous.

Conflicting motivations. Another set of misclassifications arises from cases with mixed or conflicting motivations. For example, one narrative discusses a tax increase intended primarily for deficit reduction and solvency of the Highway Trust Fund, both exogenous motives, but also references inflation concerns and contemporaneous spending increases. While the latter suggest endogenous intent, the original classification deems the measure exogenous with noted ambiguity. Such nuanced judgment

is difficult for LLMs to replicate without additional guidance on prioritizing primary over secondary motivations.

”Package” related measure. Finally, there are highly specific edge cases that hinge on bespoke decision rules. A salient example is the Tax Adjustment Act of 1966. Despite being implemented to counteract high growth following an exogenous tax cut, Romer and Romer classify it as exogenous, arguing that treating it as endogenous would yield nonsensical results. This type of reasoning, context-specific and precedent-based, poses a significant challenge for LLMs, particularly in the absence of similar examples in the prompt.

F.3 Affine GMM Endpoint Diagnostic

The main text reports the high-support $p = 1$ endpoint of the baseline affine GMM estimator. Figure 15 reports the corresponding low-support $p = 0$ endpoint as a diagnostic. This contrast is useful for interpreting the affine specification: the contractionary medium-run response is concentrated at the high-support endpoint, while the low-support endpoint is much flatter and eventually positive. The $p = 0$ series should not be read as a separately observed class of purely endogenous policies, however, because the policy-level probabilities are mostly interior and contain few observations close to the boundary. It is best interpreted as the low-support extrapolation implied by the affine moment model.

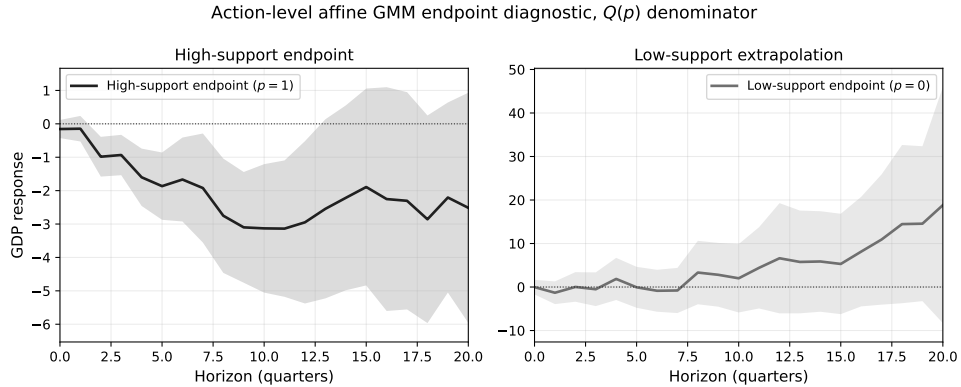


Figure 15: Affine GMM Endpoint Diagnostic: $p = 0$ and $p = 1$

Note: The figure reports the U.S. budget affine GMM endpoints for real GDP. The black curve is the main $p = 1$ endpoint and the gray curve is the implied $p = 0$ endpoint. Both use the same full-quarter GMM estimator, residualization, affine denominator $Q_h(p)$ evaluated at the corresponding endpoint, and 68% HAC confidence bands. The panels use separate vertical scales because the $p = 0$ endpoint is a low-support extrapolation, not a separately observed policy class.

F.4 Measurement Diagnostics for the U.S. Budget Endpoint

This appendix reports two diagnostics for the generated support score used in the U.S. budget application. The first asks how sensitive the GDP and inflation endpoints are to direct violations of the projected moment condition. The second asks whether the endpoints are driven by one evaluator model family.

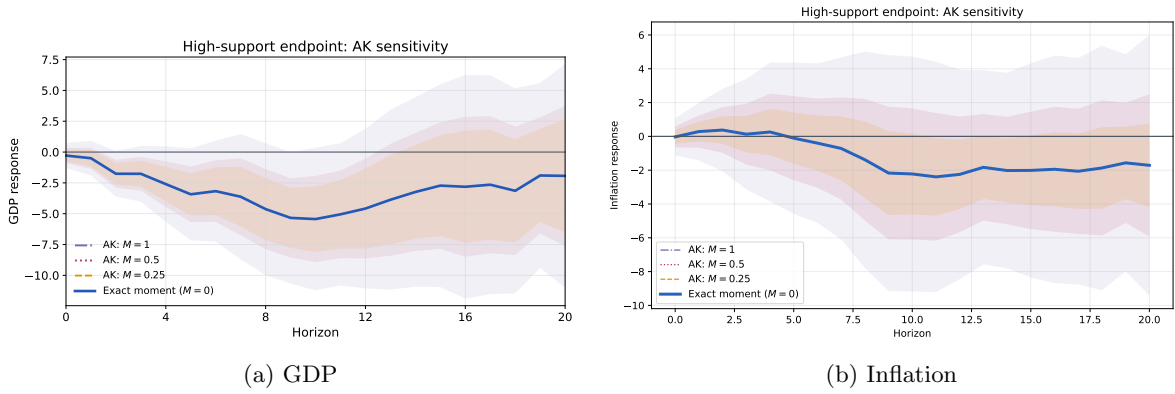


Figure 16: U.S. Budget Endpoints: Approximate-Moment Sensitivity

Note: Each panel reports bias-aware endpoint intervals under approximate moment validity. $M = 0$ corresponds to exact projected moment validity. Larger values of M allow direct moment violations measured relative to the Newey–West sampling-error scale.

Figure 16 shows that moderate approximate-moment violations widen the bands. The GDP contraction remains visible under small violations, while the inflation endpoint is more sensitive once larger direct moment failures are allowed. This exercise should be read as a sensitivity diagnostic, not as a correction for a specific measurement-error model.

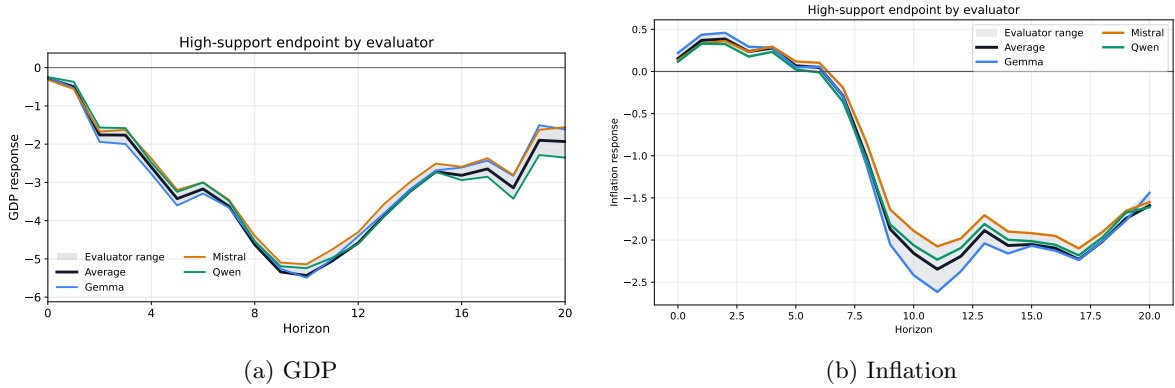


Figure 17: U.S. Budget Endpoints: Evaluator-Model Stability

Note: Each line recomputes the same endpoint using support scores from one evaluator model family. The exercise isolates sensitivity to the language-model representation used to construct the support score.

Figure 17 shows that the model-specific endpoints are not identical, but they preserve similar medium-run patterns for both GDP and inflation. This diagnostic indicates that the main pattern is not driven by a single evaluator family.

F.5 Component Diagnostics and Romer–Romer Comparison

Figure 18 reports the remaining component panels corresponding to Figure 14 in Romer and Romer (2010). The comparison is useful for separating the mechanism that carries over from the original

RR exercise from the margins that are less stable in the document-based budget application. Romer–Romer find that durable consumption falls much more than nondurables and services, that both nonresidential and residential fixed investment decline, and that net exports improve, with exports rising and imports falling. In the support-indexed GMM estimates, the domestic investment channel remains the clearest common element: nonresidential fixed investment falls strongly and persistently. The remaining margins are less aligned. Durables fall early but later rebound, nondurables and services move little, residential investment turns positive at longer horizons, and the external-sector responses go in the opposite direction of the original RR pattern. I therefore treat these panels as component diagnostics rather than as the central mechanism evidence.

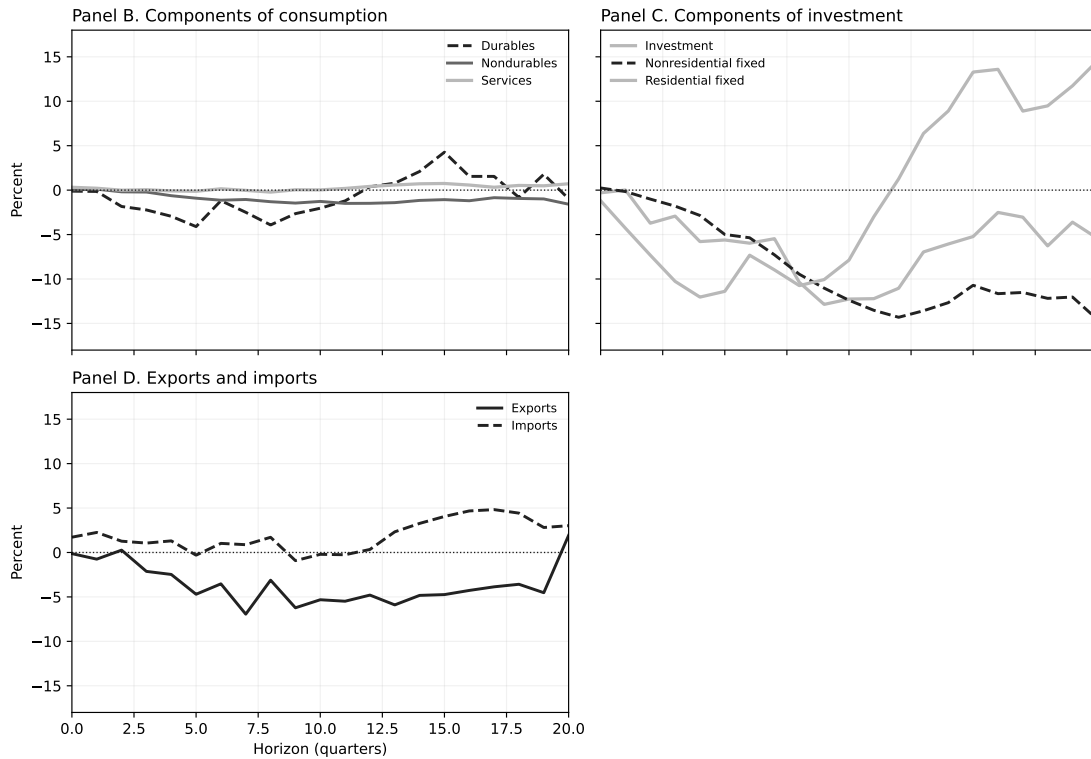


Figure 18: Additional RR-Style Component Responses Under the Baseline GMM Estimator

Note: The figure reports affine $p = 1$ endpoint responses using the same budget-measure probabilities, quarterly tax shocks, endpoint denominator, and HAC inference as Figure 5. Components are log quantity indexes, so responses are percent changes. The panels correspond to Panels B–D in Figure 14 of Romer & Romer (2010).

Tables 17–19 decompose the mechanism outcomes using the same additive object as Table 6. The pattern is consistent with the main-text interpretation. The medium-run decline in inflation and the large fall in investment are mainly associated with income-tax components, especially individual income taxes. Consumption moves in the same direction but less sharply. The nonstandard consumption/excise residual continues to behave differently from a clean indirect-tax object, reinforcing the decision to treat the UK–France VAT evidence as the sharper consumption-tax contrast.

Table 17: Composition and Additive Inflation Contributions to the U.S. Budget Endpoint

Instrument	Event share	Shock-size share	Mean p	Inflation $h = 4$	Inflation $h = 8$	Inflation $h = 12$
Individual income	30.6%	37.0%	0.65	0.21 (0.41)	-0.46 (0.69)	-1.07 (0.66)
Corporate income	32.2%	22.7%	0.64	0.13 (0.18)	0.02 (0.19)	-0.43 (0.36)
Clean standard excise	8.9%	11.2%	0.35	0.04 (0.06)	0.04 (0.08)	0.07 (0.07)
Nonstandard consumption/excise	11.1%	15.5%	0.40	-0.25 (0.26)	-0.26 (0.26)	0.15 (0.18)
Other low-support taxes	17.2%	13.6%	0.32	0.02 (0.14)	-0.08 (0.18)	0.10 (0.24)

Note: Entries are additive contributions to the aggregate affine $p = 1$ budget endpoint for year-over-year GDP-deflator inflation. Parentheses report HAC standard errors. Event shares, shock-size shares, and support statistics are defined as in Table 6.

Table 18: Composition and Additive Consumption Contributions to the U.S. Budget Endpoint

Instrument	Event share	Shock-size share	Mean p	Cons. $h = 4$	Cons. $h = 8$	Cons. $h = 12$
Individual income	30.6%	37.0%	0.65	-1.08 (0.82)	-1.54 (1.18)	-1.11 (1.56)
Corporate income	32.2%	22.7%	0.64	-0.51 (0.34)	-0.62 (0.43)	-0.40 (0.43)
Clean standard excise	8.9%	11.2%	0.35	0.08 (0.09)	-0.05 (0.17)	-0.18 (0.22)
Nonstandard consumption/excise	11.1%	15.5%	0.40	0.72 (0.49)	0.94 (0.62)	1.09 (0.72)
Other low-support taxes	17.2%	13.6%	0.32	0.04 (0.13)	0.15 (0.26)	0.28 (0.28)

Note: Entries are additive contributions to the aggregate affine $p = 1$ budget endpoint for log private consumption. Parentheses report HAC standard errors. Event shares, shock-size shares, and support statistics are defined as in Table 6.

Table 19: Composition and Additive Investment Contributions to the U.S. Budget Endpoint

Instrument	Event share	Shock-size share	Mean p	Invest. $h = 4$	Invest. $h = 8$	Invest. $h = 12$
Individual income	30.6%	37.0%	0.65	-5.68 (2.02)	-9.73 (3.67)	-9.51 (3.56)
Corporate income	32.2%	22.7%	0.64	-2.94 (1.43)	-3.86 (2.02)	-4.13 (1.66)
Clean standard excise	8.9%	11.2%	0.35	0.08 (0.38)	-0.37 (0.55)	-1.02 (0.69)
Nonstandard consumption/excise	11.1%	15.5%	0.40	2.77 (1.13)	3.75 (1.99)	3.22 (2.04)
Other low-support taxes	17.2%	13.6%	0.32	-0.02 (0.71)	-0.16 (1.09)	0.39 (1.18)

Note: Entries are additive contributions to the aggregate affine $p = 1$ budget endpoint for log private investment. Parentheses report HAC standard errors. Event shares, shock-size shares, and support statistics are defined as in Table 6.

G VAT Shocks and Macro Transmission in the United Kingdom and France

G.1 Additional VAT Channel Diagnostics

Figure 19 reports the fuller expenditure-side decomposition. The figure keeps the same outcome transformation as the main mechanism figure: each component is expressed as a change from the pre-event quarter in percentage points of trend GDP. The main text focuses on private consumption and investment because they are the central domestic-absorption channels for a VAT shock. The additional panels show that government consumption is muted at short horizons, while net exports partly offset the contraction.

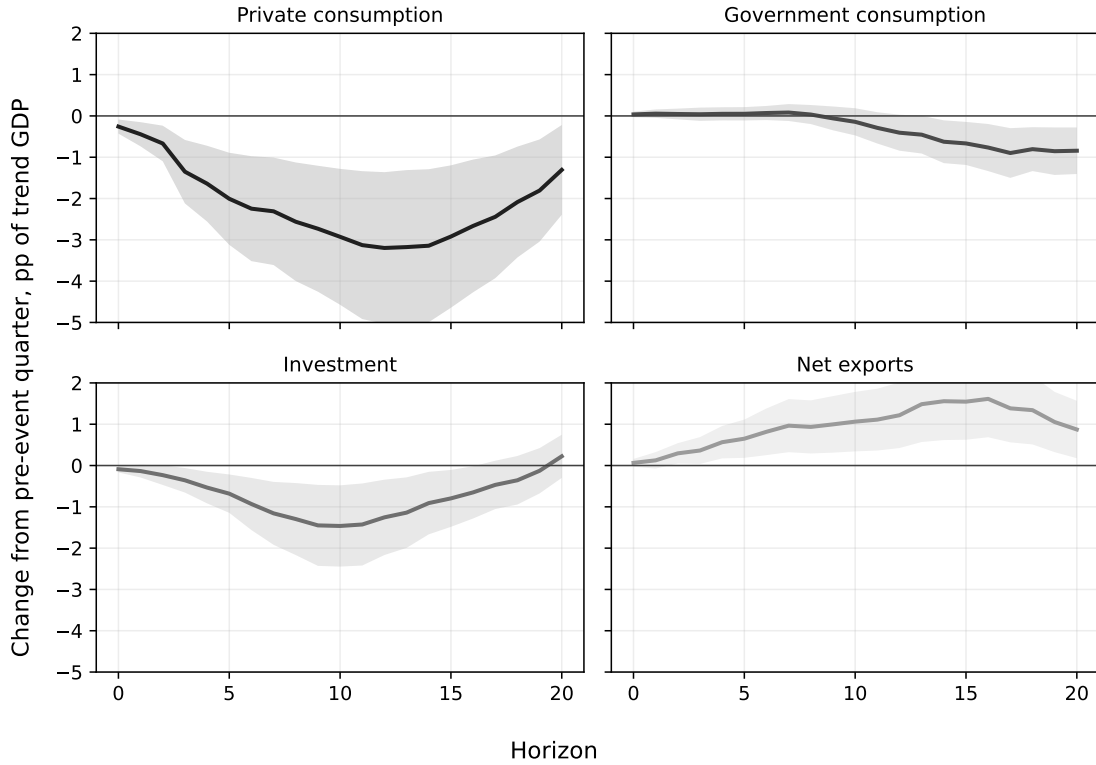


Figure 19: UK-France VAT: Additional Expenditure-Side Responses

Note: Outcomes are expressed as $y_{c,t+h} - y_{c,t-1}$ in percentage points of trend GDP. The figure reports the pooled VAT endpoint for private consumption, government consumption, investment, and net exports on a common vertical scale. Shaded areas are 68 percent HAC confidence bands.

Table 20 reports the corresponding key-horizon estimates. Decomposing net exports shows that the external offset is mostly import compression rather than a robust export boom: exports are small and imprecise, while the import contribution is positive. Since imports enter the accounting decomposition as $-M$, a positive import contribution means that imports fall.

Table 20: VAT Additional Channels at Key Horizons

Outcome	$h = 4$	$h = 8$	$h = 12$
GDP deviation from trend	-2.54	-4.72	-4.22
Private consumption response	-1.64	-2.56	-3.20
Government consumption response	0.05	0.03	-0.41
Investment response	-0.53	-1.30	-1.25
Exports response	0.15	-0.38	0.23
Imports contribution ($-M$)	0.42	1.32	0.99
Net exports response	0.56	0.93	1.22

Note: Entries are $y_{c,t+h} - y_{c,t-1}$ in percentage points of trend GDP. Imports enter as $-M$, so a positive import contribution means import compression. Net exports are exports plus the $-M$ import contribution.

G.2 VAT Packages and the Nguyen–Onnis–Rossi Concern

Nguyen et al. (2021) emphasize that UK income-tax and consumption-tax changes frequently occur together. In their proxy-SVAR, the broad consumption-tax object is an average consumption tax rate built from VAT and consumption duties. They find modest effects for this broad object once income-tax and consumption-tax shocks are jointly identified, while single-proxy specifications can make consumption taxes look much more powerful. This appendix asks whether the same concern jeopardizes the VAT endpoint reported in the main text.

The checks keep the UK–France sample fixed at 1955Q1–2007Q4. They vary only the tax actions entering the VAT object. The first diagnostic is a co-movement audit. Same-quarter overlap is high, but the VAT-income correlation is small: the pooled correlation between the VAT shock and the income-like shock is 0.02 over all quarters and 0.04 over policy quarters. An explicit two-instrument denominator diagnostic gives nearly identical GDP endpoints. At horizons 4, 8, and 12, the naive VAT endpoints are -2.47 , -4.60 , and -4.10 , while the VAT/income adjusted endpoints are -2.48 , -4.61 , and -4.12 .

Table 21 reports the main sensitivity estimates. Dropping UK 1979Q3, the VAT unification episode highlighted by Nguyen et al. (2021), weakens precision but preserves the negative GDP response. Dropping both UK 1973Q2 and UK 1979Q3 weakens the result more substantially, which shows that large UK VAT transition episodes are empirically important. By contrast, dropping VAT actions in quarters where income-like taxes move in the opposite direction and are at least half as large as the VAT move makes the response stronger, not weaker. Thus the VAT result is not driven by compensating VAT-income packages. The main attenuation occurs when the VAT object is broadened to include duties and other excise-like measures, closer to the Nguyen–Onnis–Rossi average-consumption-tax-rate object.

Table 21: VAT Sensitivity to Tax Packages and Broad Consumption-Tax Definitions

Specification	$h = 4$	$h = 8$	$h = 12$
Baseline VAT	-2.54 (0.104)	-4.72 (0.115)	-4.22 (0.130)
Drop UK 1979Q3 VAT	-2.36 (0.078)	-3.42 (0.087)	-2.78 (0.136)
Drop UK 1973Q2 and 1979Q3 VAT	-1.64 (0.269)	-3.17 (0.177)	-4.22 (0.106)
Drop opposite-sign VAT–income packages	-3.15 (0.110)	-5.33 (0.128)	-5.19 (0.118)
Drop full opposite-sign package quarters	-3.14 (0.117)	-5.37 (0.114)	-5.03 (0.092)
VAT plus duties/excise-like measures	-0.72 (0.114)	-1.62 (0.123)	-1.62 (0.163)

Note: Entries are pooled GDP endpoint responses at $p = 1$, expressed as $y_{c,t+h} - y_{c,t-1}$ in percentage points of trend GDP; p-values are in parentheses. The sample remains 1955Q1–2007Q4 in all rows. Opposite-sign VAT–income packages are country-quarters where VAT and income-like taxes move in opposite directions and the absolute income-like tax change is at least 50 percent of the VAT change. The broad consumption-tax row treats VAT, duties, and excise-like measures as one consumption-tax object.

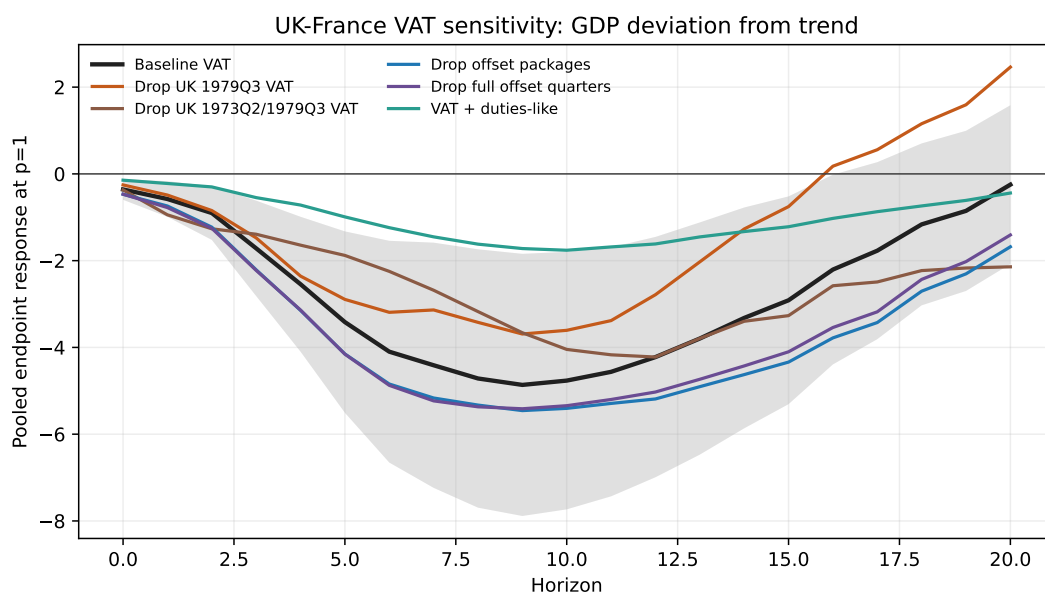


Figure 20: VAT GDP Response Under Package Sensitivity Checks

Note: The figure reports pooled GDP endpoint responses under the same sensitivity variants as Table 21. The shaded area is the 68 percent HAC band for the baseline VAT endpoint.

G.3 Country-Specific VAT Diagnostics

Figure 21 reports the same output and price endpoints separately for France and the United Kingdom on the common comparison sample. These country-specific estimates are supporting diagnostics rather than the main estimand. They show that the pooled GDP response is not averaging two opposite country patterns: GDP falls in both countries after high-support VAT shocks. The country-specific inflation responses are less precisely estimated, but both countries display a positive short-run price response.

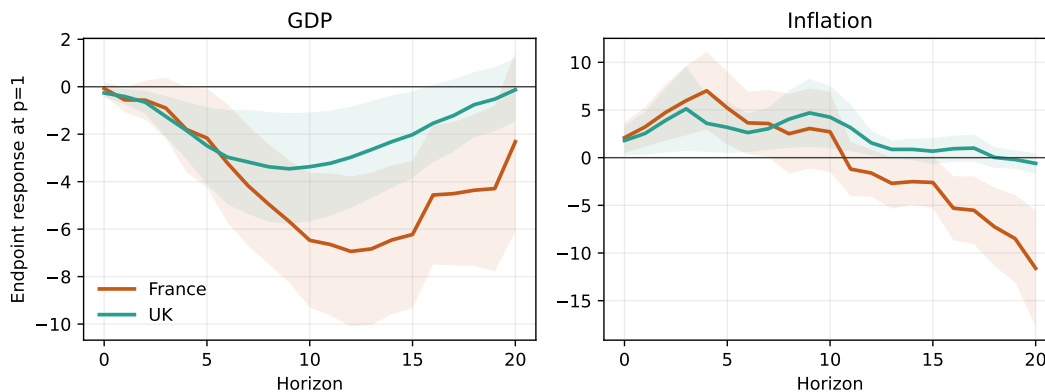


Figure 21: Country-Specific VAT Endpoint Responses

Note: The figure estimates the VAT endpoint separately for France and the United Kingdom on the common comparison sample, 1955Q1–2007Q4. GDP is expressed in percentage points of trend GDP; inflation is in percentage points. Responses are normalized by the country-specific linear-in- p denominator evaluated at $p = 1$. Shaded areas are 68 percent HAC confidence bands.

Table 22: Country-Specific VAT Endpoint Responses at Key Horizons

Country	Outcome	$h = 4$	$h = 8$	$h = 12$
France	GDP deviation from trend	-1.80	-4.95	-6.94
United Kingdom	GDP deviation from trend	-1.84	-3.37	-2.97
France	Inflation	7.02	2.51	-1.61
United Kingdom	Inflation	3.60	4.04	1.56

Note: Entries are country-specific high-support VAT endpoint responses. GDP is in percentage points of trend GDP; inflation is in percentage points.

G.4 VAT Adoption Timing

Table 23: VAT (or GST-equivalent) Adoption by Year in Developed Economies

Year	Countries
1954	France
1967	Denmark
1968	Germany
1969	Netherlands; Sweden
1970	Luxembourg; Norway
1971	Belgium
1972	Ireland
1973	Austria; Italy; United Kingdom
1977	South Korea
1980	Mexico
1985	Turkey
1986	New Zealand
1989	Japan
1990	Iceland
1991	Canada
1994	Finland
1995	Switzerland
2000	Australia